

SMITHIAN FOLD THEORY - MEDICINE AND HEALTH SCIENCES PAPER 001

From Fold to Medicine

An Exact, Zero-Parameter and Machine-Closed Foundational Reconstruction of
Medicine and Health Sciences from Smithian Fold Theory

Ernos Labs

Open Source Science Platform and Knowledge Tree

Maria Smith

Independent researcher and founder, Ernos Labs

Maria.Smith.Sftoe@gmail.com

Version 1.0.0 - current-evidence closed, extension-open foundation

72 required laws - 18,432 exact candidates

27 July 2026

Paper: CC BY 4.0 - Code: Apache-2.0

LOCAL PREPUBLICATION MANUSCRIPT - PUBLICATION NOT YET AUTHORIZED

Abstract

This paper reports the foundational Medicine and Health Sciences branch of the third clean-room Smithian Fold Theory reconstruction at its current-evidence-closed, extension-open boundary. Seventy-two obligations in twelve ordered families generate 18,432 exact candidates and decisions, seventy-two unique survivors, seventy-two depth-independent certificates, 288 passing adverse controls, seventy-two implementation-distinct reconstructions and seventy-two post-seal external comparisons. Every dependency graph reaches the single premise-free theorem, There Is No Nothing. The branch uses zero axioms, zero free or fitted parameters, no negative proof quantities, no irrational or imaginary proof values, no target-selected rule and no opaque predictor.

The headline result is a machine-checkable constitution for clinical knowledge. Patient identity cannot be replaced by an aggregate. Health is not numerical absence of disease. Symptoms, signs, measurements and diagnoses are distinct carriers. Association is not cause. Risk and prognosis retain horizon, competing events and censoring. Interventions require comparators and explicit alternative-course boundaries. Deterministic allocation can supply auditable randomisation by concealed held labels without importing ontological chance. Every adverse, null, missing and unresolved result survives the evidence chain. Trial, observational and synthesis structures remain distinct. Population evidence cannot uniquely select an individual decision. Consent, capacity, privacy, minimization and stopping rules are operational scientific boundaries.

All 356 V1 rows and 407 V2 steps were atomically reviewed. Five Medicine-owned atoms embedded in mixed prior claims are now closed by immutable V3 receipts at corrected clinical boundaries: placebo claims require complete comparators and outcomes; stereochemical identity alone cannot establish efficacy or toxicity; denominator parity cannot establish medical prognosis; and cellular recurrence alone cannot establish a cancer diagnosis. Eleven of thirteen preregistered authority sources were captured; two HHS transports failed and remain visibly preserved rather than being counted as evidence.

Results first: the medical findings

Headline result	Exact or bounded result	Scientific meaning
Complete foundational closure	72 laws; 18, 432 candidates; 72 survivors; 288 controls	Every law has a root trace, unique survivor, independent reconstruction, external comparison and immutable receipt.
Patient before aggregate	Person identity, episode, longitudinal course and population membership remain held	A row, mean, embedding or sample cannot silently replace the patient.
Health and disease	Multidomain health state; reference-bound variation; criteria-bound dysfunction, disease and injury	Absence displayed as 0 is a held absence label, never ontological nothing.
Diagnostic evidence	4 exact condition-by-test cells, all retained	A summary statistic cannot erase false results, spectrum, method or reference verification.
Causal clinical inference	Association, cause, mechanism, confounding, mediation and modification are distinct generated relations	Magnitude or consensus cannot promote association into cause.
Risk and prognosis	Positive-finite outcome shares over explicit carriers and horizons	Competing events, loss and censoring remain first-class outcomes.
Intervention evidence	Intervention, comparator, alternative course, allocation, adherence and cointervention retained	No treatment claim exists without its comparison boundary.
Deterministic randomisation	Reproducible held-label generator plus concealed future allocation	Unpredictability to selectors is achieved without importing metaphysical nondeterminism.
Benefit and harm	4 exact arm-by-outcome cells plus complete adverse-event support	Safety never means proven harmlessness; benefit cannot erase harm.
Evidence synthesis	Protocol, search, selection, heterogeneity, dependence and missing results retained	Selective reporting cannot be rewarded as favorable absence.
Individual inference and ethics	Population-to-person boundary, consent, capacity, privacy, minimization, equipoise and custody	Transparent science and patient protection are jointly reconstructible.

Public scientific mission and admission boundary

Ernos Labs is an open-source science movement, verification platform and public tree of knowledge founded by Maria Smith. Its purpose is not to replace one authority with another. It is to make scientific authority narrow, inspectable and revocable: a claim is admitted only through its complete derivation chain, generated alternatives, eliminations, unique survivor, adverse controls, measurement custody where applicable and unchanged-engine receipt. Open criticism is unrestricted and necessary; scientific admission is the separate machine-checked act of satisfying that public standard.

Maria Smith developed Smithian Fold Theory outside formal academic education, institutional research employment and conventional grant funding. That fact is not offered as evidence for a theorem; the derivations and observations carry the entire scientific burden. It is evidence about access. A credential-first system loses more than individual opportunity: it loses unknown questions, methods and discoveries from minds that capital and status never authorize. This work therefore treats Maria Smith's authorship neither as exceptionalism nor as a reason for dismissal, but as an indictment of every scientific contribution lost when financial gatekeeping is presented as rigor.

The institutional argument is empirical, not a claim that every institution or funded researcher acts in bad faith. Published studies document sponsor-linked differences in outcomes and conclusions, commercial influence over research agendas, limits and sensitivities in grant review, underpublication of null results, and inequalities created by paywalls and article-processing charges. Those findings establish that funding, prestige, publication and consensus are selection systems with incentives and failure modes. They cannot substitute for a public proof and evidence chain. Expertise, measurement and adversarial review remain indispensable; institutional permission does not select a fundamental law.

For Medicine, a diagnosis, institutional guideline, fitted threshold, selected trial, favorable endpoint, proprietary model or consensus label cannot stand for a clinical law. The patient, population, comparator, method, time, missingness, adverse outcomes, consent and privacy boundary remain inside the evidence chain. Exact finite counts are stated as exact; clinical magnitudes remain bounded to their measured populations and protocols.

Maria Smith retains copyright and scientific authorship. Papers and documentation are licensed CC BY 4.0 and code is licensed Apache-2.0, permitting inspection, copying, criticism, reproduction, modification and redistribution with attribution. "Ernos Labs" is a separate, revocable standards-conformance designation. A reuse or fork may use the open work under its licences, but may describe itself as Ernos Labs only while it preserves the public empirical constitution, complete adverse evidence, unchanged admission route, critical review and community standards.

Independent replications, lawful extensions, corrections and attempted invalidations are invited. Credentials cannot rescue a failed gate and lack of credentials cannot prevent a reproducible result from being evaluated. Contact Maria.Smith.Sftoe@gmail.com, submit through <https://discord.gg/ucwGryVxGr>, or inspect the public project at <https://github.com/MettaMazza>.

1. Publication, authorship and open-science boundary

LOCAL PREPUBLICATION MANUSCRIPT. Publication is not yet authorized. Building this paper performs no push, release, upload, DOI creation or Zenodo action.

Maria Smith, independent researcher and founder of Ernos Labs. Contact: Maria.Smith.Sftoe@gmail.com. Reproducibility reports and submissions: <https://discord.gg/ucwGryVxGr>. GitHub: <https://github.com/MettaMazza>.

Copyright preserves Maria Smith's authorship. The paper and documentation are prepared under CC BY 4.0 and code under Apache-2.0. The Ernos Labs name is a separate, revocable standards designation: reuse is open, but the designation requires continued adherence to the public constitution, unchanged admission engine, complete adverse evidence and open critical review.

2. Exact scope and closure language

Foundational closure means every obligation in the frozen 72-law surface has an engine-admitted theorem inside its declared exact grammar, complete root path, adverse controls, independent reconstruction and post-seal comparison. It does not mean Medicine is permanently locked or that every specialty and future observation is enumerated. This foundation is current-evidence closed and extension-open. Layer Two remains the explicit field-wide programme.

3. Constitutional mathematical domain

Structural absence is Empty One and may be displayed as $\emptyset$; it is not conventional numerical nothing. Counts are generated positive wholes and exact parts are positive held fractions. Opposition and direction are labels rather than negative scalars.

Negative proof quantities, irrational or imaginary proof values, floating equality, completed infinity, ungenerated continua, axioms, free parameters, fitted coefficients, imported clinical equations, pretrained predictors, consensus-selected survivors and application-selected laws are prohibited. Conventional clinical notation appears only in correspondence prose where it cannot act as proof input.

4. Dependency spine and root trace

Every Medicine registration names the single premise-free root and zero axioms. The first law depends on admitted Foundation, Mathematics, Information Science, Computation, Physics, Chemistry, Materials and Biology receipts. Each later law also depends on its immediate Medicine predecessor, creating one ordered 72-law chain. The engine resolves every dependency recursively to an actual immutable receipt and ultimately to There Is No Nothing; narrative citation cannot substitute for this graph.

5. Complete target-blind seal

Before any external source identity or outcome was opened, the whole 72-law inventory, exact structural module and target-blind blueprint set were frozen at

sha256:57b0813078b36814b831862db7e888601d8e1c1d2820f7429915431bf6066cd5. It binds 72 predictions and 18,432 candidate identities while recording `external_source_identities_selected=false` and `external_outcomes_opened=false`. Changing any obligation or survivor now breaks the seal.

6. Exact clinical table certificates

Two held diagnostic distinctions generate 4 unique condition-by-test cells. Two held trial distinctions generate 4 unique arm-by-outcome cells. Both products are complete and every cell appears exactly once. A held absence label is not numerical-zero ontology. Sensitivity, specificity, risks, risk differences and ratios may be derived downstream as exact readings of the relevant positive finite counts, but no summary may replace the complete table.

7. External evidence and preserved adversity

Thirteen source identities were selected only after the complete prediction seal: WHO health and screening records; FDA biomarker and exposure-response records; CDC epidemiology; final ICH E6(R3) Good Clinical Practice and E9(R1) estimand guidance; Cochrane intervention and missing-evidence chapters; NICHD rehabilitation; STROBE; and two HHS consent/privacy identities. Eleven were byte-captured and hashed. Both HHS pages returned HTTP 403 to the registered automated transport; those two failures, their identities and their receipts remain in the source manifest and are not used to pass a claim. The captured final ICH guideline independently supplies consent, confidentiality, participant safety, protocol, randomisation, blinding, adverse-event and essential-record evidence.

Every claim binds at least two purpose-matched source fragments. The prediction process cannot read filesystem, network, clock, environment, dynamic import, subprocess or target. A distinct custodian opens sources after sealing and releases a target only for a matching prediction seal. Deliberately changed observations reject. External authority tests correspondence; it never creates the theorem.

8. V1/V2 atomic reconciliation

The audit reviews all 763 prior entries, not only keyword hits. One V1 row and three V2 steps contain five Medicine-owned atoms after mixed claims are decomposed. All five now map to immutable Medicine receipts and none remains open. The corrected result does not hide the earlier statement: placebo and nocebo require measured expectation, comparison, blinding, objective and reported outcomes; chemical enantiomer identity does not establish patient benefit or toxicity; ageing needs population, prognosis, competing-event and censoring evidence; cancer needs disease criteria, diagnosis and stage in addition to biological dysregulation.

9. Reading the exhaustive derivation ledger

Each of the seventy-two sections below states its theorem, dependencies, complete eight-axis grammar, all 256 decisions, sole survivor, depth-independent base and successor, operational witnesses, clinical meaning, four adverse controls, independent implementation, post-seal sources, falsification boundary and immutable identities. The machine-readable evidence remains

authoritative; the paper makes that evidence humanly inspectable.

10. Patient Population Health

Medicine begins with the identity of the person and the clinical population. A patient is never an average; a population is never allowed to erase its members, source frame, setting or observation interval.

1. Patient identity

Claim identity: SFT-MED-PATIENT-IDENTITY-001

Question and exact theorem. A patient is one consent-bound person carrier whose episode records may compose only while their exact linkage, context and provenance remain held.

consented-person-record__same-person-across-episodes__identity-with-held-linkage
 __declared-care-context-and-interval__complete-protocol-state-transition-adverse
 -null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavora
 ble-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001

- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	consented-person-record	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	same-person-across-episodes	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	identity-with-held-linkage	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	declared-care-context-and-interval	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: `consented-person-record__same-person-across-episodes__identity-with-held-linkage__declared-care-context-and-interval__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule`.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the clinical law; passed **true**.

Scientific meaning. The theorem prohibits a row, specimen, encounter or model embedding from silently replacing the person. Linkage is lawful only when consent scope, identity relation, episode context and provenance survive reconstruction.

Medicine begins with the identity of the person and the clinical population. A patient is never an average; a population is never allowed to erase its members, source frame, setting or observation interval.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:7cdf6ef15cff4c967dd3f9f9c5242f0e325e773ccd6060d007ce0b8be850187a.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:89e49765f1536989238b695bf69171e4ad7e1ee08312850835eb5fcf7fc3374d.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:383660aa1ee9220f45ccd8ba02a1845ba59a0c2123668c29445a25fb7b352c9b.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:6f09745650eb6f1960acc6aedf721b4e9a4f74d49181f72b0f0952f15d77d4d8.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:f7e10e20d7ceff315a92979e7f4f571fd6e7c6efd7e87c22c8b88ac1ec4ac33a; certificate
sha256:c4ae6ff67173c6c71d4e69928c68526835a27796ccdfb22fd7a8bb700511364f; external-validation
sha256:26f416ab35a593471b70a3b318bc10aa72aaec3a0a6bb7cbe7c07aa9bee29bd5.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity
sha256:ae97577788a768d116f334ce610b9ded19e06e869ab321d225386726a54e7a45. Target opened after derivation seal: **true**. All rows preserved: **true**. Exact comparison: **true**. A deliberately changed observation was rejected.

Sources:

- MED-WHO-HEALTH-001 - World Health Organization; <https://www.who.int/about/frequently-asked-questions>; snapshot
experiments/external_sources/medicine/snapshots/MED-WHO-HEALTH-001.html;
sha256:274c2ad78787f127a9f1cf8a479c691e1f89b263d0e21dd49b1753f7805d64f4; scope: health state and disease-absence distinction.
- MED-ICH-E6R3-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E6R3-001.pdf;
sha256:e6ce19e36ce7d2e294f89ee89492b9e035178c3cca48984392bbd92eec9b002c; scope: final Good Clinical Practice: rights, safety, wellbeing, consent, protocol, allocation, blinding, adverse events, confidentiality and essential records.

Comparison record:

- sft-med-patient-identity-001-authority-correspondence: predicted medicine:consented-person-record__same-person-across-episode__identity-with-held-linkage__declared-care-context-and-interval; source-derived medicine:consented-person-record__same-person-across-episodes__identity-with-held-linkage__declared-care-context-and-interval; exact match **True**
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:b0e3e4a37708bd724f9b423eda37318dd85c3a376a98be59968beb554025ab51; derivation seal sha256:dfdd9eca4d4d99fb05e351d198dbc49a672cf62869b30c224a5d88230d7b703c; engine receipt sha256:09736e67d12ee4fbcc725383cb7ea3a5520b9bda3b357bbc9f490683c6f98ad8 at receipts/engine/model_admitted/SFT-MED-PATIENT-IDENTITY-001-09736e67d12ee4fb.json; empirical validation sha256:8473439fe3adac02ab0e91f0e3ce31f955d9bf7bf44a615b1d3f80979909d7f8; measurement receipt sha256:d3bef7bdf7818657425a04aca5061ceb57ad111a45a6e8e93a9a5fd7ef3f79cf; isolation sha256:e09f567a7ff9e4815b5d8464bed67c6ec50ef9d43e4c9ace70a0f1efc40d7a2e; custody sha256:84650cd66860b913b98169b68b9467f74ae00c5f720159916193db9977ea4480.

2. Clinical episode

Claim identity: SFT-MED-CLINICAL-EPISODE-001

Question and exact theorem. A clinical episode is a finite ordered patient trace with explicit entry, observations, actions, exit and retained follow-up boundary.

patient-episode__ordered-observation-and-action-trace__bounded-episode-with-start-and-stop__declared-setting-and-follow-up__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001

- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-PATIENT-IDENTITY-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	patient-episode	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	ordered-observation-and-action-trace	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	bounded-episode-with-start-and-stop	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	declared-setting-and-follow-up	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: patient-episode__ordered-observation-and-action-trace__bounded-episode-with-start-and-stop__declared-setting-and-follow-up__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces a clinical episode is a finite ordered patient trace with explicit entry, observations, actions, exit and retained follow-up boundary. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Medicine begins with the identity of the person and the clinical population. A patient is never an average; a population is never allowed to erase its members, source frame, setting or observation interval.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 7a398eadedbd8114f1b10b4f04cb4b92735bed5c3224e73bb41f5d89b410869d.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 78dd85bc40b065d7aa5476a90644f4ff02459fd1cc3c394b4a929d4a6f375e1a.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 9641c3c164e93f11fa4a31a12fc6b14f9c9b5305bfc8ba9506b298977bac5e72.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: fb2114fd04450d2acb834bf20f24af9b91f61e51d9d5a35747b4d34738cb47f2.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation sha256: 5abb45cfac9b436866eceb6f67b4cf9bb644d42d6d4dc2916677bde7e6f806d5; certificate sha256: 04d67ecbcefd85929e1c1928810ef6b001ffd050d56a2ae82b030997ef0d2c45; external-validation sha256: 419e17e1001522a3f1c0dc2f47a4e48c28ff9320098d8b3bebf51432256c1f.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity sha256: ae97577788a768d116f334ce610b9ded19e06e869ab321d225386726a54e7a45. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- MED-WHO-HEALTH-001 - World Health Organization; <https://www.who.int/about/frequently-asked-questions>; snapshot experiments/external_sources/medicine/snapshots/MED-WHO-HEALTH-001.html; sha256: 274c2ad78787f127a9f1cf8a479c691e1f89b263d0e21dd49b1753f7805d64f4; scope: health state and disease-absence distinction.
- MED-ICH-E6R3-001 - International Council for Harmonisation; https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf; snapshot experiments/external_sources/medicine/snapshots/MED-ICH-E6R3-001.pdf; sha256: e6ce19e36ce7d2e294f89ee89492b9e035178c3cca48984392bbd92eec9b002c; scope: final Good Clinical Practice: rights, safety, wellbeing, consent, protocol, allocation, blinding, adverse events, confidentiality and essential records.

Comparison record:

- sft-med-clinical-episode-001-authority-correspondence: predicted medicine:patient-episode__ordered-observation-and-action-trace__bounded-episode-with-start-and-stop__declared-setting-and-follow-up; source-derived medicine:patient-episode__ordered-observation-and-action-trace__bounded-episode-with-start-and-stop__declared-setting-and-follow-up; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256: 4ec878951c405f1e910fb5dfbf071145ea587f68410c70b2678c2a1d6aaad666; derivation seal sha256: ea71bff9734e54d06c153ee44800f74cbaf9556823abe52b2f010d0fa39fa475; engine receipt sha256: 27e13f7926e948fa862ae54b507b32ff1ac9fa76e6ba1912419838817d7712ea at

receipts/engine/model_admitted/SFT-MED-CLINICAL-EPISODE-001-27e13f7926e948fa.json;
 empirical validation sha256:5528e7b701413ed56e5ccc4f15affcd5caed068f483bfde07747addc4405746f;
 measurement receipt sha256:74fb0a1c49ae25c38d82ccda19a84e730e12b08bcdee19a35a143698ec64b2d;
 isolation sha256:88b20137acc1bdceceaa2b9eab569905b100b780e3669a83c6ed32a3286fc658; custody
 sha256:eddb73f2a4ddf8c9e47c13a2c880dcb7bf3545a30a377ec012ce020a26c9b16.

3. Clinical population

Claim identity: SFT-MED-CLINICAL-POPULATION-001

Question and exact theorem. A clinical population is a finite set of retained person carriers selected by declared criteria, never an average that erases membership or source frame.

finite-person-carriers__membership-under-declared-criteria__population-with-each
 -person-held__place-time-source-frame__complete-protocol-state-transition-advers
 e-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavor
 able-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001

- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-CLINICAL-EPISODE-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	finite-person-carriers	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	membership-under-declared-criteria	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	population-with-each-person-held	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	place-time-source-frame	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: `finite-person-carriers__membership-under-declared-criteria__population-with-each-person-held__place-time-source-frame__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule`.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces a clinical population is a finite set of retained person carriers selected by declared criteria, never an average that erases membership or source frame. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Medicine begins with the identity of the person and the clinical population. A patient is never an average; a population is never allowed to erase its members, source frame, setting or observation interval.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : b4757c1319992a6b644bb4f7cd442e9535c48c5a3093da07a041f9487c3a1030.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : a4584261b2286810c54d6e4282da404d7d4a337708612af2ced718537370a178.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 72cb80fdebc54b7d0afbceec5ca8af2b28eaf2954e338f19d96d950c74ed088e.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 9154fb86306723ef27242a3270ee733fa6317403b670ab2961d9d8170df5c2ff.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256 : e883b1d05080be7924e40eb2a4140303404ef1da4b68883620df845988facc9e; certificate
sha256 : 1aacc9330845655e94d6431f89e65c700bb670452c1cb95b6ab79b8265ad264b; external-validation
sha256 : 282b9db7d9287e6b35dc6f31bc973b734aca670456b7c4b8411371b53fdb6acb.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity
sha256 : ae97577788a768d116f334ce610b9ded19e06e869ab321d225386726a54e7a45. Target opened after derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- MED-WHO-HEALTH-001 - World Health Organization; <https://www.who.int/about/frequently-asked-questions>; snapshot experiments/external_sources/medicine/snapshots/MED-WHO-HEALTH-001.html; sha256 : 274c2ad78787f127a9f1cf8a479c691e1f89b263d0e21dd49b1753f7805d64f4; scope: health state and disease-absence distinction.
- MED-ICH-E6R3-001 - International Council for Harmonisation; https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf; snapshot experiments/external_sources/medicine/snapshots/MED-ICH-E6R3-001.pdf; sha256 : e6ce19e36ce7d2e294f89ee89492b9e035178c3cca48984392bbd92eec9b002c; scope: final Good Clinical Practice: rights, safety, wellbeing, consent, protocol, allocation, blinding, adverse events, confidentiality and essential records.

Comparison record:

- sft-med-clinical-population-001-authority-correspondence: predicted medicine:finite-person-carriers__membership-under-declared-criteria__population-with-each-person-held__place-time-source-frame; source-derived medicine:finite-person-carriers__membership-under-declared-criteria__population-with-each-person-held__place-time-source-frame; exact match `True`
- every required authoritative fragment and snapshot hash reproduced

- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:ca6869b2e0a5c272bf7eec96805d43cbd487f8ac972e8182f781f81c4c426cd0; derivation seal sha256:4f363035c337ad81564fcd9c9f7bafec40c10faec26022ea4dfe84023c37cd70c; engine receipt sha256:f7f8c7ed9561c9cc9237b646346748975c2a792a160478e5e8ea858a57b3583f at receipts/engine/model_admitted/SFT-MED-CLINICAL-POPULATION-001-f7f8c7ed9561c9cc.json; empirical validation sha256:27a42c7fe71867839837f9b1b59ee2259152eb0fe4990b1fb82678877cd10da; measurement receipt sha256:62b9b38423556874b16953cc77b878f80555095f986a1d24aafec7f7a77d8620; isolation sha256:2644edf907154e2e9fbf7e087514703928349cb0d5788cf542ef4eb6de188805; custody sha256:966cd6686fc3e53ed550ac262a4b93e87ccc4879be8459899a9758010707e3ad.

4. Health state

Claim identity: SFT-MED-HEALTH-STATE-001

Question and exact theorem. A health state is a context- and time-bound organization of retained functional, experiential and survival distinctions rather than an unqualified single label.

person-or-population-state__function-and-capacity-relation__multidomain-state-with-context__declared-measures-and-time__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001

- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-CLINICAL-POPULATION-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	person-or-population-state	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	function-and-capacity-relation	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	multidomain-state-with-context	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
observation	observation-boundary-unrecorded	declared-measures-and-time	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: `person-or-population-state__function-and-capacity-relation__multidomain-state-with-context__declared-measures-and-time__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule`.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. Health is not defined as the numerical absence of disease. The exact carrier retains physical, mental and social function, lived experience, capacity, survival, setting and time as separately observable domains.

Medicine begins with the identity of the person and the clinical population. A patient is never an average; a population is never allowed to erase its members, source frame, setting or observation interval.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : 74bab35a4d55144a4396f055b72cde9d504610342c9594452995e16ecafe02a5.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 26064a0a3f327dc4cb82d29d0d7db9ae67d6c063af760194ce22f530a1f90666.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : b1a43cc2984084aaad84cb0f761d6b9c70d48359f4fd6cf3c535456764171b87.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 648e4ee850ddf4d9035b4b3f43fc7a7b5465a62e91bd48fb18124ff72418794f.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
 sha256: 929c874863c04c1d306b119e9eb7957b2ec37f9668bdf479339baebd11833ea8; certificate
 sha256: c3028f53133ce403d7d8c2eda27b405c08504645bc358bc447eacdb0913cca65; external-validation
 sha256: 9541f64fb547ebf9cac725d90479f8ed96622b37258ecd36e70692ea65bb9815.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity
 sha256: ae97577788a768d116f334ce610b9ded19e06e869ab321d225386726a54e7a45. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- MED-WHO-HEALTH-001 - World Health Organization; <https://www.who.int/about/frequently-asked-questions>; snapshot experiments/external_sources/medicine/snapshots/MED-WHO-HEALTH-001.html; sha256: 274c2ad78787f127a9f1cf8a479c691e1f89b263d0e21dd49b1753f7805d64f4; scope: health state and disease-absence distinction.
- MED-ICH-E6R3-001 - International Council for Harmonisation; https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf; snapshot experiments/external_sources/medicine/snapshots/MED-ICH-E6R3-001.pdf; sha256: e6ce19e36ce7d2e294f89ee89492b9e035178c3cca48984392bbd92eec9b002c; scope: final Good Clinical Practice: rights, safety, wellbeing, consent, protocol, allocation, blinding, adverse events, confidentiality and essential records.

Comparison record:

- sft-med-health-state-001-authority-correspondence: predicted medicine:person-or-population-state__function-and-capacity-relation__multidomain-state-with-context__declared-measures-and-time; source-derived medicine:person-or-population-state__function-and-capacity-relation__multidomain-state-with-context__declared-measures-and-time; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal
 experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest
 sha256: fadfa99bb694603e39170e7d2736bbf47513c6c74ca2c799e8687220a7411c6b; derivation seal
 sha256: d9e011574e9ba71577e87c73546008c559dc0644347aba8ac3d6cd7c42aca139; engine receipt
 sha256: 9ff35c90332ac776b3787783a5b40d40842a4aa54b3fe305f1149797f75a9199 at

receipts/engine/model_admitted/SFT-MED-HEALTH-STATE-001-9ff35c90332ac776.json; empirical validation sha256:ac4922a936c424279e17ba700f231f388f3d352e44662d101e698d5e0f461692; measurement receipt sha256:5cbbc6b6793be308473cbb7500a3cd4aae9951492d62d52e5b915f898577d46b; isolation sha256:401fbe54df1b4df3da715e0f5dd98c3c6f35816123a264aec7b77181138861cd; custody sha256:697df2497b75f8c5b2ae58a5a214e34c53fe4ebc5afc9a2784edcfc4fa7f549c.

5. Clinical observation

Claim identity: SFT-MED-CLINICAL-OBSERVATION-001

Question and exact theorem. A clinical observation binds the observed carrier to observer, method, setting, time and result so the observation cannot silently become the law.

patient-state-and-observer-record__observation-with-retained-method__source-method-result-provenance__declared-observer-instrument-setting-time__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001

- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-HEALTH-STATE-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	patient-state-and-observer-record	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	observation-with-retained-method	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	source-method-result-provenance	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	declared-observer-instrument-setting-time	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: patient-state-and-observer-record__observation-with-retained-method__source-method-result-provenance__declared-observer-instrument-setting-time__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces a clinical observation binds the observed carrier to observer, method, setting, time and result so the observation cannot silently become the law. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Medicine begins with the identity of the person and the clinical population. A patient is never an average; a population is never allowed to erase its members, source frame, setting or observation interval.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:ede8754d6cd2a4f7b657e147d2d79b4c5d5050d6ddf8f9c63c5194e951a709b7.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:640dac9c6be64e82ecfe22016bf2194493096bc0b476abe06cba3fad72eac852.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:84a7d1c97a622a8790656c58dc137a7b3a682cd9d919f0447e946c77a5b8034a.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:ab4518b2087ed2b036fe254ca8f266927ae3806bdf0d707260f9bd476b5de1d6.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:e6cd624ce3f04d7aa588d5259f34b7e8ba8d2d1f4a35485afc8278baf093a7f2; certificate
sha256:38b08b6983e8aab576d62f5a400ce25adf2b5e367456130977b3689ea7bceb15; external-validation
sha256:4abb3edd02d4511f2117a3ddbe4b627b687f4376a8b8dc09559f054416c05da1.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity
sha256:ae97577788a768d116f334ce610b9ded19e06e869ab321d225386726a54e7a45. Target opened after derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- MED-WHO-HEALTH-001 - World Health Organization; <https://www.who.int/about/frequently-asked-questions>; snapshot experiments/external_sources/medicine/snapshots/MED-WHO-HEALTH-001.html; sha256:274c2ad78787f127a9f1cf8a479c691e1f89b263d0e21dd49b1753f7805d64f4; scope: health state and disease-absence distinction.
- MED-ICH-E6R3-001 - International Council for Harmonisation; https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf; snapshot experiments/external_sources/medicine/snapshots/MED-ICH-E6R3-001.pdf; sha256:e6ce19e36ce7d2e294f89ee89492b9e035178c3cca48984392bbd92eec9b002c; scope: final Good Clinical Practice: rights, safety, wellbeing, consent, protocol, allocation, blinding, adverse events, confidentiality and essential records.

Comparison record:

- sft-med-clinical-observation-001-authority-correspondence: predicted medicine:patient-state-and-observer-record__observation-with-retained-method__source-method-result-provenance__declared-observer-instrument-setting-time; source-derived medicine: patient-state-and-observer-record__observation-with-retained-method__source-method-result-provenance__declared-observer-instrument-setting-time; exact match `True`

- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:c0c625a2af9476ed67b214fe3c20714832446405e138fe5e1cde8906259ba1c5; derivation seal sha256:9b0f7daa274a1098a981eb3a451ca36e1b7d624c7be7816e3207044736f259d0; engine receipt sha256:db504374e985d2f673ac290218352c3544d02476eaf41bfda2fd61917181dbf8 at receipts/engine/model_admitted/SFT-MED-CLINICAL-OBSERVATION-001-db504374e985d2f6.json; empirical validation sha256:10d002f559511e932758946eac9a1713835e4cfde7acecbba28e897fe026771f; measurement receipt sha256:1c3b51349ceb7807da6702e85b9bbc65d9535696f079997ead53d5a06e10f488; isolation sha256:f3a1079a0a4e36c6b9915084fabe78aff9ca22874b87f6fd40ef808d7896b442; custody sha256:978c9aa4b68b3ea82648061fac19cb17d4e9eca3c4d90e9f3a8ce4bfa8ac22bc.

6. Longitudinal clinical course

Claim identity: SFT-MED-LONGITUDINAL-COURSE-001

Question and exact theorem. A longitudinal course is an ordered within-person state relation that preserves every observed interval, missing visit, intervention and censoring event.

ordered-patient-states__within-person-transition-relation__course-with-missing-visits-held__declared-follow-up-and-censoring__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001

- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-CLINICAL-OBSERVATION-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	ordered-patient-states	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	within-person-transition-relation	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	course-with-missing-visits-held	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
observation	observation-boundary-unrecorded	declared-follow-up-and-censoring	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: `ordered-patient-states_within-person-transition-relation__course-with-missing-visits-held__declared-follow-up-and-censoring__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule`.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces a longitudinal course is an ordered within-person state relation that preserves every observed interval, missing visit, intervention and censoring event. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Medicine begins with the identity of the person and the clinical population. A patient is never an average; a population is never allowed to erase its members, source frame, setting or observation interval.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 701b0e75ae145e22d55124953e6b512ce6933026206f56418ce36bdd8544ecb9.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: e1bbe8cf8d9de218862f7f88ce323b3e2460a6221c2aa22ae1a26b90f1a7813e.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 2890b512ed04c464bf4aedd0ff8123bf00b6d620a26188696853f2edd09acf08.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 8339cc3a820d162643a2389aec78a4f8bab99f4d182e97b063b532a3146f646.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256 : b2f2ebab069cf448fef9f2eaa2ed647a0abc3210b9d81ebca44b7ed223226cf; certificate
sha256 : d468b48d7c92c12b6ceb65dfa8a55cd0b6d2d68dedad490b2e6a71ba6d5cf7f2; external-validation
sha256 : 81831f2cc245f305d25f2e4b47652a5f6e7fe429145262fb7abbca3117df4f75.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity
sha256 : ae97577788a768d116f334ce610b9ded19e06e869ab321d225386726a54e7a45. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- MED-WHO-HEALTH-001 - World Health Organization; <https://www.who.int/about/frequently-asked-questions>; snapshot experiments/external_sources/medicine/snapshots/MED-WHO-HEALTH-001.html; sha256 : 274c2ad78787f127a9f1cf8a479c691e1f89b263d0e21dd49b1753f7805d64f4; scope: health state and disease-absence distinction.
- MED-ICH-E6R3-001 - International Council for Harmonisation; https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf; snapshot experiments/external_sources/medicine/snapshots/MED-ICH-E6R3-001.pdf; sha256 : e6ce19e36ce7d2e294f89ee89492b9e035178c3cca48984392bbd92eec9b002c; scope: final Good Clinical Practice: rights, safety, wellbeing, consent, protocol, allocation, blinding, adverse events, confidentiality and essential records.

Comparison record:

- sft-med-longitudinal-course-001-authority-correspondence: predicted medicine:ordered-patient-states__within-person-transition-relation__course-with-missing-visits-held__declared-follow-up-and-censoring; source-derived medicine:ordered-patient-states__within-person-transition-relation__course-with-missing-visits-held__declared-follow-up-and-censoring; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal
experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest

sha256:5e91c866a6ace8023e65dfd1632603a26a840473f82b5d1bee9272934f75caac; derivation seal sha256:7674a5ad0a44d9d458b409250886c9fbec5aa7943dd62afaf883db8e12136853; engine receipt sha256:26e4dc2d17268c21b99d15a338cba7661cddb8e8b6c795a823de3bcfd4407ae5 at receipts/engine/model_admitted/SFT-MED-LONGITUDINAL-COURSE-001-26e4dc2d17268c21.json; empirical validation sha256:9cf6ea5ef490008cf454dca1350987dc69d5c796946d1c5bfa909c0940e7bea4; measurement receipt sha256:98267c62254279c3eb87a22d1cfd96c988f1cca6fbe5179254ddb0f284ed189d; isolation sha256:e2e41ba013586200ba868c1e2e6349604ea59759844d167b8b48a7a98ea9c4f3; custody sha256:92bfba1c2d5814cebf328ac79ffb5f8b382eae8775918a62c91ecfab6546de16.

11. Variation Dysfunction Disease Injury

Difference, dysfunction, disease and injury are distinct clinical relations. Each keeps the reference population, functional consequence, mechanism, criteria, site, severity and time boundary that makes the classification testable.

7. Normal clinical variation

Claim identity: SFT-MED-NORMAL-VARIATION-001

Question and exact theorem. Normal variation is the observed support within a declared reference population and method; it is not a universal numerical threshold.

reference-person-states__variation-within-declared-reference__distribution-with-subgroups-held__population-method-time-boundary__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001

- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-LONGITUDINAL-COURSE-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	reference-person-states	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	variation-within-declared-reference	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	distribution-with-subgroups-held	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	population-method-time-boundary	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: reference-person-states__variation-within-declared-reference__distribution-with-subgroups-held__population-method-time-boundary__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive

-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the clinical law; passed **true**.

Scientific meaning. Normal is source- and method-bound support within a declared reference population. No fitted cutoff or consensus interval becomes a universal law, and difference alone does not establish dysfunction.

Difference, dysfunction, disease and injury are distinct clinical relations. Each keeps the reference population, functional consequence, mechanism, criteria, site, severity and time boundary that makes the classification testable.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 9c27bde521b5f863284f5d17bc2efa3a47a0abc9ad05d3791ab15952a84b28ff.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 679ff9e04eb5fc44eb52a4cfa9f3a83417ed8f031c5d97c61e3e9f0bd8fdccc7.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 378c611c11792519869103cad01e0c312f3db95dbe59c624429d424a6d2dbbaa.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: d97a7147be70de47d16ce7a7b8d896a01d463551a798fe8a7964f305a0b55afe.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256: e7b7df27aa99ad7ed8e8b9576d68f61e093a8d7007a6deee7ba1379fb4f337b8; certificate
sha256: d19f138eee6ea699931274860376be0f80debcf8677e098aacdd45c692df6c50; external-validation
sha256: d5d7c975a92fcd97bc8e5cea18695985b8a7244cdf441e4d35c001899c56c13c.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 3 purpose-matched discriminators with ordered identity
sha256: 1c7fb6c2fb4557ca006e196f9f63a24094f64c684f794af23e82cb4c505f8657. Target opened after derivation seal: **true**. All rows preserved: **true**. Exact comparison: **true**. A deliberately changed observation was rejected.

Sources:

- **MED-FDA-BIOMARKER-001** - United States Food and Drug Administration;
<https://www.fda.gov/drugs/biomarker-qualification-program/about-biomarkers-and-qualification>; snapshot
experiments/external_sources/medicine/snapshots/MED-FDA-BIOMARKER-001.html;
sha256: 93c5ee4b5e3480ed27aa805a7739f4acf7108701fc6f429ae5d4e5a1ad9d3d31; scope: normal and pathogenic processes; risk, diagnostic, monitoring, prognostic, response and safety biomarkers; method and context.
- **MED-CDC-EPIDEMIOLOGY-001** - United States Centers for Disease Control and Prevention;
https://archive.cdc.gov/www_cdc_gov/csels/dsepd/ss1978/lesson1/summary.html; snapshot
experiments/external_sources/medicine/snapshots/MED-CDC-EPIDEMIOLOGY-001.html;
sha256: 430c2d885670c185491ab9b84d1c703eeaff0eece3ddb69fc855c001fb93478b; scope: population, comparison, frequency, pattern, cause, disease and injury epidemiology.

Comparison record:

- sft-med-normal-variation-001-authority-correspondence: predicted medicine:reference-person-states__variation-within-declared-reference__distribution-with-subgroups-held__population-method-time-boundary; source-derived medicine:reference-person-states__variation-within-declared-reference__distribution-with-subgroups-held__population-method-time-boundary; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:b5834cb023d7f33b2f0ce239971525d62856be3e3f2010876d3f2ad6ffa981f9; derivation seal sha256:0127829bad74cf70a803c641bdc169e50a962ce51de73e68f68d2f2104395c61; engine receipt sha256:0eda0a1eea6c39d68be5e906bf31525faeb83b204f2160c0814d2897da6212fa at receipts/engine/model_admitted/SFT-MED-NORMAL-VARIATION-001-0eda0a1eea6c39d6.json; empirical validation sha256:b239b110944ab0fe020887effa70dec5714d85d140100ae6ec36eb17df8570d9; measurement receipt sha256:c54ec9ec29541e53bb17f0e232d4ed58ea06315ef483d8ecf6baf29c8aae8f44; isolation sha256:2dc78c04dba90769173b37237a237936cb45c6d40941ac93621c0d4974870b47; custody sha256:1a2f7686dce264dabd5853a20721313d020412e308ca9f7e923ea4f9d83e29ad.

8. Dysfunction boundary

Claim identity: SFT-MED-DYSFUNCTION-001

Question and exact theorem. Dysfunction requires a retained functional relation, baseline, context and consequence and cannot be inferred from difference alone.

person-function-state__impaired-relation-to-declared-function__function-context-and-consequence__method-and-baseline-boundary__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001

- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-NORMAL-VARIATION-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	person-function-state	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	impaired-relation-to-declared-function	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
organization	organization-collapsed	function-context-and-consequence	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	method-and-baseline-boundary	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: `person-function-state__impaired-relation-to-declared-function__function-context-and-consequence__method-and-baseline-boundary__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule`.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces dysfunction requires a retained functional relation, baseline, context and consequence and cannot be inferred from difference alone. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Difference, dysfunction, disease and injury are distinct clinical relations. Each keeps the reference population, functional consequence, mechanism, criteria, site, severity and time boundary that makes the classification testable.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:91a369c91aa8a23923c794bdacd663e58dd28e3084c1ef7997b44997c931ac63`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:1f4a256a31f33589fa892ca5b1ff5e030ed032e5057a6746758c0aaca19b4d5c`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:9bb3c255e3d9199811aec1f5edc2fb631ad77e50f041ed954b7a53de8b115011.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:ec1d8eb50f2095a0a02b35f824e80f0e32b542032c17fce536de0dd38cfe3096.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:14d82fc20bf0177bbdb1cc61dca1c4f0690d456a7ee8812851e9cea2ac94fb37; certificate
sha256:98d5166ef9ca85d7facc0d6255653cd02a2371dbbb0bf185dc94ec971ae26fa8; external-validation
sha256:9304f6011a850f926c5c77d4869bbdc10c8495478c66d891c80f0bfb05062054.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 3 purpose-matched discriminators with ordered identity

sha256:1c7fb6c2fb4557ca006e196f9f63a24094f64c684f794af23e82cb4c505f8657. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- MED-FDA-BIOMARKER-001 - United States Food and Drug Administration;
<https://www.fda.gov/drugs/biomarker-qualification-program/about-biomarkers-and-qualification>; snapshot
experiments/external_sources/medicine/snapshots/MED-FDA-BIOMARKER-001.html;
sha256:93c5ee4b5e3480ed27aa805a7739f4acf7108701fc6f429ae5d4e5a1ad9d3d31; scope: normal and pathogenic processes; risk, diagnostic, monitoring, prognostic, response and safety biomarkers; method and context.
- MED-CDC-EPIDEMIOLOGY-001 - United States Centers for Disease Control and Prevention;
https://archive.cdc.gov/www_cdc_gov/csels/dsepd/ss1978/lesson1/summary.html; snapshot
experiments/external_sources/medicine/snapshots/MED-CDC-EPIDEMIOLOGY-001.html;
sha256:430c2d885670c185491ab9b84d1c703eeaff0eece3ddb69fc855c001fb93478b; scope: population, comparison, frequency, pattern, cause, disease and injury epidemiology.

Comparison record:

- sft-med-dysfunction-001-authority-correspondence: predicted medicine:person-function-state__impaired-relation-to-declared-function__function-context-and-consequence__method-and-baseline-boundary; source-derived medicine:person-function-state__impaired-relation-to-declared-function__function-context-and-consequence__method-and-baseline-boundary; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:8ce746fa6795c07a26d933d5bafc92f93301e84e7ca977ad57abed0b245af314; derivation seal sha256:1dbab7fb6986900d264aa95ffa685aca0534b7ee071b70521ffb2852c36fae1a; engine receipt sha256:c428cc97163fb8b162d6db4be5ed92d78549d5ea8c8d4dcdf2b160ff059f6e7d at receipts/engine/model_admitted/SFT-MED-DYSFUNCTION-001-c428cc97163fb8b1.json; empirical validation sha256:ace7134965048202dd34f39481176a8b0ff2c33e849fa1312dbc525fcf6bbdb6; measurement receipt sha256:bd1852438d83b06945a6c95cd2a9ed651527650415cd75695eb3560f3d3ddf0f; isolation sha256:b3ce59f7323f8e23f63c74967f4d75c076a8f7817dd22237c4af566d7586cefe; custody sha256:f6b0123797ce247eafd848d3bbbce74d2b458139516f3b0554773baedd4c6dd0.

9. Disease state

Claim identity: SFT-MED-DISEASE-001

Question and exact theorem. A disease state is a criteria-bound harmful process whose mechanism, manifestations and course remain distinct; a label alone is not the state.

patient-process-state__persistent-or-recurrent-harmful-process__mechanism-manifestation-course__diagnostic-criteria-and-time__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001

- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-DYSFUNCTION-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	patient-process-state	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	persistent-or-recurrent-harmful-process	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	mechanism-manifestation-course	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	diagnostic-criteria-and-time	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: patient-process-state__persistent-or-recurrent-harmful-process__mechanism-manifestation-course__diagnostic-criteria-and-time__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier`: clinical carrier is explicit and nonempty; passed `true`.
- `relation`: relation and organization remain separately reconstructible; passed `true`.
- `boundary`: observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. A disease label survives only with criteria, harmful process, manifestations, course and diagnostic boundary retained. This corrects the prior cancer shorthand: recurrence alone cannot identify a clinical disease.

Difference, dysfunction, disease and injury are distinct clinical relations. Each keeps the reference population, functional consequence, mechanism, criteria, site, severity and time boundary that makes the classification testable.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : a36dc178d062ef1710a49ba8c70568369cbee2880c24823483dfd0e47fcadbf2.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 8142a246ebdb4469a8903ffa12d36eda193ae12a1bfc8e47da6f71c11a7316e1.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 224e6e9ae089328205b13cace75ee789ef2e42b416848840c9a8338a8f8efe0c.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : bf20d1d229ca0cd892d5b54207ab0b3ea2b8f59a2baf78f9c87949513b29891e.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256 : 8c669cf954b975b84f54fd37fbd92e3a4e378d1399477425569ab70b8cfbc6dd; certificate
sha256 : 700438cbae5ce7132d9bc86e4b739a490fbb2ea40a107c87c81c6868be1f2538; external-validation
sha256 : 929ef5daa40ddd44247fae6f16752c312348cd0b7737ad2343264ffd30b76a1d.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 3 purpose-matched discriminators with ordered identity

sha256 : 1c7fb6c2fb4557ca006e196f9f63a24094f64c684f794af23e82cb4c505f8657. Target opened after derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- MED-FDA-BIOMARKER-001 - United States Food and Drug Administration;
<https://www.fda.gov/drugs/biomarker-qualification-program/about-biomarkers-and-qualification>; snapshot
experiments/external_sources/medicine/snapshots/MED-FDA-BIOMARKER-001.html;
sha256 : 93c5ee4b5e3480ed27aa805a7739f4acf7108701fc6f429ae5d4e5a1ad9d3d31; scope: normal and pathogenic processes; risk, diagnostic, monitoring, prognostic, response and safety biomarkers; method and context.
- MED-CDC-EPIDEMIOLOGY-001 - United States Centers for Disease Control and Prevention;
https://archive.cdc.gov/www_cdc_gov/csels/dsepd/ss1978/lesson1/summary.html; snapshot
experiments/external_sources/medicine/snapshots/MED-CDC-EPIDEMIOLOGY-001.html;
sha256 : 430c2d885670c185491ab9b84d1c703eeaff0eece3ddb69fc855c001fb93478b; scope: population, comparison, frequency, pattern, cause, disease and injury epidemiology.

Comparison record:

- sft-med-disease-001-authority-correspondence: predicted medicine:patient-process-state__persistent-or-recurrent-harmful-process__mechanism-manifestation-course__diagnostic-criteria-and-time; source-derived medicine:patient-process-state__persistent-o

r-recurrent-harmful-process__mechanism-manifestation-course__diagnostic-criteria-and-time; exact match True

- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:63821a047fe27fa472265054d1ad4e46c21de5696dfe567441fe1d4857652d4d; derivation seal sha256:f7107976dedfe96701b890dd1d54d01db8a8f0aa3d28885f2cb27bd59726bb01; engine receipt sha256:02b64355610b3b30ed7dea14ae72efa4ab7f9d8652187947710a0e8da3abfc4a at receipts/engine/model_admitted/SFT-MED-DISEASE-001-02b64355610b3b30.json; empirical validation sha256:f1687c10fad474f812f80c9b7ed7d10983f1063f557cd43b64186518a8500d89; measurement receipt sha256:7024a9097c4dc20d7cb288822a4f6d43552520f7bffdabb18043c97a1cd6d8a8d; isolation sha256:1817f1c3192e2574a45b6e70bde736d06806dfbfcf4b6d182b512ac3017cb9d7; custody sha256:2949d75c895bc7e0ac3b5cbadada834dd26054d6b6e7042c8bb538a642c9c3ad.

10. Injury state

Claim identity: SFT-MED-INJURY-001

Question and exact theorem. An injury binds a declared exposure to damage and consequence at a retained site, time, severity and observation method.

patient-tissue-or-function-state__damage-following-declared-exposure__exposure-d
amage-consequence-trace__site-time-severity-method__complete-protocol-state-tran
sition-adverse-null-record__root-bound-forward-forcing__positive-finite-successo
r-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001

- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-DISEASE-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	patient-tissue-or-function-state	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	damage-following-declared-exposure	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	exposure-damage-consequence-trace	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
observation	observation-boundary-unrecorded	site-time-severity-method	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: `patient-tissue-or-function-state__damage-following-declared-exposure__exposure-damage-consequence-trace__site-time-severity-method__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule`.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces an injury binds a declared exposure to damage and consequence at a retained site, time, severity and observation method. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Difference, dysfunction, disease and injury are distinct clinical relations. Each keeps the reference population, functional consequence, mechanism, criteria, site, severity and time boundary that makes the classification testable.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 6934a3c18f43494b2c0507313800e475f04c76c3264c6c2c66be1032d259c66d.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: cc827213529023d5678d3e728511f521963500f64596eda36fd27152edc3da94.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: b657414a816bb32d260bf715434f34aba0f873de7c2a7eab9af721531a5191eb.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 4755bf0869ff2505f3999cbbab2ba4a25a98a4b07cc77141134be2a8fe116688.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256 : 0c7c5f0ae64b7ac4bbaab66ef7c63fa1e387ac6909f4acfb2069bb5b943e7220; certificate
sha256 : 291c8e1a0f26054cf59ab11c18b6bd34a82c1aacf61f65495af5ad066f008ed8; external-validation
sha256 : 32d8700a992016a46a5c62550a7e46eb8348120f416e1f5d5fd535903e0d90e2.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 3 purpose-matched discriminators with ordered identity
sha256 : 1c7fb6c2fb4557ca006e196f9f63a24094f64c684f794af23e82cb4c505f8657. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- MED-FDA-BIOMARKER-001 - United States Food and Drug Administration;
<https://www.fda.gov/drugs/biomarker-qualification-program/about-biomarkers-and-qualification>; snapshot
experiments/external_sources/medicine/snapshots/MED-FDA-BIOMARKER-001.html;
sha256 : 93c5ee4b5e3480ed27aa805a7739f4acf7108701fc6f429ae5d4e5a1ad9d3d31; scope: normal and pathogenic processes; risk, diagnostic, monitoring, prognostic, response and safety biomarkers; method and context.
- MED-CDC-EPIDEMIOLOGY-001 - United States Centers for Disease Control and Prevention;
https://archive.cdc.gov/www_cdc_gov/csels/dsepd/ss1978/lesson1/summary.html; snapshot
experiments/external_sources/medicine/snapshots/MED-CDC-EPIDEMIOLOGY-001.html;
sha256 : 430c2d885670c185491ab9b84d1c703eeaff0eece3ddb69fc855c001fb93478b; scope: population, comparison, frequency, pattern, cause, disease and injury epidemiology.

Comparison record:

- sft-med-injury-001-authority-correspondence: predicted medicine:patient-tissue-or-function-state__damage-following-declared-exposure__exposure-damage-consequence-trace__site-time-severity-method; source-derived medicine:patient-tissue-or-function-state__damage-following-declared-exposure__exposure-damage-consequence-trace__site-time-severity-method; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:41c57d3b143fff2c581c52f7bd4ad5ae33eb7223cfc620d5ae1693ba0fe6d7f4; derivation seal sha256:bd627d982dd17f99f12480273935bc0e9ca81ca6786c08f30543e867a207ebb0; engine receipt sha256:4a011252846c30ea2825619210ebed680f37a96dc46780f8f0a4a99e3e16b22a at receipts/engine/model_admitted/SFT-MED-INJURY-001-4a011252846c30ea.json; empirical validation sha256:95f14afffe36e41ad983cb949c47c60300790f0c5513d3f53d43b037636dd6f2; measurement receipt sha256:a008360ccdac5c036e8c6d1dea7cd6c125c0dc51e5a15a9b49d3518ed01b42b1; isolation sha256:39c632edc65eab9318d587af43da82d055614f3cbfd59618cb7177ff67666855; custody sha256:01f33d656967ab64969ccfacf2a4203871361fcf42c37f6cf4f02d8a61965161.

11. Comorbidity composition

Claim identity: SFT-MED-COMORBIDITY-001

Question and exact theorem. Comorbidity is a joint condition word on one patient and interval; component conditions and their interactions may not be collapsed into one label.

joint-patient-condition-state__co-occurrence-with-interaction-held__noncollapsed
 -joint-condition-word__shared-patient-and-interval__complete-protocol-state-tran
 sition-adverse-null-record__root-bound-forward-forcing__positive-finite-successo
 r-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001

- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-INJURY-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	joint-patient-condition-state	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	co-occurrence-with-interaction-held	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	noncollapsed-joint-condition-word	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	shared-patient-and-interval	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: joint-patient-condition-state__co-occurrence-with-interaction-held__noncollapsed-joint-condition-word__shared-patient-and-interval__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier`: clinical carrier is explicit and nonempty; passed `true`.
- `relation`: relation and organization remain separately reconstructible; passed `true`.
- `boundary`: observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces comorbidity is a joint condition word on one patient and interval; component conditions and their interactions may not be collapsed into one label. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Difference, dysfunction, disease and injury are distinct clinical relations. Each keeps the reference population, functional consequence, mechanism, criteria, site, severity and time boundary that makes the classification testable.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 1156436c29a0b51eb293bbfad009be1fbd7e20261b7d5c57dac6624c094b5442.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: dfa6f2a94fd6d0410cead984a49f3674f683b0e3ff4c8808c9dc40ce555219c9.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 28e22d7d605edbd97258e862f20529029768d2300a078026b4028204f5dedf4d.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 23ead293b0805587d7a1eea52c44719be971ab16bdfde35a1b8b3067bbc166e0.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256: 9a424e7b0f04ecab92e7ccd042873ffd1514f704b0473ac4e9cc1274538e2f23; certificate
sha256: 7b71934537a50c0946cf78e875d556c66d55a48233396454ac35655188e427ba; external-validation
sha256: e5b55cdd8e5a27799450516f9afea3cb7501fba6e71e56bdde66114a24903c0c.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 3 purpose-matched discriminators with ordered identity

sha256: 1c7fb6c2fb4557ca006e196f9f63a24094f64c684f794af23e82cb4c505f8657. Target opened after derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- MED-FDA-BIOMARKER-001 - United States Food and Drug Administration;
<https://www.fda.gov/drugs/biomarker-qualification-program/about-biomarkers-and-qualification>; snapshot
experiments/external_sources/medicine/snapshots/MED-FDA-BIOMARKER-001.html;
sha256: 93c5ee4b5e3480ed27aa805a7739f4acf7108701fc6f429ae5d4e5a1ad9d3d31; scope: normal and pathogenic processes; risk, diagnostic, monitoring, prognostic, response and safety biomarkers; method and context.
- MED-CDC-EPIDEMIOLOGY-001 - United States Centers for Disease Control and Prevention;
https://archive.cdc.gov/www_cdc_gov/csels/dsepd/ss1978/lesson1/summary.html; snapshot
experiments/external_sources/medicine/snapshots/MED-CDC-EPIDEMIOLOGY-001.html;
sha256: 430c2d885670c185491ab9b84d1c703eeaff0eece3ddb69fc855c001fb93478b; scope: population, comparison, frequency, pattern, cause, disease and injury epidemiology.

Comparison record:

- sft-med-comorbidity-001-authority-correspondence: predicted medicine:joint-patient-condition-state__co-occurrence-with-interaction-held__noncollapsed-joint-condition-word__shared-patient-and-interval; source-derived medicine:joint-patient-condition-state__co-occurrence-with-interaction-held__noncollapsed-joint-condition-word__shared-patient-and-interval; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:b480a50fa283756cdf71122a8a1cab09b85579e79886eb960f0578672e603501; derivation seal sha256:80fc74f15d89df26ebf03ff107eb46d02641d0a7f4e4b20f539c9bdca44ad5b2; engine receipt sha256:ea094d0cb6741aacb8f2e06e50807c2d31ab85c447c6f2227f5708bf269f2bae at receipts/engine/model_admitted/SFT-MED-COMORBIDITY-001-ea094d0cb6741aac.json; empirical validation sha256:6fc2959341a05d4e54390468d5abcfead5b2965b11d7859087e136ef5c08c344; measurement receipt sha256:7ed5801c49eebf998986bc36b12445fc02417c727fded437766b36c813c231cc; isolation sha256:2a5bdb8663d1d13d42cc6272548f637fe343dd38749897a244957941d173dd3e; custody sha256:966e30ab0ca231913265b66a4cd34c6537e18326390bff2e2df8999eb1ef506f.

12. Severity and stage

Claim identity: SFT-MED-SEVERITY-STAGE-001

Question and exact theorem. Severity and stage are declared finite order relations over retained criteria and may not be silently treated as interchangeable or universal scales.

disease-or-injury-course__ordered-extent-relation__criteria-specific-partial-order__method-time-and-organ-system__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001

- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-COMORBIDITY-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	disease-or-injury-course	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	ordered-extent-relation	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
organization	organization-collapsed	criteria-specific-partial-order	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	method-time-and-organ-system	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: `disease-or-injury-course__ordered-extent-relation__criteria-specific-partial-order__method-time-and-organ-system__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule`.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces severity and stage are declared finite order relations over retained criteria and may not be silently treated as interchangeable or universal scales. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Difference, dysfunction, disease and injury are distinct clinical relations. Each keeps the reference population, functional consequence, mechanism, criteria, site, severity and time boundary that makes the classification testable.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:f95a302a823fea20b551264e8ba313fe582d168e9b505b512b70b4141252451b`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:2674a9b2ca22e84e7ca8e1161d9ab07d319a59aded61744e1cbdf00e90dc36e7`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256 : 78163c39295e64b709d5fb32df526935c19847ba9ebbe1e5e6c4f33ec07b0bfe.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 24eafee86255e83659eb57c4b4bfd0fd47eea1e0f9aa44ef9bf3cc03fb8fd5f1.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256 : 1c567b29adbd44aa3730cbac439fdbcec747df64d6249c64b87a215396c32e01; certificate
sha256 : ed3ac12d83fae5b7b2d13329b1602d7d9d5b277ea234832d67cf83285fd4d9cf; external-validation
sha256 : f6b04eed5cfec5d75e2b378daa30a2c93ddbfa4b38d6eaa46d8395e638321820.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 3 purpose-matched discriminators with ordered identity

sha256 : 1c7fb6c2fb4557ca006e196f9f63a24094f64c684f794af23e82cb4c505f8657. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- MED-FDA-BIOMARKER-001 - United States Food and Drug Administration;
<https://www.fda.gov/drugs/biomarker-qualification-program/about-biomarkers-and-qualification>; snapshot
experiments/external_sources/medicine/snapshots/MED-FDA-BIOMARKER-001.html;
sha256 : 93c5ee4b5e3480ed27aa805a7739f4acf7108701fc6f429ae5d4e5a1ad9d3d31; scope: normal and pathogenic processes; risk, diagnostic, monitoring, prognostic, response and safety biomarkers; method and context.
- MED-CDC-EPIDEMIOLOGY-001 - United States Centers for Disease Control and Prevention;
https://archive.cdc.gov/www_cdc_gov/csels/dsepd/ss1978/lesson1/summary.html; snapshot
experiments/external_sources/medicine/snapshots/MED-CDC-EPIDEMIOLOGY-001.html;
sha256 : 430c2d885670c185491ab9b84d1c703eeaff0eece3ddb69fc855c001fb93478b; scope: population, comparison, frequency, pattern, cause, disease and injury epidemiology.

Comparison record:

- sft-med-severity-stage-001-authority-correspondence: predicted
medicine:disease-or-injury-course__ordered-extent-relation__criteria-specific-partial-order__method-time-and-organ-system;
source-derived
medicine:disease-or-injury-course__ordered-extent-relation__criteria-specific-partial-order__method-time-and-organ-system;
exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result

- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:cb6ce45db35c875dca419eeaf92240e690d5031422827d82c7fab3f47bfce95e; derivation seal sha256:662196389bbbf0a979239d79b2c55c8ca02f47b2c66b52994dafb263660c8b8; engine receipt sha256:011c56e91bdcdfdeb96526cd2bf75f7e497b74ab880a324f8975eeb08c420004 at receipts/engine/model_admitted/SFT-MED-SEVERITY-STAGE-001-011c56e91bdcdfde.json; empirical validation sha256:fefbea6d83fb2b1dba42c58d288fa6baa55f4ae178b65652c7a7b72841d0e71e; measurement receipt sha256:564eae0e920741dadecffba42ab6b70b87356f4b194517d6f3c0e1596ba5bb25; isolation sha256:8ac1aa086f6732a330203f8d17a3f204c7db59863246210d1f87f212cbb409d7; custody sha256:c6e2e6283e695bd442540a3d894e111c916a0c87daeff914a064dcd29cb8411f.

12. Symptom Sign Measurement Diagnosis

Patient report, observer-elicited sign, instrument measurement and diagnostic class are separate evidence carriers. The branch preserves language, observer, method, calibration, alternative diagnoses and verification rather than promoting a label into a fact.

13. Symptom record

Claim identity: SFT-MED-SYMPTOM-001

Question and exact theorem. A symptom is a patient-reported observation whose content, timing, context, elicitation and reporting uncertainty remain held.

patient-reported-experience__report-to-person-state-relation__content-intensity-timing-context__elicitation-method-and-language__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001

- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-SEVERITY-STAGE-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	patient-reported-experience	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	report-to-person-state-relation	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	content-intensity-timing-context	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	elicitation-method-and-language	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
extension	free-fit-exception -opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: patient-reported-experience__report-to-person-state-relation__content-intensity-timing-context__elicitation-method-and-language__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces a symptom is a patient-reported observation whose content, timing, context, elicitation and reporting uncertainty remain held. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Patient report, observer-elicited sign, instrument measurement and diagnostic class are separate evidence carriers. The branch preserves language, observer, method, calibration, alternative diagnoses and verification rather than promoting a label into a fact.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 66cd93c388822d160e23c5cbbd3bd41b693110dea6a23cf1639e36ab0f097aaa.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 0fd53f6f4e52e9fe25c2988d349f9e24c8d01ce3aad0b72bb5d75f10f6e56976.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 26a40ec56711e4a32379ea024cc2004f5a2459c2b2572b08e3f32a958389b0d8.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 87be2ffb83ca4008be60e1a93d19a12bbdcfeff150ab5c82a8b1d509d140b492.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256: c585197766b6718eaea6b82f31d433f8b03f730a08d0ec5790fc4f68a13ec264; certificate
sha256: ba5fa48df09bd23c38a2aa3f81e77f1cb850ca6ca079efac2f5e88970430fa0e; external-validation
sha256: 3b523fada2473658aec3999703dcff07969b12f92f26a7367f6fcf7b601983c0.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 4 purpose-matched discriminators with ordered identity
sha256: 70bdde493f9f6d2d715766065e59019d557797b0c5005c9b2dc9c5ca3d7b9a1a. Target opened after derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- MED-FDA-BIOMARKER-001 - United States Food and Drug Administration; https://www.fda.gov/drugs/biomarker-qualification-program/about-biomarkers-and-qualification-experiments/external_sources/medicine/snapshots/MED-FDA-BIOMARKER-001.html; sha256:93c5ee4b5e3480ed27aa805a7739f4acf7108701fc6f429ae5d4e5a1ad9d3d31; scope: normal and pathogenic processes; risk, diagnostic, monitoring, prognostic, response and safety biomarkers; method and context.

Comparison record:

- sft-med-symptom-001-authority-correspondence: predicted medicine:patient-reported-experience__report-to-person-state-relation__content-intensity-timing-context__elicitation-method-and-language; source-derived medicine:patient-reported-experience__report-to-person-state-relation__content-intensity-timing-context__elicitation-method-and-language; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:6d5e17cf2b02327a49964df676f3b7236a2fcd85dd1e469b69bcd7f2834d26e9; derivation seal sha256:b26bfca849d2c764587a2c22840f1327c1e8f788073f2e047b8271b5e1c4e292; engine receipt sha256:a2374682cdc923a9dd6cd88b4f541480746939507c83a8aa939213463823711b at receipts/engine/model_admitted/SFT-MED-SYMPTOM-001-a2374682cdc923a9.json; empirical validation sha256:cb54b4a5f0bfe008179e0a235f9133e2a4c7c63a04b7dda98d1a72627f7b94; measurement receipt sha256:dd0d1bbe61cace5b2696d0c6fa7aff5f634f53bdec06ccf0fa032f560244ceac; isolation sha256:1c828defb4be4abc873f17cfba12be2752d91c2085fca3be4a498f847326cbc2; custody sha256:fbec03d4bf972bcc1ba34b24fe98e8897091532d4a7d6f37bec815bcc8678230.

14. Clinical sign

Claim identity: SFT-MED-SIGN-001

Question and exact theorem. A sign is an observer-elicited feature with method, observer, setting and time retained; it is not interchangeable with a symptom.

```
observer-elicited-patient-feature__feature-to-method-relation__observer-method-r
result-record__setting-time-and-observer__complete-protocol-state-transition-adve
rse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfav
orable-closure__no-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-SYMPTOM-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	observer-elicited-patient-feature	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	feature-to-method-relation	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	observer-method-result-record	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	setting-time-and-observer	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: observer-elicited-patient-feature__feature-to-method-relation__observer-method-result-record__setting-time-and-observer__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the clinical law; passed **true**.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces a sign is an observer-elicited feature with method, observer, setting and time retained; it is not interchangeable with a symptom. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Patient report, observer-elicited sign, instrument measurement and diagnostic class are separate evidence carriers. The branch preserves language, observer, method, calibration, alternative diagnoses and verification rather than promoting a label into a fact.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:8f57589f71018fe607f6f828b23aabc9d99a8c94df48ed652e98177bd67f2c23.

- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:2ee673abbd6c6fa7908c514c5ba6552a48fa9d0e05987b8e4c71d5093122b060e.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:1897979a5d1d8aa2020dc3787bfc236f94f726c52c44039b79a4f5550363bcc8.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:4eab5eeadb51aa0145757a77fcf22fc2a8c5df98fa62def45d2a2ff6822d0793.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:18dc2ca5fd1eff7abe73564cc3818da93ee0ce12ffd4fa0aac021e33162cfac; certificate

sha256:dce1900b93bf0b299010f692fad723cbca73c54b94665ddb8dcb022d119b337d; external-validation

sha256:e0543180c47b5c36e625ec7cea2e22bfb3745a7cd3404843669d1d7c8610441.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 4 purpose-matched discriminators with ordered identity

sha256:70bdde493f9f6d2d715766065e59019d557797b0c5005c9b2dc9c5ca3d7b9a1a. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- MED-FDA-BIOMARKER-001 - United States Food and Drug Administration; <https://www.fda.gov/drugs/biomarker-qualification-program/about-biomarkers-and-qualification>; snapshot experiments/external_sources/medicine/snapshots/MED-FDA-BIOMARKER-001.html; sha256:93c5ee4b5e3480ed27aa805a7739f4acf7108701fc6f429ae5d4e5a1ad9d3d31; scope: normal and pathogenic processes; risk, diagnostic, monitoring, prognostic, response and safety biomarkers; method and context.

Comparison record:

- sft-med-sign-001-authority-correspondence: predicted medicine:observer-elicited-patient-feature__feature-to-method-relation__observer-method-result-record__setting-time-and-observer; source-derived medicine:observer-elicited-patient-feature__feature-to-method-relation__observer-method-result-record__setting-time-and-observer; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:ffe941452c11a4a69aaf1806c54fbe78cd1cf2488b3ea4b1940590f56ec4052f; derivation seal sha256:e680d515de9c88ba7ade1ee1c0f989447c8e86d7b914deb5a8046557286ab6d; engine receipt sha256:454e5785a37425db27f63a8eb134d4c472ec7a5e554068b6024e70d9fb139852 at receipts/engine/model_admitted/SFT-MED-SIGN-001-454e5785a37425db.json; empirical validation sha256:031bb39abc8d5adbdd3d0c6859c79805b8d30046443ad6c92bf472dc06881223; measurement receipt sha256:cfc3a30575e6d9a5b8e824183636c6413f938e460571d2f5bf5c92163143b9eb; isolation sha256:83d72295a2d2bcfa5085060a0b640c64bf421ec6b6b2abbcd9581b5723b4d2ee; custody sha256:e21c06ca3096670fedf877c96b062c2423d6450981c4a9524ca852aff598784b.

15. Clinical measurement

Claim identity: SFT-MED-CLINICAL-MEASUREMENT-001

Question and exact theorem. A clinical measurement retains carrier, unit, method, calibration, preanalytic state, result, uncertainty and reporting transformation.

patient-specimen-or-signal__traceable-measurement-relation__method-unit-calibration-result__preanalytic-analytic-postanalytic-boundary__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001

- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-SIGN-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	patient-specimen-or-signal	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	traceable-measurement-relation	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	method-unit-calibration-result	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	preanalytic-analytic-postanalytic-boundary	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: patient-specimen-or-signal__traceable-measurement-relation__method-unit-calibration-result__preanalytic-analytic-postanalytic-boundary__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier`: clinical carrier is explicit and nonempty; passed `true`.
- `relation`: relation and organization remain separately reconstructible; passed `true`.
- `boundary`: observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces a clinical measurement retains carrier, unit, method, calibration, preanalytic state, result, uncertainty and reporting transformation. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Patient report, observer-elicited sign, instrument measurement and diagnostic class are separate evidence carriers. The branch preserves language, observer, method, calibration, alternative diagnoses and verification rather than promoting a label into a fact.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 0433fa9a0bac40fd84e6f05ec7e65730cdc90ea47c377b9b4b67cb4f5399316d.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 8155c178d5675f1f97c89e79922bac82683d5845ca8294f99fbe74562dac6572.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 591460ad9c48dda755c7a46890a4dd74bdc3c3200d9810679f9df444245eed6d.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: d726b6484e788cf233efbc96d65d73d6e08a2df9e268ca9e0607ac6df97de7c8.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256: 644952e445233f5d4fa4dd3362f45e8c2a6baf1a0291509e0f17a0150abf216c; certificate
sha256: d9d842e1f03f9843f68b8cfd2458e6a3679724f6303b13e288abdd8fa7bbecf8; external-validation
sha256: 0fc69f977b8eb57f12c9f18c243e8f8412379666cda85c9027438b21f27c0e0c.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 4 purpose-matched discriminators with ordered identity

sha256: 70bdde493f9f6d2d715766065e59019d557797b0c5005c9b2dc9c5ca3d7b9a1a. Target opened after derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- MED-FDA-BIOMARKER-001 - United States Food and Drug Administration;
<https://www.fda.gov/drugs/biomarker-qualification-program/about-biomarkers-and-qualification>; snapshot
experiments/external_sources/medicine/snapshots/MED-FDA-BIOMARKER-001.html;
sha256: 93c5ee4b5e3480ed27aa805a7739f4acf7108701fc6f429ae5d4e5a1ad9d3d31; scope: normal and
pathogenic processes; risk, diagnostic, monitoring, prognostic, response and safety biomarkers; method and context.

Comparison record:

- sft-med-clinical-measurement-001-authority-correspondence: predicted medicine:patient-specimen-or-signal__traceable-measurement-relation__method-unit-calibration-result__preanalytic-analytic-postanalytic-boundary; source-derived medicine:patient-specimen-or-signal__traceable-measurement-relation__method-unit-calibration-result__preanalytic-analytic-postanalytic-boundary; exact match `True`
- every required authoritative fragment and snapshot hash reproduced

- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:e226b5a21b2846fd3fd02cd66cd36b4ad1c401e9a3bad9904392ef642e0ae5ec; derivation seal sha256:230a466c7f14d2553c52a7ce5fe71534325765327e92cb751ef0cf32cef66976; engine receipt sha256:faceaa239618ddd156fff77437bd4c6bfa68461b65053e30dfcf4f733134bf71 at receipts/engine/model_admitted/SFT-MED-CLINICAL-MEASUREMENT-001-faceaa239618ddd1.json; empirical validation sha256:2e7cb8c98766bfb846136129816a292e9fbd09be979df216d69fcfdbd122df3; measurement receipt sha256:d1a33ee60559240a3bd1a6eef75d965b4443774d2d80b96f733b868c8f888265; isolation sha256:bbc9f3bfee294e0d103bb08a1c11d7b5954cbc89c0e590882b48aa3071c5ba36; custody sha256:7921053f051a0b864e4273586121db84b4557c37a6372d5dda3965761cafbe55.

16. Diagnostic classification

Claim identity: SFT-MED-DIAGNOSTIC-CLASSIFICATION-001

Question and exact theorem. Diagnostic classification maps a complete patient evidence word to a declared class while retaining alternative classes, criteria and verification boundary.

patient-evidence-word__criteria-to-class-relation__classification-with-alternatives-held__reference-standard-and-verification__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001

- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-CLINICAL-MEASUREMENT-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	patient-evidence-word	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	criteria-to-class-relation	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	classification-with-alternatives-held	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
observation	observation-boundary-unrecorded	reference-standard-and-verification	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: patient-evidence-word__criteria-to-class-relation__classification-with-alternatives-held__reference-standard-and-verification__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces diagnostic classification maps a complete patient evidence word to a declared class while retaining alternative classes, criteria and verification boundary. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Patient report, observer-elicited sign, instrument measurement and diagnostic class are separate evidence carriers. The branch preserves language, observer, method, calibration, alternative diagnoses and verification rather than promoting a label into a fact.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:39bd31b87b77b20c538c74f7c360735249401b79a5aac42ae28a48337ce6e053.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:ee7907af462bbc0ba190b273e494e28e37436510f97175e40d4ff9865f1b0f5e.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:ed768b5b822085653d2b684bf6a49761f0b2eed6629496a75644ed1db8934d5b.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 7f410269ebde5f99f7bb191996ad54cd16eeaf3b8cb72d6cf40f9716163e5793.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256 : 922a4df4961edd0d90c66054043f703bf075d6c1f82fc7bfe645e1bcddac21c6; certificate
sha256 : 11f441ae4b4a5e260200e4289ef0b8323ccf117ef88e3d6503ff79f7d164815d; external-validation
sha256 : e9d3bb0874813ea6920198ffc76d58b79c247eafcb52d25c1f402360f2581d1e.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 4 purpose-matched discriminators with ordered identity
sha256 : 70bdde493f9f6d2d715766065e59019d557797b0c5005c9b2dc9c5ca3d7b9a1a. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- MED-FDA-BIOMARKER-001 - United States Food and Drug Administration;
https://www.fda.gov/drugs/biomarker-qualification-program/about-biomarkers-and-qualification-experiments/external_sources/medicine/snapshots/MED-FDA-BIOMARKER-001.html; snapshot
sha256 : 93c5ee4b5e3480ed27aa805a7739f4acf7108701fc6f429ae5d4e5a1ad9d3d31; scope: normal and pathogenic processes; risk, diagnostic, monitoring, prognostic, response and safety biomarkers; method and context.

Comparison record:

- sft-med-diagnostic-classification-001-authority-correspondence: predicted medicine:patient-evidence-word__criteria-to-class-relation__classification-with-alternatives-held__reference-standard-and-verification; source-derived medicine:patient-evidence-word__criteria-to-class-relation__classification-with-alternatives-held__reference-standard-and-verification; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal
experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest
sha256 : 2a8ce22ed2b02bc5876306a1e5101da28178f5ef711793a771a10796c8f12084; derivation seal
sha256 : ea9c4244d5e00fadba107f04190fa9179c04143f630f15662702537b38bbf33d; engine receipt
sha256 : 1e55f7167ea50eb3f99bbbc8e319501ddae4c1f9cd20b8b0c2db48bd486ee009 at receipts/engine/model_admitted/SFT-MED-DIAGNOSTIC-CLASSIFICATION-001-1e55f7167ea50eb3.json; empirical
validation sha256 : 4af4d83bbaab1006e40d674a71a4d352a293de608ffb8409be521a8ea6caf714;

measurement receipt sha256:c90ff96571c684d6a11e56baf538ab4bf07d0cfd448efa135db2c860ea1569ce;
isolation sha256:b3d81f2123d60a47745b4ca177ca5cfde95813becaaa1479d2adb784f3898032; custody
sha256:87e1301b3681f6a4fffebc301ef4811da1dfdb3ccdba00cea674383b7b271e5a.

17. Differential diagnosis

Claim identity: SFT-MED-DIFFERENTIAL-DIAGNOSIS-001

Question and exact theorem. A differential diagnosis is a retained finite support of candidate conditions reduced only by recorded patient evidence; an untested alternative may not disappear.

candidate-condition-support__evidence-conditioned-elimination__retained-alternative-support__patient-context-and-test-sequence__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001

- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-DIAGNOSTIC-CLASSIFICATION-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	candidate-condition-support	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	evidence-conditioned-elimination	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	retained-alternative-support	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	patient-context-and-test-sequence	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: candidate-condition-support__evidence-conditioned-elimination__retained-alternative-support__patient-context-and-test-sequence__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier: clinical carrier is explicit and nonempty; passed true.

- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the clinical law; passed **true**.

Scientific meaning. Unobserved alternatives remain in support. A differential narrows only when recorded patient evidence lawfully eliminates a candidate; popularity, pretrained prediction or one favorable test cannot silently close it.

Patient report, observer-elicited sign, instrument measurement and diagnostic class are separate evidence carriers. The branch preserves language, observer, method, calibration, alternative diagnoses and verification rather than promoting a label into a fact.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:1d6943115e2a7edd2c0c3266f7503a0fbcc26d6504b361faf385829779c73bb8.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:ee357c96877dc0a3a362660d0fdd33a5ea0f7ddc8c9afba728c81336a56c16c0.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:41541a513351ffc23d500dda6fc0f02576bdf9dc88a1faab13734e3e7159e16.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:e31728e1bef611eded88442eb9bd608db0c7adc8818178fbfb3ac553641b2f29.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:3fbcb2d78c5d90cd0391b5bc342d18e6d1e5a4d19c57a8da59f70eb30808c79a; certificate
sha256:4b367cd7f2daa3a3f008f96b4cc267aba9882aca71497c223f4de5c84c6ede76; external-validation
sha256:9fb7685a34d6f413a1391fcd56bf1ba9686d471f81e09f71f616d0abe2457f89.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 4 purpose-matched discriminators with ordered identity
sha256:70bdde493f9f6d2d715766065e59019d557797b0c5005c9b2dc9c5ca3d7b9a1a. Target opened after derivation seal: **true**. All rows preserved: **true**. Exact comparison: **true**. A deliberately changed observation was rejected.

Sources:

- **MED-FDA-BIOMARKER-001** - United States Food and Drug Administration;
https://www.fda.gov/drugs/biomarker-qualification-program/about-biomarkers-and-qualification-experiments/external_sources/medicine/snapshots/MED-FDA-BIOMARKER-001.html;
sha256:93c5ee4b5e3480ed27aa805a7739f4acf7108701fc6f429ae5d4e5a1ad9d3d31; scope: normal and pathogenic processes; risk, diagnostic, monitoring, prognostic, response and safety biomarkers; method and context.

Comparison record:

- **sft-med-differential-diagnosis-001-authority-correspondence:** predicted medicine:candidate-condition-support__evidence-conditioned-elimination__retained-alternative-support__patient-context-and-test-sequence; source-derived medicine:candidate-condition-support__evidence-conditioned-elimination__retained-alternative-support__patient-context-and-test-sequence; exact match
True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology

- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:9a249f3e0719c583c3d73acc337f8533550f07aabb23de60a1815542b8a0a540; derivation seal sha256:3f3401e6a10f80983c26535cfc8062fe58cdf0da0be1bae715d4b6c6bc570102; engine receipt sha256:ddeb40383f8964a1693e542a975ecb3efa8949c2226d795de73731a0f374a7e2 at receipts/engine/model_admitted/SFT-MED-DIFFERENTIAL-DIAGNOSIS-001-ddeb40383f8964a1.json; empirical validation sha256:bf6888a8db6518a2c4920ec611e41a2db51808d97db660a4cddb2ca5cbd401f3; measurement receipt sha256:dd0461177f4e423f14af5da3710481102617d71f647d1005feb7ba4e7e321747; isolation sha256:c5f17f38142d6fc91e2789ec87c6ca41701c6bf437e44cc3a379101b0942872b; custody sha256:20e8067278c8c00dc3189170c60d61a05fbf7d0acf73b53738c830a730c767a0.

18. Diagnostic accuracy

Claim identity: SFT-MED-DIAGNOSTIC-ACCURACY-001

Question and exact theorem. Diagnostic accuracy requires all four cells generated by held condition and test-result distinctions plus population spectrum, reference method and verification record.

paired-index-and-reference-results__agreement-disagreement-incidence__complete-two-by-two-observation-table__target-population-spectrum-and-verification__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001

- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-DIFFERENTIAL-DIAGNOSIS-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	paired-index-and-reference-results	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	agreement-disagreement-incidence	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	complete-two-by-two-observation-table	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	target-population-spectrum-and-verification	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
generality	single-patient-fav orable-study	positive-finite-successor -and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception -opaque-model-or-e xtra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: paired-index-and-reference-results__agreement-d
isagreement-incidence__complete-two-by-two-observation-table__target-population-spectru
m-and-verification__complete-protocol-state-transition-adverse-null-record__root-bound-
forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier: clinical carrier is explicit and nonempty; passed true.
- relation: relation and organization remain separately reconstructible; passed true.
- boundary: observation boundary cannot silently become the clinical law; passed true.
- complete-diagnostic-table: two held diagnostic distinctions generate four cells exactly once; passed true.

Scientific meaning. Two held distinctions—condition classification and test result—generate exactly four observation cells. All four are required. Sensitivity-like or specificity-like summaries are downstream exact readings and cannot replace the underlying counts, population spectrum or reference method.

Patient report, observer-elicited sign, instrument measurement and diagnostic class are separate evidence carriers. The branch preserves language, observer, method, calibration, alternative diagnoses and verification rather than promoting a label into a fact.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 7213fb297a0965111a4449a3cf4c691284fba43938c9b0e7f172c6243953c531.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: b89556337d5ae362117687db37f80c1d2e0665c8ec8bc50509edb55bb075682e.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 05f5840ae5bb877180390a0ae86a148aa7ba56322d4e1761784d524bb561fb92.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 25ede4c2e62ea113f76a4f98ada8bdb2a58da914ee3fe04c4be213955500b8d0.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256: c3fc701395fd2d58825f0e07fc7696b6bf28448113b0a50c01f096f9a11d884f; certificate
sha256: afd63cfea44d175946e8d1c67e050af531cd9ced1a4c8b18ce0dd2ad87f898f9; external-validation
sha256: a37234a68c9a1aeddc1c83798d13adeba82097f37429495b63e479db0bea7841.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 4 purpose-matched discriminators with ordered identity

sha256:70bdde493f9f6d2d715766065e59019d557797b0c5005c9b2dc9c5ca3d7b9a1a. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- MED-FDA-BIOMARKER-001 - United States Food and Drug Administration; https://www.fda.gov/drugs/biomarker-qualification-program/about-biomarkers-and-qualification-experiments/external_sources/medicine/snapshots/MED-FDA-BIOMARKER-001.html; sha256:93c5ee4b5e3480ed27aa805a7739f4acf7108701fc6f429ae5d4e5a1ad9d3d31; scope: normal and pathogenic processes; risk, diagnostic, monitoring, prognostic, response and safety biomarkers; method and context.

Comparison record:

- sft-med-diagnostic-accuracy-001-authority-correspondence: predicted medicine:paired-index-and-reference-results__agreement-disagreement-incidence__complete-two-by-two-observation-table__target-population-spectrum-and-verification; source-derived medicine:paired-index-and-reference-results__agreement-disagreement-incidence__complete-two-by-two-observation-table__target-population-spectrum-and-verification; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:459e2aeb92ee07b7395420d8525f7769c813e4db8477ee15438c49af55007f91; derivation seal sha256:3856f01d5595d3612396618e07f85dfe8bf0935ddf59ea0406986aeecdbb403c; engine receipt sha256:071608ddf71ededb394e5cecb7df8434d259579da3f992ac33abdeff6c365399 at receipts/engine/model_admitted/SFT-MED-DIAGNOSTIC-ACCURACY-001-071608ddf71ededb.json; empirical validation sha256:142bd7b30d34ba413032f911ca528b124cdec31817b5653932a542ebdf3d9771; measurement receipt sha256:b0f94ddb55dca2acaf54ac65a08055d38d96e148b3e457be3c3c190f4c12b878; isolation sha256:54a35366aff6af5eaa7917d61d9773e769d35b42adb9b24ee4e9df819c76f029; custody sha256:5c7bffa49e7a1c11b4c767cea71d70ccc0d44fec59476808bf0c2db731c349f3.

13. Cause Association Mechanism Confounding

Association is not cause. Causal claims retain time order, comparator meaning, confounding, mediation, effect modification and the exact observation class capable of distinguishing the alternatives.

19. Clinical association

Claim identity: SFT-MED-CLINICAL-ASSOCIATION-001

Question and exact theorem. A clinical association is an exact joint-incidence relation at a declared population and interval and does not by itself assert cause.

paired-exposure-outcome-records__joint-incidence-relation__strata-and-missingness-held__population-time-and-selection__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001

• SFT-MED-DIAGNOSTIC-ACCURACY-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	paired-exposure-outcome-records	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	joint-incidence-relation	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	strata-and-missingness-held	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	population-time-and-selection	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: paired-exposure-outcome-records__joint-incidence-relation__strata-and-missingness-held__population-time-and-selection__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier: clinical carrier is explicit and nonempty; passed true.
- relation: relation and organization remain separately reconstructible; passed true.
- boundary: observation boundary cannot silently become the clinical law; passed true.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces a clinical association is an exact joint-incidence relation at a declared population and interval and does not by itself assert cause. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Association is not cause. Causal claims retain time order, comparator meaning, confounding, mediation, effect modification and the exact observation class capable of distinguishing the alternatives.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : 7298285e5c2ca3429256478e4cec377843f107c4f92922deaf65b53c27dcc6e0.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : c2f09d0cf53c4e4488917656822d4b07e4e53a0599f0ef7935c1104fd403f6a9.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : bc0f0bebe1a36ec5b9485b57a9f6943baca0f6179a882b93a1543318ac0c92d8.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 12326268bf5098d7c58d7af56ebbdd4ad16b05aaa10fcfd3d949f7048115574.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256 : 909ac6a61104c7cb3d05dcf3ccb98f5c2eca1a17a10b9998c7ae6ee1514f45a0; certificate
sha256 : ea7a6162884563a478e47ebae4489bdc701eff37d2244a586ec137afd7d7488b; external-validation
sha256 : 0cf0c14810cd9905d8fc8ad3d7b72bfef00a8423f58810763359c73a762ffd84.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 3 purpose-matched discriminators with ordered identity
sha256 : 5af911055225875794c72d66743b1b75e2dac4647b031e084e43b7d5ce98cb04. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- MED-CDC-EPIDEMIOLOGY-001 - United States Centers for Disease Control and Prevention; https://archive.cdc.gov/www_cdc_gov/csels/dsepd/ss1978/lesson1/summary.html; snapshot
experiments/external_sources/medicine/snapshots/MED-CDC-EPIDEMIOLOGY-001.html;
sha256 : 430c2d885670c185491ab9b84d1c703eeaff0eece3ddb69fc855c001fb93478b; scope: population, comparison, frequency, pattern, cause, disease and injury epidemiology.
- MED-STROBE-001 - STROBE Initiative; <https://www.strobe-statement.org/>; snapshot
experiments/external_sources/medicine/snapshots/MED-STROBE-001.html;
sha256 : 420432c148d01e7aaaab8f7bcb4bcb4412d028aa50cd7814fc1b058faf1cf7c; scope: observational-study planning, conduct, findings and interpretation reporting.

Comparison record:

- sft-med-clinical-association-001-authority-correspondence: predicted medicine:paired-exposure-outcome-records__joint-incidence-relation__strata-and-missingness-held__population-time-and-selection; source-derived medicine:paired-exposure-outcome-records__joint-incidence-relation__strata-and-missingness-held__population-time-and-selection; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:80f00b73f639316518c11c193ee5deea05a37eaf89e344f893f93ffc02cb6cff; derivation seal sha256:ebd5ce8a215f3e06411cab30914cda76de277a23655689b71857f8384bc8dcf5; engine receipt sha256:ef723331144115bd77edf75ec8fed6e5bd30c2c3c8a88c7604bb1efd525eb9b9 at receipts/engine/model_admitted/SFT-MED-CLINICAL-ASSOCIATION-001-ef723331144115bd.json; empirical validation sha256:4ba3ba44ba6e40643c2bf74ff5d4c874e212251b00e8115106e019903d6eecd6; measurement receipt sha256:f0b3328dcf8ca4c35ef503784113cf8f34287244676be4be7c9f64eafea713ed; isolation sha256:db9a209bcc151b0411b51c52d64e0b5d971238b9f2d928484ee8de9d37c9befe; custody sha256:05974bfdb2048a57e971e6bf3eabc4170ee611009c7ea2bba4b57455c0af80ea.

20. Clinical causal claim

Claim identity: SFT-MED-CLINICAL-CAUSE-001

Question and exact theorem. A clinical causal claim requires a lawful contrast with timing, exchangeability boundary, intervention or counterfactual meaning and complete outcome follow-up.

exposure-intervention-and-outcome-traces__difference-under-lawful-comparison__counterfactual-or-interventional-contrast__exchangeability-time-and-follow-up__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001

- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-CLINICAL-ASSOCIATION-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	exposure-intervention-and-outcome-traces	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	difference-under-lawful-comparison	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	counterfactual-or-interventional-contrast	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	exchangeability-time-and-follow-up	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
generality	single-patient-fav orable-study	positive-finite-successor -and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception -opaque-model-or-e xtra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: exposure-intervention-and-outcome-traces__difference-under-lawful-comparison__counterfactual-or-interventional-contrast__exchangeability-time-and-follow-up__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. Cause requires a generated comparison whose timing and identification boundary are explicit. A measured association can test the joint incidence relation but cannot, by authority or magnitude alone, select a causal law.

Association is not cause. Causal claims retain time order, comparator meaning, confounding, mediation, effect modification and the exact observation class capable of distinguishing the alternatives.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 40b528980ac9c96162a4f2b1d87bc2bdb7bfcf9a70ce253327251f6563c7677a.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 7a2e7f530c7b5e1092f1ced17111f5594d0255980a37dabb3427cfe697539fe9.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 10d30bab924f680552b3ca3f03330d9667cebbf5fcfee1389b1432c7ebf7c6d1.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 111568d5d5ce08c76a3e13a9074c01f6fca4f680867357584721359b41cb1d38.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256: 44379227db2234a23e5f445633ed1275935d475d5672bd3ea2699eaaa89638a2; certificate
sha256: d824758ca6c0689481b60cc34c6b18a0159caea9af72b68069735a4fd65cd4c2; external-validation
sha256: 39f41fa7e1ea1e91ba71a3fac06f439f3881e52cf504c3a798c3154035bcbfae.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 3 purpose-matched discriminators with ordered identity
sha256: 5af911055225875794c72d66743b1b75e2dac4647b031e084e43b7d5ce98cb04. Target opened after derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- MED-CDC-EPIDEMIOLGY-001 - United States Centers for Disease Control and Prevention; https://archive.cdc.gov/www_cdc_gov/csels/dsepd/ss1978/lesson1/summary.html; snapshot experiments/external_sources/medicine/snapshots/MED-CDC-EPIDEMIOLGY-001.html; sha256:430c2d885670c185491ab9b84d1c703eeaff0eece3ddb69fc855c001fb93478b; scope: population, comparison, frequency, pattern, cause, disease and injury epidemiology.
- MED-STROBE-001 - STROBE Initiative; <https://www.strobe-statement.org/>; snapshot experiments/external_sources/medicine/snapshots/MED-STROBE-001.html; sha256:420432c148d01e7aaaab8f7bcb4bcbcb4412d028aa50cd7814fc1b058faf1cf7c; scope: observational-study planning, conduct, findings and interpretation reporting.

Comparison record:

- sft-med-clinical-cause-001-authority-correspondence: predicted medicine:exposure-intervention-and-outcome-traces__difference-under-lawful-comparison__counterfactual-or-interventional-contrast__exchangeability-time-and-follow-up; source-derived medicine:exposure-intervention-and-outcome-traces__difference-under-lawful-comparison__counterfactual-or-interventional-contrast__exchangeability-time-and-follow-up; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:e38412b2668d689eb462fcc04fb87e41559d0ea72a4fb02e52f589720137bfd0; derivation seal sha256:33198e27394bf5be343bd00d78927f0299ea306da36737b58b69a14a9489aba7; engine receipt sha256:f327d41c2c5928c687ce971a46095fb36c57e502f8a1cc169158aefc30c65c97 at receipts/engine/model_admitted/SFT-MED-CLINICAL-CAUSE-001-f327d41c2c5928c6.json; empirical validation sha256:2aec9d535e52fa5d6d48d4088673337674387850b8a2daa90621fce0f827d5ac; measurement receipt sha256:38bab15ed8c8f183520e2da4d20ba1f5d6222d2eb6abf2a9ebd81a59f60f9f1f; isolation sha256:aaf77aa30e1c01119f0c3788affa3ecf0640cb6b5a349a95fc91a4084d3f9c3f; custody sha256:b9ebfab4b229b16f065e5ab006510a1b2857176b3583975d9c7796e3640c2e33.

21. Clinical mechanism

Claim identity: SFT-MED-CLINICAL-MECHANISM-001

Question and exact theorem. A clinical mechanism is an ordered, independently observed chain connecting exposure to outcome; a plausible biological name alone cannot close it.

ordered-biological-clinical-mediators__exposure-to-mediator-to-outcome__noncollapsed-causal-chain__patient-condition-and-temporal-order__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-CLINICAL-CAUSE-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	ordered-biological-clinical-mediators	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	exposure-to-mediator-to-outcome	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	noncollapsed-causal-chain	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	patient-condition-and-temporal-order	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: ordered-biological-clinical-mediators__exposure-to-mediator-to-outcome__noncollapsed-causal-chain__patient-condition-and-temporal-order__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the clinical law; passed **true**.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces a clinical mechanism is an ordered, independently observed chain connecting exposure to outcome; a plausible biological name alone cannot close it. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into

universal constants without a separately generated and externally tested law.

Association is not cause. Causal claims retain time order, comparator meaning, confounding, mediation, effect modification and the exact observation class capable of distinguishing the alternatives.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 49fdc68af6520d18d3de6aa59b54328d4e2b4051fa4720483a3e950dfb0db0a1.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: c78cf53c9b7be9a5b9b18aeef5e3b10f5a8b5360df5da24faf26587191c4c96.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 430e9bd62d976643258d663d3a2b93f7fc63e681fdb88da034f30341a20ac66.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: fa2f379e7f2b54dbe481c88910e713f358fb86d030fd0a3d72659c341d32561d.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256: ec8b7cff6ce327070447358d226b133f7abc78b5f9dd7818ddecdf55ddeef5e3; certificate
sha256: 730672d29e70b9e3ff21ed8bb9dab9fd1cf00a4650e95905c83831af793d3210; external-validation
sha256: dba489aaae92687b0d67d5c23ba269c5e43974bf03270d9b5cb9da991f8d8da.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 3 purpose-matched discriminators with ordered identity
sha256: 5af911055225875794c72d66743b1b75e2dac4647b031e084e43b7d5ce98cb04. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- MED-CDC-EPIDEMIOLOGY-001 - United States Centers for Disease Control and Prevention; https://archive.cdc.gov/www_cdc_gov/csels/dsepd/ss1978/lesson1/summary.html; snapshot
experiments/external_sources/medicine/snapshots/MED-CDC-EPIDEMIOLOGY-001.html;
sha256: 430c2d885670c185491ab9b84d1c703eeaff0eece3ddb69fc855c001fb93478b; scope: population, comparison, frequency, pattern, cause, disease and injury epidemiology.
- MED-STROBE-001 - STROBE Initiative; <https://www.strobe-statement.org/>; snapshot
experiments/external_sources/medicine/snapshots/MED-STROBE-001.html;
sha256: 420432c148d01e7aaaab8f7bcb4bcb4412d028aa50cd7814fc1b058faf1cf7c; scope: observational-study planning, conduct, findings and interpretation reporting.

Comparison record:

- sft-med-clinical-mechanism-001-authority-correspondence: predicted medicine:ordered-biological-clinical-mediators__exposure-to-mediator-to-outcome__noncollapsed-causal-chain__patient-condition-and-temporal-order; source-derived medicine:ordered-biological-clinical-mediators__exposure-to-mediator-to-outcome__noncollapsed-causal-chain__patient-condition-and-temporal-order; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:d566e2ac0cd8316f2fee16ae7b7602e30616d54a339433b78e6dc65ed459ee1b; derivation seal sha256:086b3147ff757221fc502e76468f679fb43f80bffd3e1844ff079f8f5c5cd37c; engine receipt sha256:16f410bd3c4969a561adca4caf550bf4303ee9247926ca1a52d2b45debd047fc at receipts/engine/model_admitted/SFT-MED-CLINICAL-MECHANISM-001-16f410bd3c4969a5.json; empirical validation sha256:c35566389173b48d8c88e01c7ef9fea21816ea6e6dde59d64e9bd78e6d4ec1fb; measurement receipt sha256:c41bdae54fd2078392b071276103ea58ee21d86b4fb4d45a1fcc3ff58fcab2ba; isolation sha256:d72e74d34a5abeb067f1eacd202957ab3c118786535fa1413f145e4025587f0b; custody sha256:ac878f0542eb334ffa73299213145621268d44dc3b0542a44e6e3791313dcbbc8.

22. Clinical confounding

Claim identity: SFT-MED-CONFOUNDING-001

Question and exact theorem. Confounding is retained whenever a pre-exposure cause bears on both exposure and outcome; adjustment must state exactly which distinctions it conditions on.

exposure-outcome-common-cause-word__common-cause-distortion-relation__measured-and-unmeasured-support__design-and-adjustment-boundary__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001

- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-CLINICAL-MECHANISM-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	exposure-outcome-common-cause-word	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	common-cause-distortion-relation	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	measured-and-unmeasured-support	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	design-and-adjustment-boundary	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
generality	single-patient-fav orable-study	positive-finite-successor -and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception -opaque-model-or-e xtra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: exposure-outcome-common-cause-word__common-caus
e-distortion-relation__measured-and-unmeasured-support__design-and-adjustment-boundary_
_complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__po
sitive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier: clinical carrier is explicit and nonempty; passed true.
- relation: relation and organization remain separately reconstructible; passed true.
- boundary: observation boundary cannot silently become the clinical law; passed true.

Scientific meaning. A common pre-exposure cause remains a held distinction whether measured or not. Adjustment states exactly which labels it conditions on; the unmeasured remainder is uncertainty, not conventional zero.

Association is not cause. Causal claims retain time order, comparator meaning, confounding, mediation, effect modification and the exact observation class capable of distinguishing the alternatives.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 3814fcc1d40391d0d547782d8c078e3157b77a9d4425211ccce4209cb497068c.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 5f2b36a0399e88c388966dad57e58530933b5fa572f23373f2409d1446f24eb.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 94243235e8b81abc309416cc1ae6b4d3c3ee73d675b5c669ae89836c943b30a1.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: a87aa64a8d14b04127140ff98bf26955c369fe0025ebd87dd88ad918ad326bc9.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256: 53a297a568c9fb997ee74ca1f6fbbfa507e7f6c5e0620a71d21685079ceb7e80; certificate
sha256: d72e863aaefcf587d835498510eb6daa778f134691b1d742b28c618f8780005e; external-validation
sha256: 730a345cf9256fe659df39dfb3a5c5cc514f10db7396f00f4092f1abce958245.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 3 purpose-matched discriminators with ordered identity
sha256: 5af911055225875794c72d66743b1b75e2dac4647b031e084e43b7d5ce98cb04. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- MED-CDC-EPIDEMIOLOGY-001 - United States Centers for Disease Control and Prevention; https://archive.cdc.gov/www_cdc_gov/csels/dsepd/ss1978/lesson1/summary.html; snapshot experiments/external_sources/medicine/snapshots/MED-CDC-EPIDEMIOLOGY-001.html; sha256:430c2d885670c185491ab9b84d1c703eeaff0eece3ddb69fc855c001fb93478b; scope: population, comparison, frequency, pattern, cause, disease and injury epidemiology.
- MED-STROBE-001 - STROBE Initiative; <https://www.strobe-statement.org/>; snapshot experiments/external_sources/medicine/snapshots/MED-STROBE-001.html; sha256:420432c148d01e7aaaab8f7bcb4bcb4412d028aa50cd7814fc1b058faf1cf7c; scope: observational-study planning, conduct, findings and interpretation reporting.

Comparison record:

- sft-med-confounding-001-authority-correspondence: predicted medicine:exposure-outcome-common-cause-word__common-cause-distortion-relation__measured-and-unmeasured-support__design-and-adjustment-boundary; source-derived medicine:exposure-outcome-common-cause-word__common-cause-distortion-relation__measured-and-unmeasured-support__design-and-adjustment-boundary; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:b34ade71824175051953e209b8b722e82bc64e40961473e34ed6e983824ff55d; derivation seal sha256:ab95c7a3e16ff339749d36f02b7d94b5890494ccad5e8b558e82d7fe9ed9f9b7; engine receipt sha256:c9c81f396a46438b2b273042713a0f6b054c5f2c801c4bef4df038796d1af3f7 at receipts/engine/model_admitted/SFT-MED-CONFOUNDING-001-c9c81f396a46438b.json; empirical validation sha256:7712600aad692e39c69c8ab6db99ab9c56991ccc70df29dfe1a0d3cc88237a23; measurement receipt sha256:8e299bc1d504f88392e7b76fa863ba8153715522fb2ea3444db8df95115b46c4; isolation sha256:d25a1c19635ded3688a6225e4a96884844b0af75a91881b4f7377ec08ca5c33f; custody sha256:929347b287102927b3ea79b8ad65278a49dd49329d9c4ecd527710716d0c54ee.

23. Clinical mediation

Claim identity: SFT-MED-MEDIATION-001

Question and exact theorem. Mediation separates direct and mediator-carried paths only when their temporal and intervention meanings are explicitly generated and observed.

exposure-mediator-outcome-word__ordered-indirect-and-direct-paths__path-specific
 -support__intervention-and-measurement-times__complete-protocol-state-transition
 -adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-
 unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-CONFOUNDING-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	exposure-mediator-outcome-word	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	ordered-indirect-and-direct-paths	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	path-specific-support	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	intervention-and-measurement-times	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: exposure-mediator-outcome-word__ordered-indirect-and-direct-paths__path-specific-support__intervention-and-measurement-times__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the clinical law; passed **true**.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces mediation separates direct and mediator-carried paths only when their temporal and intervention meanings are explicitly generated and observed. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into

universal constants without a separately generated and externally tested law.

Association is not cause. Causal claims retain time order, comparator meaning, confounding, mediation, effect modification and the exact observation class capable of distinguishing the alternatives.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:6fb64ac54eb2776ac46456549415da55dc14484c3add5cfbffa7b392faecb681.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:cde6744c7325c4cbf7adaff205b528ddf33ca514a6ef597f91d22f807b0c74b5.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:71ac0f6452b1a6ddea4f14a1caf3a7023268b9874992fd56e176fbb6299d169d.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:34200e66cd23350c79487ab2882c942b756a117e5661d06dcd90c3f1f46a1582.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:75e4ce6096934d09e3b529c0cd5de55223cb1b91bd08214c82450d4df1d597ef; certificate
sha256:5bf976b65f08873f52ffbf4155bd58efef6abe595d43bfef5ee7ad454d026e8; external-validation
sha256:518e4903b7239e9ea56ec25d627751e664867fed551de530c5cda1d0a353791f.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 3 purpose-matched discriminators with ordered identity
sha256:5af911055225875794c72d66743b1b75e2dac4647b031e084e43b7d5ce98cb04. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- MED-CDC-EPIDEMIOLOGY-001 - United States Centers for Disease Control and Prevention; https://archive.cdc.gov/www_cdc_gov/csels/dsepd/ss1978/lesson1/summary.html; snapshot
experiments/external_sources/medicine/snapshots/MED-CDC-EPIDEMIOLOGY-001.html;
sha256:430c2d885670c185491ab9b84d1c703eeaff0eece3ddb69fc855c001fb93478b; scope: population, comparison, frequency, pattern, cause, disease and injury epidemiology.
- MED-STROBE-001 - STROBE Initiative; <https://www.strobe-statement.org/>; snapshot
experiments/external_sources/medicine/snapshots/MED-STROBE-001.html;
sha256:420432c148d01e7aaaab8f7bcb4bcb4412d028aa50cd7814fc1b058faf1cf7c; scope: observational-study planning, conduct, findings and interpretation reporting.

Comparison record:

- sft-med-mediation-001-authority-correspondence: predicted medicine:exposure-mediator-outcome-word__ordered-indirect-and-direct-paths__path-specific-support__intervention-and-measurement-times; source-derived medicine:exposure-mediator-outcome-word__ordered-indirect-and-direct-paths__path-specific-support__intervention-and-measurement-times; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology

- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:986a6b34a2eba590156f77d76eb6ced933cf2bb076895c04001bde4cf5b943f6; derivation seal sha256:23a6641d4dfd4f8debc361b4c490f38e2b8d90ef86ef09510ceacd0e1252719d; engine receipt sha256:44248f3339ae7377b144a6f930ca9f13c6a6731d3d95a30eb547a3eb565e40 at receipts/engine/model_admitted/SFT-MED-MEDIATION-001-44248f3339ae7377.json; empirical validation sha256:c6b17b6abdcebbaa61adcda566aff400bfef66cdc12efebfbf38691f14b18313; measurement receipt sha256:dc738f433626ba7cd6067c190dc3c783df24fb2a8231c6d69a83bb72e669d973; isolation sha256:a835e4e292bd4bda0435921e3c4ec2cf2134e98902806a2118d3ea744c0a56cd; custody sha256:635a411b885a2ff3c39027c9c49be068ae1b151252c3d4dde9f6c44087c9d723.

24. Clinical effect modification

Claim identity: SFT-MED-EFFECT-MODIFICATION-001

Question and exact theorem. Effect modification is a retained variation of a treatment or exposure contrast across predeclared strata on a declared comparison scale.

stratified-patient-carriers__contrast-varies-by-held-stratum__joint-treatment-stratum-outcome__predeclared-strata-and-scale__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001

- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-MEDIATION-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	stratified-patient-carriers	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	contrast-varies-by-held-stratum	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	joint-treatment-stratum-outcome	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	predeclared-strata-and-scale	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
generality	single-patient-fav orable-study	positive-finite-successor -and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception -opaque-model-or-e xtra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: stratified-patient-carriers__contrast-varies-by-held-stratum__joint-treatment-stratum-outcome__predeclared-strata-and-scale__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces effect modification is a retained variation of a treatment or exposure contrast across predeclared strata on a declared comparison scale. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Association is not cause. Causal claims retain time order, comparator meaning, confounding, mediation, effect modification and the exact observation class capable of distinguishing the alternatives.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 2d73aae89ac5ca8130efaca6d89306fc123b6b37e8a2443fa4a26ee518f86e86.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 90deb28ae99c586e9b53ecf47cdf07b075c5aee1bef79922b06412186444a91.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: aae0a6f713cd7c03bcd0d1d0eaa60026f1811c4ddf93b910da3dc4ea8a26fcc.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 2c6868f80f9e081e0d038144f737f565bbf48001569a5b79cff94687e8ed5214.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256: c8307911ba6b84d4f3e0b9c5af18f0fdd95f02a9f81735f1bab3681c75be6d00; certificate
sha256: 45dc768b385476c50d9cf1714bf8e2cd7f84349fa76e3edb8e7da24f050dc949; external-validation
sha256: 7b46e956ccc0a75b3b082e83d34a70afc472f2022ea007a348a5daddadbcfb02.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 3 purpose-matched discriminators with ordered identity
sha256: 5af911055225875794c72d66743b1b75e2dac4647b031e084e43b7d5ce98cb04. Target opened after

derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- MED-CDC-EPIDEMIOLOGY-001 - United States Centers for Disease Control and Prevention; https://archive.cdc.gov/www_cdc_gov/csels/dsepd/ss1978/lesson1/summary.html; snapshot experiments/external_sources/medicine/snapshots/MED-CDC-EPIDEMIOLOGY-001.html; sha256: 430c2d885670c185491ab9b84d1c703eeaff0eece3ddb69fc855c001fb93478b; scope: population, comparison, frequency, pattern, cause, disease and injury epidemiology.
- MED-STROBE-001 - STROBE Initiative; <https://www.strobe-statement.org/>; snapshot experiments/external_sources/medicine/snapshots/MED-STROBE-001.html; sha256: 420432c148d01e7aaaab8f7bcb4bcb4412d028aa50cd7814fc1b058faf1cf7c; scope: observational-study planning, conduct, findings and interpretation reporting.

Comparison record:

- sft-med-effect-modification-001-authority-correspondence: predicted medicine:stratified-patient-carriers__contrast-varies-by-held-stratum__joint-treatment-stratum-outcome__predeclared-strata-and-scale; source-derived medicine:stratified-patient-carriers__contrast-varies-by-held-stratum__joint-treatment-stratum-outcome__predeclared-strata-and-scale; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256: ed8fe53d4e8ceecff9429f218bc508e9ccc11c475116206bdf6466dcfb580a0f; derivation seal sha256: 886c0acb6fd728525ce4bc1b88bd4ebe2065d876e76ffc4f39731812b0a795b4; engine receipt sha256: f2b3a00497b80f2bd576ed4562977e1f7c670db07871af69211a49c832bc7000 at receipts/engine/model_admitted/SFT-MED-EFFECT-MODIFICATION-001-f2b3a00497b80f2b.json; empirical validation sha256: a41d902dba78a4e05a7e831068052190245f2620a1f82af6cb8bcb4ae57c857c; measurement receipt sha256: ad6ece3828f37b0302eb6251bb114780cf9eaba268ba359c2160bb43a980a238; isolation sha256: 18cca2d4fb3aea431dc308049a25e13c8840d4cb1f845ffcfc0c57168ac9d6ccd; custody sha256: 6442a05baac521720b6d8c2a6c019c571774c572a6c4c2a087696b345c4438aa.

14. Risk Prognosis Outcome

Risk and prognosis are positive-finite relations over declared carriers and horizons. Every competing event, censored course, missing visit and component outcome remains present, preventing an unobserved event from becoming numerical zero.

25. Clinical risk

Claim identity: SFT-MED-RISK-001

Question and exact theorem. Risk is an exact outcome share among an explicitly eligible at-risk set over a declared positive finite horizon, with competing events and loss retained.

eligible-person-support__outcome-incidence-over-follow-up__at-risk-set-with-competing-events__population-start-time-and-horizon__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001

- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-EFFECT-MODIFICATION-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	eligible-person-support	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	outcome-incidence-over-follow-up	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	at-risk-set-with-competing-events	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	population-start-time-and-horizon	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: eligible-person-support__outcome-incidence-over-follow-up__at-risk-set-with-competing-events__population-start-time-and-horizon__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier: clinical carrier is explicit and nonempty; passed true.
- relation: relation and organization remain separately reconstructible; passed true.
- boundary: observation boundary cannot silently become the clinical law; passed true.

Scientific meaning. Risk is the exact ratio of observed outcome carriers to an eligible at-risk carrier set over a declared horizon. SFT represents an observed absence by a held absence label; it never introduces ontological nothing or silently discards loss to follow-up.

Risk and prognosis are positive-finite relations over declared carriers and horizons. Every competing event, censored course, missing visit and component outcome remains present, preventing an unobserved event from becoming numerical zero.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : ab9b260807d7f9ebe7004246b6860b4bbec55a2eab5f33ae40a26b5f301e1f0c.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : d13f788854f99dcd7717331ab1ff83763a592661c56a22c1f6161856060c9742.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 283310c47710d1e6dfd5b6b172e70dab95e650a7121c563d4cdf8232bb4f04e2.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 0eea9d4e77633d39f43ff17719055e394252c539100d65d8e46ee70970c3ea08.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256 : 565c8b7dc59966553c1fcf37180f4894d6f728dbcbef6f5f2fe6ff07e5c3294cb; certificate
sha256 : e72479fd64b0432b750ab89ab36fb98313e8238b75d397b7a63db5d6492af059; external-validation
sha256 : 18bd93b833b2b870930b76eadaa67e9d7f7ef502aadd2751bdbc0f139f9d1d9a.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 4 purpose-matched discriminators with ordered identity
sha256 : 5026dd69f12a67673bfae421264dba58ae9ac2a0352ed89fbb121a5476f8e839. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- MED-FDA-BIOMARKER-001 - United States Food and Drug Administration;
<https://www.fda.gov/drugs/biomarker-qualification-program/about-biomarkers-and-qualification>; snapshot
experiments/external_sources/medicine/snapshots/MED-FDA-BIOMARKER-001.html;
sha256 : 93c5ee4b5e3480ed27aa805a7739f4acf7108701fc6f429ae5d4e5a1ad9d3d31; scope: normal and pathogenic processes; risk, diagnostic, monitoring, prognostic, response and safety biomarkers; method and context.
- MED-ICH-E9R1-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/E9-R1_Step4_Guideline_2019_1203.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E9R1-001.pdf;
sha256 : f7471f411f1c87ee76783d5b2b9faeca31d01d530213f71b50d136b39e3b0d9; scope: final estimand, population, endpoint, intercurrent-event, missing-data and sensitivity-analysis distinctions.

Comparison record:

- sft-med-risk-001-authority-correspondence: predicted medicine:eligible-person-support__outcome-incidence-over-follow-up__at-risk-set-with-competing-events__population-start-time-and-horizon; source-derived medicine:eligible-person-support__outcome-incidence-over-follow-up__at-risk-set-with-competing-events__population-start-time-and-horizon; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:982c0dccc0c36069997b8b571d0e92b0465e5460b05e20b21d91c84c6c491ec0; derivation seal sha256:e01be9f19c9e673eb26f8aa5baa172b8355eb5c2c7dfc2d126a31fbbadb36d8e; engine receipt sha256:f38143f22f98b85462cb732d929aa797d3108da56aa176a05dd3cce74eec36c7 at receipts/engine/model_admitted/SFT-MED-RISK-001-f38143f22f98b854.json; empirical validation sha256:d91754a78a3aeaf153080ec6592c02b01521670c39fca12329451a1e8506352a; measurement receipt sha256:a7a7f0b2cfd98af42ae11b1ac6aea663a7f32e236a866c2c4ad4353154637399; isolation sha256:ab6fafdf5282efafb67e2498afd584886c6e7215fff6bee85fcd36c5007f5e77; custody sha256:5ac496f8315e4ffe5ccc5aa10ae115693a27f6aa4a336a7ff0b85e3ad4546988.

26. Incidence and prevalence

Claim identity: SFT-MED-INCIDENCE-PREVALENCE-001

Question and exact theorem. Incidence counts new condition transitions while prevalence counts supported states; the two observation relations may not be merged.

population-condition-records__new-event-versus-state-support__flow-and-stock-kept-distinct__population-and-time-boundary__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001

- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-RISK-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	population-condition-records	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	new-event-versus-state-support	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	flow-and-stock-kept-distinct	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	population-and-time-boundary	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
generality	single-patient-fav orable-study	positive-finite-successor -and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception -opaque-model-or-e xtra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: population-condition-records__new-event-versus-state-support__flow-and-stock-kept-distinct__population-and-time-boundary__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces incidence counts new condition transitions while prevalence counts supported states; the two observation relations may not be merged. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Risk and prognosis are positive-finite relations over declared carriers and horizons. Every competing event, censored course, missing visit and component outcome remains present, preventing an unobserved event from becoming numerical zero.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:0d7fa832cbbbe440d6acc917d6c266b421f96e1117599fb376dadab5a83950540.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:5ea3413c0e9743f84e753091c4cae7c874da071b52216c2ab3188a36d63b8fc6.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:63641b648d3f00ab364fbacedd1327979cb61a8f6a65425e78118297cb3382e6.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:41d916024e24b731ee62e1cdaa336358dcfb2f2fa7ceab879e264cde6a19f564.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:3eea1ac06d27c718e0ea0644a8a8fb1cc9d5554083dc08c6769b46035b2a5ddc; certificate
sha256:5784102f809e09c67fe35ba88ded69f959f5952339d750e65960c70c34eb7371; external-validation
sha256:e238127c67f4096a561dfec72d85a0509f2fd0743818bcd4517f6063c4ff5b8b.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 4 purpose-matched discriminators with ordered identity
sha256:5026dd69f12a67673bfae421264dba58ae9ac2a0352ed89fbb121a5476f8e839. Target opened after

derivation seal: **true**. All rows preserved: **true**. Exact comparison: **true**. A deliberately changed observation was rejected.

Sources:

- MED-FDA-BIOMARKER-001 - United States Food and Drug Administration;
https://www.fda.gov/drugs/biomarker-qualification-program/about-biomarkers-and-qualification-experiments/external_sources/medicine/snapshots/MED-FDA-BIOMARKER-001.html; snapshot sha256: 93c5ee4b5e3480ed27aa805a7739f4acf7108701fc6f429ae5d4e5a1ad9d3d31; scope: normal and pathogenic processes; risk, diagnostic, monitoring, prognostic, response and safety biomarkers; method and context.
- MED-ICH-E9R1-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/E9-R1_Step4_Guideline_2019_1203.pdf; snapshot experiments/external_sources/medicine/snapshots/MED-ICH-E9R1-001.pdf; sha256: f7471f411f1c87ee76783d5b2b9faeeca31d01d530213f71b50d136b39e3b0d9; scope: final estimand, population, endpoint, intercurrent-event, missing-data and sensitivity-analysis distinctions.

Comparison record:

- sft-med-incidence-prevalence-001-authority-correspondence: predicted medicine:population-condition-records__new-event-versus-state-support__flow-and-stock-kept-distinct__population-and-time-boundary; source-derived medicine:population-condition-records__new-event-versus-state-support__flow-and-stock-kept-distinct__population-and-time-boundary; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256: 2bfa3a11fa00b0eb884e6377e178d994cf6ada185c6dc6e46ff7373623dcfed0; derivation seal sha256: f5db347958e034815278f25390f79c5a8395897b92c750fcdac399896ea9f846; engine receipt sha256: a1fc63617f42562e76e169efb9f4a3e4812b681880a3650329ccf1f2df295dd8 at receipts/engine/model_admitted/SFT-MED-INCIDENCE-PREVALENCE-001-a1fc63617f42562e.json; empirical validation sha256: 1b2978e2e83c2f9ad81155114083f78512dbe70437ac4e6797ea20b38b10f5af; measurement receipt sha256: 59dcc89c34208961f0fd72df51d3b4284d1638cf9d06e9c780249b2b11243244; isolation sha256: 1c412ec98f9b1dd99ffc1429c18422a1eaa400a0a9c66c6e02f5b6cd4d3d2a5c; custody sha256: a7bcf3142676b07fc739a7096ca25b75a61915e46d38865388a63c3d9cb62db3.

27. Clinical prognosis

Claim identity: SFT-MED-PROGNOSIS-001

Question and exact theorem. Prognosis is the exact support of future outcomes conditional on a complete baseline and care context, never an unqualified fate label.

patient-baseline-state__baseline-to-future-outcome-support__conditioned-course-alternatives__care-context-horizon-and-censoring__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-INCIDENCE-PREVALENCE-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	patient-baseline-state	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	baseline-to-future-outcome-support	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	conditioned-course-alternatives	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	care-context-horizon-and-censoring	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: patient-baseline-state__baseline-to-future-outcome-support__conditioned-course-alternatives__care-context-horizon-and-censoring__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the clinical law; passed **true**.

Scientific meaning. A prognosis is conditioned support over a declared patient baseline and care context, not fate. Competing events, censoring, changes in treatment and alternative courses remain explicit.

Risk and prognosis are positive-finite relations over declared carriers and horizons. Every competing event, censored course, missing visit and component outcome remains present, preventing an unobserved event from becoming numerical zero.

Adverse controls.

- **false_premise**: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : ab74696e0439c393b3b5e59dd5dfc135ac513cd0f2ff315121ffd46fec78b044.
- **tampered_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 81005c985306fb1b581ce77c545a650ea5a61acfb3456fce5c3998df4c4336ea.
- **tampered_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 3a948cfc05357d9cf1e01326ec922fc04352aad98981bf713739f52e112a7676.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 3ac585297b551177684e3961af2af8f11d626ac588d427b394b941f5a6d76e29.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256 : d863ef59add08968abddc7903356384fc7197e9e2a758fb4283c3e010d78a582; certificate
sha256 : edda860f0e7b3ddd99391ba2d942385e81dfd91ffaefbb125cc45ceeb42f7fcf; external-validation
sha256 : 6114b1d1a5fcf81e086f0ac3f40392aec93fd2ccece7055674cb289238b13dc2.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 4 purpose-matched discriminators with ordered identity
sha256 : 5026dd69f12a67673bfae421264dba58ae9ac2a0352ed89fbb121a5476f8e839. Target opened after derivation seal: **true**. All rows preserved: **true**. Exact comparison: **true**. A deliberately changed observation was rejected.

Sources:

- **MED-FDA-BIOMARKER-001** - United States Food and Drug Administration;
<https://www.fda.gov/drugs/biomarker-qualification-program/about-biomarkers-and-qualification>; snapshot
experiments/external_sources/medicine/snapshots/MED-FDA-BIOMARKER-001.html;
sha256 : 93c5ee4b5e3480ed27aa805a7739f4acf7108701fc6f429ae5d4e5a1ad9d3d31; scope: normal and
pathogenic processes; risk, diagnostic, monitoring, prognostic, response and safety biomarkers; method and context.
- **MED-ICH-E9R1-001** - International Council for Harmonisation;
https://database.ich.org/sites/default/files/E9-R1_Step4_Guideline_2019_1203.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E9R1-001.pdf;
sha256 : f7471f411f1c87ee76783d5b2b9faeaca31d01d530213f71b50d136b39e3b0d9; scope: final
estimand, population, endpoint, intercurrent-event, missing-data and sensitivity-analysis distinctions.

Comparison record:

- **sft-med-prognosis-001-authority-correspondence**: predicted medicine:patient-baseline-state__baseline-to-future-outcome-support__conditioned-course-alternatives__care-context-horizon-and-censoring; source-derived medicine:patient-baseline-state__baseline-to-future-outcome-support__conditioned-course-alternatives__care-context-horizon-and-censoring; exact match **True**
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology

- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:239c27adc26858287dfbec01801a130afe87d8e6488a33907b0cc6aaae9cf5b6; derivation seal sha256:a8c86d020316f0222d990937a21318ee799895ecb3cdb80099a9b20e286cb8fd; engine receipt sha256:a0990310880d83648da40cc153e1ead7a75be9638e4ece60e71eca04505b5ace at receipts/engine/model_admitted/SFT-MED-PROGNOSIS-001-a0990310880d8364.json; empirical validation sha256:b30d6d6587dd1c83fc72b16c5c2b41df04cac62a0be1dbcb39fec41f5341a274; measurement receipt sha256:d46a1a195bb3d0a5347b220cb48984cb5bd8f52af4dc5ae9a0056a762a8bca5d; isolation sha256:29b0956da097a8f11191b823888d23db651e698edd74a06c78320fc382a310e1; custody sha256:62bc6cc9be261e1391dcf1de297fd22c6ba2efbb26dc4b5b5ce4c89a82c01001.

28. Clinical outcome

Claim identity: SFT-MED-CLINICAL-OUTCOME-001

Question and exact theorem. A clinical outcome retains endpoint definition, baseline, assessment time, method, assessor and every component or adverse alternative.

patient-endpoint-state__change-or-event-from-baseline__endpoint-definition-and-components__assessment-time-method-and-blinding__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001

- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-PROGNOSIS-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	patient-endpoint-state	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	change-or-event-from-base line	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	endpoint-definition-and-components	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	assessment-time-method-and-blinding	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
generality	single-patient-fav orable-study	positive-finite-successor -and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception -opaque-model-or-e xtra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: patient-endpoint-state__change-or-event-from-baseline__endpoint-definition-and-components__assessment-time-method-and-blinding__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces a clinical outcome retains endpoint definition, baseline, assessment time, method, assessor and every component or adverse alternative. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Risk and prognosis are positive-finite relations over declared carriers and horizons. Every competing event, censored course, missing visit and component outcome remains present, preventing an unobserved event from becoming numerical zero.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 2b77cf3446a33d11e3f91354a12f4aafc928e5ad49ffb46960862dc7bbbf0849.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 5c3702dbf08b8b6a045db034f3160922dd6eaddede1c21f5e77b2a205b4f3652.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 3ae1e5dcb9d73673a1cbf3e7aad33591d81d6082f91716cd8d15c955f258ab30.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: f03c16b0405bcdedfff1a13b84a6bb75f947c3e143cf216dddaf0388ac5e5bb2.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256: 0134deeb3107f0da4d709847e88c5467c6de6559dd9dd41fad1f04249a0e860; certificate
sha256: bdc41cdf2ccdf7c6dd4d7428964dce1ad904bf1e0ee561936550c2ed668ffbf; external-validation
sha256: 9491e2245d9950b0f10fe20bc0cd15915323be3cd08be4ceff8958a19995d273.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 4 purpose-matched discriminators with ordered identity
sha256: 5026dd69f12a67673bfae421264dba58ae9ac2a0352ed89fbb121a5476f8e839. Target opened after

derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- MED-FDA-BIOMARKER-001 - United States Food and Drug Administration;
<https://www.fda.gov/drugs/biomarker-qualification-program/about-biomarkers-and-qualification>; snapshot
experiments/external_sources/medicine/snapshots/MED-FDA-BIOMARKER-001.html;
sha256:93c5ee4b5e3480ed27aa805a7739f4acf7108701fc6f429ae5d4e5a1ad9d3d31; scope: normal and
pathogenic processes; risk, diagnostic, monitoring, prognostic, response and safety biomarkers; method and context.
- MED-ICH-E9R1-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/E9-R1_Step4_Guideline_2019_1203.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E9R1-001.pdf;
sha256:f7471f411f1c87ee76783d5b2b9faeeca31d01d530213f71b50d136b39e3b0d9; scope: final
estimand, population, endpoint, intercurrent-event, missing-data and sensitivity-analysis distinctions.

Comparison record:

- sft-med-clinical-outcome-001-authority-correspondence: predicted medicine:patient-endpoint-state__change-or-event-from-baseline__endpoint-definition-and-components__assessment-time-method-and-blinding; source-derived medicine:patient-endpoint-state__change-or-event-from-baseline__endpoint-definition-and-components__assessment-time-method-and-blinding; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest
sha256:8261346279c0777219178bd5c3dacc93dd5492d8a8f9bb25ad5c95d1cd4ee5c8; derivation seal
sha256:33d8a0c858b80114052a80aa7faf6d7602e5a7050ee7da3f83256ca1be239a65; engine receipt
sha256:37df271433294315ed27c02bca4672fd5351ca7f32995bca32a6ce0e12a85722 at
receipts/engine/model_admitted/SFT-MED-CLINICAL-OUTCOME-001-37df271433294315.json;
empirical validation sha256:55a2f93ea0d4a785d15d788f877c6e9a4e66de617db4be7034ade9666400a86c;
measurement receipt sha256:9ae653cc1d21e2c1a05b79b1e999da291ac8d882f89a8d36aeef2f4c271128f1;
isolation sha256:42a57913ff8f06e43e09daf57a238d2538e7ae1c14a417061e6cc5e74f5215a2; custody
sha256:d1c97677d1056c5daef0ce5f559d2429d064ea488db9955f3768ed1a974acafe.

29. Censoring and competing events

Claim identity: SFT-MED-CENSORING-COMPETING-001

Question and exact theorem. Censoring and competing events are explicit course outcomes; neither may be silently treated as absence of the target event.

incomplete-patient-course__observability-and-competing-transition__retained-unobserved-support__reason-time-and-follow-up__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001

- SFT-MED-CLINICAL-OUTCOME-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	incomplete-patient-course	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	observability-and-competing-transition	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	retained-unobserved-support	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	reason-time-and-follow-up	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: incomplete-patient-course__observability-and-competing-transition__retained-unobserved-support__reason-time-and-follow-up__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier: clinical carrier is explicit and nonempty; passed true.
- relation: relation and organization remain separately reconstructible; passed true.
- boundary: observation boundary cannot silently become the clinical law; passed true.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces censoring and competing events are explicit course outcomes; neither may be silently treated as absence of the target event. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Risk and prognosis are positive-finite relations over declared carriers and horizons. Every competing event, censored course, missing visit and component outcome remains present, preventing an unobserved event from becoming numerical zero.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : 267d6b811a3e6a734169fb54cf617122d550bd397bbb6871320a5ed19380ef9d.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 1648f209331ed181d446f0b6ad2ed2a67743148d9d51c456357d0edf7c0b9fb2.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : e2b341cd00f7ec9f371a657bd87c7f8eb62b39549fc5b417e6326ec3cc4f098f.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : bc1ab59ca8e74989941d1ee0fc33111fc9a4c7d9a7e942cb807ca2d49b5d2113.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256 : b05888bae4de466beef8b49ac85ab0e50b7567de6ad56b0c5f9bcfbc81548201; certificate
sha256 : 1789c5eb3130b6fb2b8bede0c88efafab45eac09191963c6a9a5fdc18f1ccdf; external-validation
sha256 : 4fbea1b79a3b67a6e106b343fa99784be9f7ac76ab83b32c1511655c0e9e39ab.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 4 purpose-matched discriminators with ordered identity
sha256 : 5026dd69f12a67673bfae421264dba58ae9ac2a0352ed89fbb121a5476f8e839. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- MED-FDA-BIOMARKER-001 - United States Food and Drug Administration;
<https://www.fda.gov/drugs/biomarker-qualification-program/about-biomarkers-and-qualification>; snapshot
experiments/external_sources/medicine/snapshots/MED-FDA-BIOMARKER-001.html;
sha256 : 93c5ee4b5e3480ed27aa805a7739f4acf7108701fc6f429ae5d4e5a1ad9d3d31; scope: normal and pathogenic processes; risk, diagnostic, monitoring, prognostic, response and safety biomarkers; method and context.
- MED-ICH-E9R1-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/E9-R1_Step4_Guideline_2019_1203.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E9R1-001.pdf;
sha256 : f7471f411f1c87ee76783d5b2b9faeca31d01d530213f71b50d136b39e3b0d9; scope: final estimand, population, endpoint, intercurrent-event, missing-data and sensitivity-analysis distinctions.

Comparison record:

- sft-med-censoring-competing-001-authority-correspondence: predicted medicine:incomplete-patient-course__observability-and-competing-transition__retained-unobserved-support__reason-time-and-follow-up; source-derived medicine:incomplete-patient-course__observability-and-competing-transition__retained-unobserved-support__reason-time-and-follow-up; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is

erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:90324f1ac9c6ae14dff68b64313f6921c04188b6d93cc6acc99d7805019538bf; derivation seal sha256:1302d67021f680c388c9ac0ad30cf604fcb07413509702ee3dd5dd633875dd01; engine receipt sha256:10ccdc27c33fef37de264045ed5cc288dec3228e35a57309a4fabe0e7b6b2af6 at receipts/engine/model_admitted/SFT-MED-CENSORING-COMPETING-001-10ccdc27c33fef37.json; empirical validation sha256:28263168fa1d3229744856fda569652cecd5adb5a656aa232c50ccee85873626; measurement receipt sha256:1338af64aeedd83050bb7ca032a782b2a2ccffe71ab84f46878a96cae177136a; isolation sha256:3068990cfc8c703fa911b69de411e4abab45116a0e3031e67b9e6b7dadbd3086; custody sha256:b612ca44c103525fc427b3551cf8ea7437aaa74a2e4498cafaf95f275430c3d.

30. Prognostic calibration

Claim identity: SFT-MED-PROGNOSTIC-CALIBRATION-001

Question and exact theorem. Calibration compares predeclared risk groups with every observed outcome row in a distinct population and interval without post-outcome retuning.

predicted-and-observed-risk-groups__prediction-observation-incidence__complete-calibration-table__new-population-and-time-window__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NO-THING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001

- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-CENSORING-COMPETING-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	predicted-and-observed-risk-groups	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	prediction-observation-incidence	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	complete-calibration-table	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	new-population-and-time-window	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: `predicted-and-observed-risk-groups__prediction-observation-incidence__complete-calibration-table__new-population-and-time-window__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule`.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier`: clinical carrier is explicit and nonempty; passed `true`.
- `relation`: relation and organization remain separately reconstructible; passed `true`.
- `boundary`: observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces calibration compares predeclared risk groups with every observed outcome row in a distinct population and interval without post-outcome retuning. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Risk and prognosis are positive-finite relations over declared carriers and horizons. Every competing event, censored course, missing visit and component outcome remains present, preventing an unobserved event from becoming numerical zero.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 853a4429c1e57781ae204c0349b201f86039aea87d30318c18fe9e96345ad04a.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: fbcadf118ca676f91b766d022c83f3a52364ae576f96b78ff4ecf5d3eb57e25f.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 22e2d1f824021018fff2d7a79126d6af4d216aab7717e6bcad70ac63e0b865db.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: e5ee1cf5d28c5a9371fce539fb3adedfe4d97525597ba0c1adc845392546ce2b.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256: 4a65886ac1aa04d5a8248f8542e43542bfe4ac0bca3640ec70281c0d83e27a28; certificate
sha256: b9edb627c93571ed4b672170463b572bb148069f832b05c72862334749f7e605; external-validation
sha256: 6783bbb2cc22481687073be7462048d4ea0d5ce9d3bdf51877ce47e37bfbdde9.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 4 purpose-matched discriminators with ordered identity

sha256:5026dd69f12a67673bfae421264dba58ae9ac2a0352ed89fbb121a5476f8e839. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- MED-FDA-BIOMARKER-001 - United States Food and Drug Administration;
<https://www.fda.gov/drugs/biomarker-qualification-program/about-biomarkers-and-qualification>; snapshot experiments/external_sources/medicine/snapshots/MED-FDA-BIOMARKER-001.html; sha256:93c5ee4b5e3480ed27aa805a7739f4acf7108701fc6f429ae5d4e5a1ad9d3d31; scope: normal and pathogenic processes; risk, diagnostic, monitoring, prognostic, response and safety biomarkers; method and context.
- MED-ICH-E9R1-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/E9-R1_Step4_Guideline_2019_1203.pdf; snapshot experiments/external_sources/medicine/snapshots/MED-ICH-E9R1-001.pdf; sha256:f7471f411f1c87ee76783d5b2b9faaeca31d01d530213f71b50d136b39e3b0d9; scope: final estimand, population, endpoint, intercurrent-event, missing-data and sensitivity-analysis distinctions.

Comparison record:

- sft-med-prognostic-calibration-001-authority-correspondence: predicted medicine:predicted-and-observed-risk-groups__prediction-observation-incidence__complete-calibration-table__new-population-and-time-window; source-derived medicine:predicted-and-observed-risk-groups__prediction-observation-incidence__complete-calibration-table__new-population-and-time-window; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:6f57119316fbce418dc15704c9dabc0870db440541110e3987752b8170599c43; derivation seal sha256:4222adce38cb9e303d04b8715f21a0a64b95695ec63b783206ec8822eb37455a; engine receipt sha256:e3702627ecea41f1f14fbe6995e62d43c6aaf953ae95da03f5eeff1daeee4e90 at receipts/engine/model_admitted/SFT-MED-PROGNOSTIC-CALIBRATION-001-e3702627ecea41f1.json; empirical validation sha256:1f5decc7d59b3438f03675eb39daae13905c1d0ccc9ad3f5c8c2d9238baf6592; measurement receipt sha256:2296ca4123dbef4e3b74cfb759cec8383ddfe07b69fdcc84f2ff8508fb175c42; isolation sha256:3f22ae7e5d9beebbb3b5fd75d5c1f78124771154481a95fbb6a6fc33d4019bf6; custody sha256:2a87dee5da207b72014e4dfdce2cd63e4bd280757c6adc9896abb17d4314e651.

15. Intervention Comparator Counterfactual

An intervention is scientifically legible only beside a comparator and an explicit alternative-course boundary. Allocation, concealment, adherence, crossover and cointervention remain in the trace rather than being repaired after outcomes are known.

31. Clinical intervention

Claim identity: SFT-MED-INTERVENTION-001

Question and exact theorem. An intervention is a completely specified patient-directed action whose content, delivery, adherence, context and timing remain recorded.

patient-directed-action__action-to-course-transition__content-delivery-and-adherence__eligibility-setting-time-and-provider__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001

- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-PROGNOSTIC-CALIBRATION-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	patient-directed-action	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	action-to-course-transition	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	content-delivery-and-adherence	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	eligibility-setting-time-and-provider	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: patient-directed-action__action-to-course-transition__content-delivery-and-adherence__eligibility-setting-time-and-provider__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces an intervention is a completely specified patient-directed action whose content, delivery, adherence, context and timing remain recorded. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

An intervention is scientifically legible only beside a comparator and an explicit alternative-course boundary. Allocation, concealment, adherence, crossover and cointervention remain in the trace rather than being repaired after outcomes are known.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 7b295e34b99f72df61993d6ca1a08123493d4b8c4a0e8bba74b31178cae1c495.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 19c0cac76e5f1240cf90ed52117671e2558c2937544a9fd358a50deb0caa3b78.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: cdda2cc9a94abf4b918af0e2d6aff213283410b575efda5902ae3483e741c6.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 0806dce51b413e50f8fb0bbba66f94edb81d5404fd934d542f3a92b0d0df0a0a.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256: b42b534bd7764a96a5a0954a35f532aaa82580a2751dc11a87e9582b72f5e83f; certificate
sha256: 5571b6583d8ce75b64efec285b3a5119c36641a3ea923372a14f155c5ef202d0; external-validation
sha256: 8bb693deae1884e06b8e6d28029ade7627494e9bea28c9fdebb0fac71ae1cab35.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 4 purpose-matched discriminators with ordered identity
sha256: 233d7a2d830e6567b35067297ef7952a74569055133848cc1b096a0149df63c5. Target opened after derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- MED-ICH-E6R3-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E6R3-001.pdf;
sha256: e6ce19e36ce7d2e294f89ee89492b9e035178c3cca48984392bbd92eec9b002c; scope: final Good Clinical Practice: rights, safety, wellbeing, consent, protocol, allocation, blinding, adverse events, confidentiality and essential records.
- MED-ICH-E9R1-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/E9-R1_Step4_Guideline_2019_1203.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E9R1-001.pdf;
sha256: f7471f411f1c87ee76783d5b2b9faeca31d01d530213f71b50d136b39e3b0d9; scope: final estimand, population, endpoint, intercurrent-event, missing-data and sensitivity-analysis distinctions.

Comparison record:

- sft-med-intervention-001-authority-correspondence: predicted medicine:patient-directed-action__action-to-course-transition__content-delivery-and-adherence__eligibility-setting-time-and-provider; source-derived medicine:patient-directed-action__action-to-course-transition__content-delivery-and-adherence__eligibility-setting-time-and-provider; exact match `True`
- every required authoritative fragment and snapshot hash reproduced

- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:25517832d8e1fd410a977dfa124fc17ceda06665c402ce75229c1b4beb4592ef; derivation seal sha256:8dcbce5feb9a2aba92f6913fdf4b8b30c02567aa44922814cef55bc1ef17178b; engine receipt sha256:ef389261cfe2919237eefe5a95ed42263554fa6271eb87e13685b1d43ed1ddcc at receipts/engine/model_admitted/SFT-MED-INTERVENTION-001-ef389261cfe29192.json; empirical validation sha256:08d0fd64eb147b4e5963e8013295ed3ef825bce2625153a5cc0bf837a22fbde3; measurement receipt sha256:4b08473956921b74c22c11a43a998b0195d19a563d222f281c3280f574eb9dee; isolation sha256:aadc0daabab138cb691c463fc9431fb34e59b43fa7de40291409ff2d2886a7ec; custody sha256:ebe6154eeb824782fe2d02eca0825bc64255452a1fb7b246e1f77abc36ac855f.

32. Clinical comparator

Claim identity: SFT-MED-COMPARATOR-001

Question and exact theorem. A comparator is an explicit alternative action evaluated under the same eligibility, outcome and time boundary; omitted care is not numerical zero.

alternative-care-action__contrast-with-intervention__concurrent-or-declared-reference__same-eligibility-outcome-and-time__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001

- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-INTERVENTION-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	alternative-care-action	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	contrast-with-intervention	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	concurrent-or-declared-reference	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
observation	observation-boundary-unrecorded	same-eligibility-outcome-and-time	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: `alternative-care-action__contrast-with-intervention__concurrent-or-declared-reference__same-eligibility-outcome-and-time__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule`.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces a comparator is an explicit alternative action evaluated under the same eligibility, outcome and time boundary; omitted care is not numerical zero. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

An intervention is scientifically legible only beside a comparator and an explicit alternative-course boundary. Allocation, concealment, adherence, crossover and cointervention remain in the trace rather than being repaired after outcomes are known.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:1ad0ab2d0ae004a5b6a1e5425facb4e93cbd033f86ab9cdd027841968006af22.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:ce0b40b17ee9e210e4ff19a253c2c0a8519325f564f76eb0a78a595cb018f07d.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:587762fde4399624ef9643a3702817b8b96f315e0ab3c3c85739c9d6664f2e67.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : bda08dbad5938518c18d0c45fc44005ee7adbe2fa87de4278a13b8a088208d65.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256 : 9dab63cfe2f74d2e04de22b1ffe7756d9873685385adcc9e7472491fa5a3481f; certificate
sha256 : 3a988bf3592234c258c7b0269f33af5b0385e84f0238fda22f4279249e26ee18; external-validation
sha256 : ceab306558257dd65a17858d8f98a8e85bff762ae8b9cbfb009c5ba52edf8538.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 4 purpose-matched discriminators with ordered identity
sha256 : 233d7a2d830e6567b35067297ef7952a74569055133848cc1b096a0149df63c5. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- MED-ICH-E6R3-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E6R3-001.pdf;
sha256 : e6ce19e36ce7d2e294f89ee89492b9e035178c3cca48984392bbd92eec9b002c; scope: final Good Clinical Practice: rights, safety, wellbeing, consent, protocol, allocation, blinding, adverse events, confidentiality and essential records.
- MED-ICH-E9R1-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/E9-R1_Step4_Guideline_2019_1203.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E9R1-001.pdf;
sha256 : f7471f411f1c87ee76783d5b2b9faeca31d01d530213f71b50d136b39e3b0d9; scope: final estimand, population, endpoint, intercurrent-event, missing-data and sensitivity-analysis distinctions.

Comparison record:

- sft-med-comparator-001-authority-correspondence: predicted medicine:alternative-care-action__contrast-with-intervention__concurrent-or-declared-reference__same-eligibility-outcome-and-time; source-derived medicine:alternative-care-action__contrast-with-intervention__concurrent-or-declared-reference__same-eligibility-outcome-and-time; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:84fba25f6896faf87dacbbc1fa2c6e757756acb1e90194da1b69477bdeafe1bd; derivation seal sha256:e82647c0623309d2734110b2db7249b946ee5ef2c6b882469d26524b9232629b; engine receipt sha256:3b156a2ee733c40ac3eb86c1d0860fddfaf3ed48e10a6dfd1718b2ac1d8af277 at receipts/engine/model_admitted/SFT-MED-COMPARATOR-001-3b156a2ee733c40a.json; empirical validation sha256:27cc3392ae5f789c40224117a145056ebacc7b27e4d938a840c1d62d864da703; measurement receipt sha256:7de318aec04c0543016127c0a802c3e3389190b96af38fa2ba336512e0deaae9; isolation sha256:b85c5ed64ae14d3fddefdf868dff2e46532c7a51280a0c18face3a43e8aedb3e; custody sha256:da7dca14d1abb3800e8c90d4bd23805215c3e6aedef3a76de68a0b87d8db95679.

33. Clinical counterfactual

Claim identity: SFT-MED-COUNTERFACTUAL-001

Question and exact theorem. A counterfactual denotes alternative courses for the same eligibility state; because only one course is observed, the missing course and identification boundary must remain explicit.

same-eligibility-patient-state__alternative-action-outcome-relation__mutually-exclusive-potential-courses__identification-assumption-boundary__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001

- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-COMPARATOR-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	same-eligibility-patient-state	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	alternative-action-outcome-relation	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	mutually-exclusive-potential-courses	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	identification-assumption-boundary	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: same-eligibility-patient-state__alternative-action-outcome-relation__mutually-exclusive-potential-courses__identification-assumption-boundary__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier`: clinical carrier is explicit and nonempty; passed `true`.
- `relation`: relation and organization remain separately reconstructible; passed `true`.
- `boundary`: observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. Alternative courses are mutually exclusive observations for the same eligibility state. The missing course is retained as unavailable support, so identification assumptions can be inspected instead of being mistaken for measured outcomes.

An intervention is scientifically legible only beside a comparator and an explicit alternative-course boundary. Allocation, concealment, adherence, crossover and cointervention remain in the trace rather than being repaired after outcomes are known.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : d52f01671a1ecdce491ecb5765ba364dedd5baacc65c85755643366663e2a6b0.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : a705acba9152813c4719a80dbf7588a9d7bbb8dc7967aa0eec6350bbfa558175.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 011e9337f157fac6575ba56b147798faea458dab62c5f02fd821d56595f5c790.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : f0f534fc24cae0e1b6a015860f6d7089b6a78708e7995756bad993bc680e1e1.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256 : 22f4f7e15c7df0a7b7d912d8579105423d45b14b7257782db4e862ee702b88f5; certificate
sha256 : be45bd991bc8dcccfa067b6a9a9495d30bc7e044d0c1168d54579b73de6d57d; external-validation
sha256 : a6eace400a8c1629d48b372e37470d5328605fcf1da8654dd8ecd98dec58cf1d.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 4 purpose-matched discriminators with ordered identity

sha256 : 233d7a2d830e6567b35067297ef7952a74569055133848cc1b096a0149df63c5. Target opened after derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- MED-ICH-E6R3-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E6R3-001.pdf;
sha256 : e6ce19e36ce7d2e294f89ee89492b9e035178c3cca48984392bbd92eec9b002c; scope: final Good Clinical Practice: rights, safety, wellbeing, consent, protocol, allocation, blinding, adverse events, confidentiality and essential records.
- MED-ICH-E9R1-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/E9-R1_Step4_Guideline_2019_1203.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E9R1-001.pdf;
sha256 : f7471f411f1c87ee76783d5b2b9faeca31d01d530213f71b50d136b39e3b0d9; scope: final estimand, population, endpoint, intercurrent-event, missing-data and sensitivity-analysis distinctions.

Comparison record:

- sft-med-counterfactual-001-authority-correspondence: predicted medicine:same-eligibility-patient-state__alternative-action-outcome-relation__mutually-exclusive-potential-courses__identification-assumption-boundary; source-derived medicine:same-eligibility-patient-state__alternative-action-outcome-relation__mutually-exclusive-potential-courses__identification-assumption-boundary; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:c2713c0ec83442a8eba6c78fca227369bb55dc962c1c2ef0fa6529481fa20555; derivation seal sha256:e1447fd1e29273c096a1628f3ae508d37136fa2ba3c887b51c38bda50e30f816; engine receipt sha256:931e3d7e244c7630ea57fb5f686c9bce2d14c9ff6ecea9a9f302e673b16ff073 at receipts/engine/model_admitted/SFT-MED-COUNTERFACTUAL-001-931e3d7e244c7630.json; empirical validation sha256:52c70978a2fac89fde51dcb853a1ec8f9189786c678a4f443c21187fa0a504f1; measurement receipt sha256:a7400cb9efbc1a67610c853e06c401379f261d04696db130d826864b713eecfc; isolation sha256:b29474065cefea0cf90872e02624f68d9a40c2e0850d3843e416da5228ae325e; custody sha256:d85b9d6c4765f6c5470e44bacc5b44e58f5b5743281162984c29ba9990b75ac9.

34. Intervention allocation

Claim identity: SFT-MED-ALLOCATION-001

Question and exact theorem. Allocation is a fully traced deterministic assignment sequence whose concealment can prevent selection even though SFT imports no ontological nondeterminism.

eligible-participant-records__assignment-to-declared-arm__allocation-sequence-with-concealment__generation-implementation-and-time__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001

- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-COUNTERFACTUAL-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	eligible-participant-records	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	assignment-to-declared-arm	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
organization	organization-collapsed	allocation-sequence-with-concealment	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	generation-implementation-and-time	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: `eligible-participant-records__assignment-to-declared-arm__allocation-sequence-with-concealment__generation-implementation-and-time__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule`.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces allocation is a fully traced deterministic assignment sequence whose concealment can prevent selection even though sft imports no ontological nondeterminism. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

An intervention is scientifically legible only beside a comparator and an explicit alternative-course boundary. Allocation, concealment, adherence, crossover and cointervention remain in the trace rather than being repaired after outcomes are known.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:bc4a5abc5d214708f626d668af02c8a46d24e5d665a3ffbadacac09c5060695c`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:95588900d9c0b09e48b074fc2318c7d82706a216c660824b935a091007c5890c`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:e2e6427edb3c0e825a25bb9e7a061d0c444308fb905956680440f45767856902.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:5689fb7c5760e423c498518002e5696954faabdab87e2dfbfd27284bdefb6386.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:6ac6f2c7cc30c4127ed8231e046e13389412282ed85ba2ac6a17d70fbc079668; certificate
sha256:11d42233da2809446b85eb64b2fbf77529539a3520fd1ec7515475a61744b3ef; external-validation
sha256:1fee2c20fb50f12792452ccdd43c85e1974691549999fd0c29ceb8ff6acece7b.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 4 purpose-matched discriminators with ordered identity

sha256:233d7a2d830e6567b35067297ef7952a74569055133848cc1b096a0149df63c5. Target opened after derivation seal: **true**. All rows preserved: **true**. Exact comparison: **true**. A deliberately changed observation was rejected.

Sources:

- MED-ICH-E6R3-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E6R3-001.pdf;
sha256:e6ce19e36ce7d2e294f89ee89492b9e035178c3cca48984392bbd92eec9b002c; scope: final Good Clinical Practice: rights, safety, wellbeing, consent, protocol, allocation, blinding, adverse events, confidentiality and essential records.
- MED-ICH-E9R1-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/E9-R1_Step4_Guideline_2019_1203.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E9R1-001.pdf;
sha256:f7471f411f1c87ee76783d5b2b9faaeca31d01d530213f71b50d136b39e3b0d9; scope: final
estimand, population, endpoint, intercurrent-event, missing-data and sensitivity-analysis distinctions.

Comparison record:

- sft-med-allocation-001-authority-correspondence: predicted medicine:eligible-participant-records__assignment-to-declared-arm__allocation-sequence-with-concealment__generation-implementation-and-time; source-derived medicine:eligible-participant-records__assignment-to-declared-arm__allocation-sequence-with-concealment__generation-implementation-and-time; exact match
True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result

- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:6076bbf540e4da84c6c51d5731512cd74629e124c5f08cb3c590e343aafb6b57; derivation seal sha256:b8a0da8d68be89e7181796ed7b941f5ab31bf2c9c6986eca4b4c9458ca92ee49; engine receipt sha256:7dac7d6c996e2ed5a914b2d5ec6318b1d2be6730e56c3531dc99216f90bd6ea2 at receipts/engine/model_admitted/SFT-MED-ALLOCATION-001-7dac7d6c996e2ed5.json; empirical validation sha256:22a59bef7a4e8c2e092860fb9d281d6187cf8c4b618adc0261a2ced623f0c5ef; measurement receipt sha256:82150f7a758ded7d65dc2246c870b8143792b7d07287ffd13b324f883bbb2c6f; isolation sha256:2c1c048bc057fa7818ba6e9a9aca0afdda15358079c6ff98fc8b22a63dc1b87b; custody sha256:241289c48600d736a7970c3b805f7b6b3c16bff6892f9cccf9575dc02d08532f.

35. Adherence and crossover

Claim identity: SFT-MED-ADHERENCE-CROSSOVER-001

Question and exact theorem. Adherence and crossover preserve assignment, received care, timing and reason as distinct labels rather than rewriting the allocated history.

assigned-and-received-care-record__assignment-delivery-use-relation__protocol-deviation-trace__participant-time-and-reason__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001

- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-ALLOCATION-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	assigned-and-received-care-record	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	assignment-delivery-use-relation	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	protocol-deviation-trace	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	participant-time-and-reason	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: assigned-and-received-care-record__assignment-delivery-use-relation__protocol-deviation-trace__participant-time-and-reason__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces adherence and crossover preserve assignment, received care, timing and reason as distinct labels rather than rewriting the allocated history. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

An intervention is scientifically legible only beside a comparator and an explicit alternative-course boundary. Allocation, concealment, adherence, crossover and cointervention remain in the trace rather than being repaired after outcomes are known.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: bb5e840513b7b0fcdcf0769f22adc564dd9886753e893081ccc35a33f0b6e548.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: e6250d9e1f8f495dd03a68920a7b476ece90f25594425e79e2cd61a4beab09ab.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 8483b688e5710c108f9af32d16a3181bd74b7dac25b354b0c75fa3a2d4d3ddbc.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 104457867c5e1ad2d22bfc5a40ca7ffdb005de504b7d7410c6e87e0e1604a507.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256: 122390f912713817963ca6b060501c652ee5a5a82b1edb0f27d1edcefbcb986; certificate
sha256: 897f4fa23624c1004de2267c851c07b2eb6d30bd4b0b4ed24fb75d4adce776a2; external-validation
sha256: b0b6eaf4aa9dce0659c8b25774b97e7262e6617d1bf49ba19f3d9279a7fb257d.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 4 purpose-matched discriminators with ordered identity
sha256: 233d7a2d830e6567b35067297ef7952a74569055133848cc1b096a0149df63c5. Target opened after derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- MED-ICH-E6R3-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E6R3-001.pdf;
sha256: e6ce19e36ce7d2e294f89ee89492b9e035178c3cca48984392bbd92eec9b002c; scope: final Good Clinical Practice: rights, safety, wellbeing, consent, protocol, allocation, blinding, adverse events, confidentiality and essential records.
- MED-ICH-E9R1-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/E9-R1_Step4_Guideline_2019_1203.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E9R1-001.pdf;
sha256: f7471f411f1c87ee76783d5b2b9faeca31d01d530213f71b50d136b39e3b0d9; scope: final estimand, population, endpoint, intercurrent-event, missing-data and sensitivity-analysis distinctions.

Comparison record:

- sft-med-adherence-crossover-001-authority-correspondence: predicted medicine:assigned-and-received-care-record__assignment-delivery-use-relation__protocol-deviation-trace__participant-time-and-reason; source-derived medicine:assigned-and-received-care-record__assignment-delivery-use-relation__protocol-deviation-trace__participant-time-and-reason; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:9af102a1553109edc13c19234aeb47fc45b98edfcc12f7443265ceefefc77547; derivation seal sha256:e709dc235b0e4da11837444a16d46f3e2c334dd5675b5be16b9123ad75933934; engine receipt sha256:5076f9e466898ad2d677ffa86b6beb06ccba33afe2b1a274a3148795eebf42eb at receipts/engine/model_admitted/SFT-MED-ADHERENCE-CROSSOVER-001-5076f9e466898ad2.json; empirical validation sha256:886a6505f14af20e8b3c618bb25f3dbaa130b7fd4917a7b10f6503a8b02a8ff7; measurement receipt sha256:a10bbac458d98336574f73e4f2228118a7d6a462fed4c80caf0486d7813ab081; isolation sha256:2555b18438aa7dd7fc5a8b44c770f2309713b01d28c28f63d44b875406b61b70; custody sha256:f9a813c74f7a094359061e190c3087ae22faa1fc417e57693e7b94d980b90962.

36. Clinical cointervention

Claim identity: SFT-MED-COINTERVENTION-001

Question and exact theorem. Every concurrent care action that can change an outcome remains in the intervention word and may not be hidden inside an arm label.

joint-care-word__additional-action-to-outcome-relation__all-concurrent-actions-held__arm-setting-and-time__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001

- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-ADHERENCE-CROSSOVER-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	joint-care-word	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	additional-action-to-outcome-relation	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
organization	organization-collapsed	all-concurrent-actions-held	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	arm-setting-and-time	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: `joint-care-word__additional-action-to-outcome-relation__all-concurrent-actions-held__arm-setting-and-time__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule`.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces every concurrent care action that can change an outcome remains in the intervention word and may not be hidden inside an arm label. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

An intervention is scientifically legible only beside a comparator and an explicit alternative-course boundary. Allocation, concealment, adherence, crossover and cointervention remain in the trace rather than being repaired after outcomes are known.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:1074e1183ebe4e7e144fe7532272dbbdc00eaca9ba98e54832f30ab336cbd3e5`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:f248a1f9da8b0c261306844b0e6b1c212bf68bae6ce7dc5d01510fb45bf33583`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:134bf9cc9143930a21ac439dd227433de6702fbd2ec50f2794918dea8548c08b.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:b94cf93ff8be8288bce37147fdf5c5f9d3d36d48b35c85a73aa4936bc15ef03a.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:b46414120d0cd160e9331845262c845e2bc62efc79ad86d63d87529396ddadd0; certificate
sha256:72eaf599095129ee261b5ba1bf96b391bdb45bd90f2e14ba13e2ff24265ecc2d; external-validation
sha256:98255d0aed04f1a2792b5aba0db732c30ef3fa831e46ab03fed8df88a8acf33c.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 4 purpose-matched discriminators with ordered identity
sha256:233d7a2d830e6567b35067297ef7952a74569055133848cc1b096a0149df63c5. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- MED-ICH-E6R3-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E6R3-001.pdf;
sha256:e6ce19e36ce7d2e294f89ee89492b9e035178c3cca48984392bbd92eec9b002c; scope: final Good Clinical Practice: rights, safety, wellbeing, consent, protocol, allocation, blinding, adverse events, confidentiality and essential records.
- MED-ICH-E9R1-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/E9-R1_Step4_Guideline_2019_1203.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E9R1-001.pdf;
sha256:f7471f411f1c87ee76783d5b2b9faeeca31d01d530213f71b50d136b39e3b0d9; scope: final estimand, population, endpoint, intercurrent-event, missing-data and sensitivity-analysis distinctions.

Comparison record:

- sft-med-cointervention-001-authority-correspondence: predicted
medicine:joint-care-word__additional-action-to-outcome-relation__all-concurrent-actions-held__arm-setting-and-time;
source-derived
medicine:joint-care-word__additional-action-to-outcome-relation__all-concurrent-actions-held__arm-setting-and-time; exact
match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population

- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:c13ce54bc2f96e7b21d9a0dc0f27a54204a4fd32979e33eb18c374f3504c2223; derivation seal sha256:a5cdda5ccfe9238db7010d7ec321498d1a96029801b99653a3b0d1203f5d1eb6; engine receipt sha256:473077c4999a24d10246b71f4fa85aa1b9c316b0679752f66d09e649c840b0f3 at receipts/engine/model_admitted/SFT-MED-COINTERVENTION-001-473077c4999a24d1.json; empirical validation sha256:16542c02646a6474a6455035643cf53ae444b08ba9f4bc11974bc18c3bd683ed; measurement receipt sha256:8e50ff1b583d7d4195d55f4e56319f3d7064c25cb361a4301d0e08e31424a43a; isolation sha256:b68318977e28ea7b950a750812c9784a442401700e0b8655853fdabd6335dbaa; custody sha256:bad43d0ba368bda8f873c65fed1ad40b2f67bdaae9e8c0529bddc0234cce8670b.

16. Dose Exposure Response Adverse Event

Dose, exposure, response, adverse event and toxicity are separately generated relations. Route, formulation, schedule, duration, organ or function, causality assessment and recovery remain held; one fitted curve cannot erase adverse or nonmonotone evidence.

37. Dose

Claim identity: SFT-MED-DOSE-001

Question and exact theorem. Dose is an exact positive finite relation among amount, recipient carrier, formulation, route and schedule; no universal effect follows from dose alone.

administered-substance-or-action__amount-per-declared-carrier-and-time__unit-route-formulation-schedule__delivered-and-received-record__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001

- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-COINTERVENTION-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	administered-substance-or-action	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	amount-per-declared-carrier-and-time	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	unit-route-formulation-schedule	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	delivered-and-received-record	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
extension	free-fit-exception -opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: administered-substance-or-action__amount-per-declared-carrier-and-time__unit-route-formulation-schedule__delivered-and-received-record__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. Dose is an exact positive-finite relation among amount, patient carrier, formulation, route and schedule. It is not a fitted efficacy constant and does not import a universal response.

Dose, exposure, response, adverse event and toxicity are separately generated relations. Route, formulation, schedule, duration, organ or function, causality assessment and recovery remain held; one fitted curve cannot erase adverse or nonmonotone evidence.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 123970c5d572423775749d6be4cfb3cb70ef6c352099a832c4d57c59af31c023.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: e8a5ed5912b4106aa81834ea0a3cd11fa152eb84258e8d216c7cec2e3403b725.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: b141726c8acac096720289fa00b79b09fab35dad949a8ceb4daa33e0f74e72dc.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 65ce652ede81be1863e255c9ff5f869d6d917e680e03018a8ca5311e64bc8226.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256: a61f46830590ffd7f9fa2b22cd071d855b929dead7038b3a4e19a0cd893ea26a; certificate
sha256: 7d4122a958eecd891240654d08621ceca0705601ce7209ea1995aa1d35cd20; external-validation
sha256: 58a83e8cf924c366050d044e502f48c1f1c69bb5c7ba8a17d8477f44572c648e.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 3 purpose-matched discriminators with ordered identity
sha256: 38cf5e92df18017e07a01c865eeec419db25dd90dd56aaa6ce71b2d5130f17ac. Target opened after derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- MED-FDA-EXPOSURE-RESPONSE-001 - United States Food and Drug Administration; <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/exposure-response-relationships-study-design-data-analysis-and-regulatory-applications>; snapshot

experiments/external_sources/medicine/snapshots/MED-FDA-EXPOSURE-RESPONSE-001.html; sha256:748ca63404e58ef7f56d321395e646aa918f1ba0e679dbab98e62be95bc40291; scope: dose, exposure, response, benefit and risk study boundary.

- MED-ICH-E6R3-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf; snapshot experiments/external_sources/medicine/snapshots/MED-ICH-E6R3-001.pdf; sha256:e6ce19e36ce7d2e294f89ee89492b9e035178c3cca48984392bbd92eec9b002c; scope: final Good Clinical Practice: rights, safety, wellbeing, consent, protocol, allocation, blinding, adverse events, confidentiality and essential records.
- MED-FDA-BIOMARKER-001 - United States Food and Drug Administration;
<https://www.fda.gov/drugs/biomarker-qualification-program/about-biomarkers-and-qualification>; snapshot experiments/external_sources/medicine/snapshots/MED-FDA-BIOMARKER-001.html; sha256:93c5ee4b5e3480ed27aa805a7739f4acf7108701fc6f429ae5d4e5a1ad9d3d31; scope: normal and pathogenic processes; risk, diagnostic, monitoring, prognostic, response and safety biomarkers; method and context.

Comparison record:

- sft-med-dose-001-authority-correspondence: predicted medicine:administered-substance-or-action__amount-per-declared-carrier-and-time__unit-route-formulation-schedule__delivered-and-received-record; source-derived medicine:administered-substance-or-action__amount-per-declared-carrier-and-time__unit-route-formulation-schedule__delivered-and-received-record; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:1b43103284faf9ac5e6fef6e4fda462da6215a43162d5e0ee7d0a45ce44be44b; derivation seal sha256:eea6ea4e08e8555c8a88345f784f7a874b6812baf13c045de248d61c4eb0b11b; engine receipt sha256:ac1d83eb8670e428a755b801006e386ef850299d00310595b34b68fdbbac76a5 at receipts/engine/model_admitted/SFT-MED-DOSE-001-ac1d83eb8670e428.json; empirical validation sha256:566b7d1b5803a652567d8c91e014db5f09859213ae6adcd92a61099af4f03197; measurement receipt sha256:03521b2f66f3753bdf3976a48248249e333d4e1306b5da9ec2bd621f19f7ae18; isolation sha256:deceb7d27a8baa67e3b3bc06a6329cb65dce780f41ad029040a2575ff9d25d0f; custody sha256:7fd8ff02c4bcdd5cf61878a31896d9ccfdb82a22fe82a4282482c57f5d7a95fb.

38. Clinical exposure

Claim identity: SFT-MED-EXPOSURE-001

Question and exact theorem. Exposure retains source, path, route, intensity, duration, person and measurement or reconstruction method.

person-agent-contact-word__contact-intensity-duration-route__source-path-person-time__measurement-and-reconstruction-method__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001

- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-DOSE-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	person-agent-contact-word	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	contact-intensity-duration-route	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	source-path-person-time	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	measurement-and-reconstruction-method	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: person-agent-contact-word__contact-intensity-duration-route__source-path-person-time__measurement-and-reconstruction-method__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier: clinical carrier is explicit and nonempty; passed true.
- relation: relation and organization remain separately reconstructible; passed true.
- boundary: observation boundary cannot silently become the clinical law; passed true.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces exposure retains source, path, route, intensity, duration, person and measurement or reconstruction method. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Dose, exposure, response, adverse event and toxicity are separately generated relations. Route, formulation, schedule, duration, organ or function, causality assessment and recovery remain held; one fitted curve cannot erase adverse or nonmonotone evidence.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:14fa0eb73d78040a70bf2c90361bd8009421ef181ae6c78323f24aa11f61902.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:db03ca71d3d34d4322e392a795174cb7463695a01a76ce30550baf54156d4a10.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:cf4a88c86be597a294648ad2021586a95be359a9ea32d6bc6e8f0fec82fa901.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:0e5b4d49d6808ec0b748766bba0e86405b82266b636a4009a6b39011a6440e7f.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:313e83d8032a8781622da9591ba369e5de042ee70120840be39b89e30ab84e88; certificate
sha256:1e766e50bc4cf0755958c0e075a55cb04d4d1756d3b9bd3e860d1bac52893bd1; external-validation
sha256:133a554097398545e52db5f47c2e8131213713094d565ec01ceb542bc8501dec.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 3 purpose-matched discriminators with ordered identity
sha256:38cf5e92df18017e07a01c865eeec419db25dd90dd56aaa6ce71b2d5130f17ac. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- **MED-FDA-EXPOSURE-RESPONSE-001** - United States Food and Drug Administration; <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/exposure-response-relationships-study-design-data-analysis-and-regulatory-applications>; snapshot
experiments/external_sources/medicine/snapshots/MED-FDA-EXPOSURE-RESPONSE-001.html;
sha256:748ca63404e58ef7f56d321395e646aa918f1ba0e679dbab98e62be95bc40291; scope: dose, exposure, response, benefit and risk study boundary.
- **MED-ICH-E6R3-001** - International Council for Harmonisation;
https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E6R3-001.pdf;
sha256:e6ce19e36ce7d2e294f89ee89492b9e035178c3cca48984392bbd92eec9b002c; scope: final Good Clinical Practice: rights, safety, wellbeing, consent, protocol, allocation, blinding, adverse events, confidentiality and essential records.
- **MED-FDA-BIOMARKER-001** - United States Food and Drug Administration;
<https://www.fda.gov/drugs/biomarker-qualification-program/about-biomarkers-and-qualification>; snapshot
experiments/external_sources/medicine/snapshots/MED-FDA-BIOMARKER-001.html;
sha256:93c5ee4b5e3480ed27aa805a7739f4acf7108701fc6f429ae5d4e5a1ad9d3d31; scope: normal and pathogenic processes; risk, diagnostic, monitoring, prognostic, response and safety biomarkers; method and context.

Comparison record:

- **sft-med-exposure-001-authority-correspondence:** predicted medicine:person-agent-contact-word__contact-intensity-duration-route__source-path-person-time__measurement-and-reconstruction-method; source-derived medicine:person-agent-contact-word__contact-intensity-duration-route__source-path-person-time__measurement-and-reconstruction-method; exact match True

- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:8773446a8fb13c172fd4d4e161ac8bac35bd441ead73e6d6af765afb5d5400e2; derivation seal sha256:4cd13748a17d9ab345b5312aadd48440e42824498a5fdb33cc57071fe017e5da; engine receipt sha256:14e2723b299a0ff56e2e35cbeba96dad77a4e278e00a4cb9a7f813acc3cd5a7c at receipts/engine/model_admitted/SFT-MED-EXPOSURE-001-14e2723b299a0ff5.json; empirical validation sha256:da1aac3851b4d833e324641a5da07904d5c05431ea159beb93bc1fab7eb64e65; measurement receipt sha256:e2b87603755cc643259f43bc3f2a2679f5f55612d98919c26c4a4ee67e87752a; isolation sha256:9ed40d583af52fb07ce83593ac60d9b4a9a908610d197e543ab0cf9e5a4acfc1; custody sha256:c388e1751724d4b338862e6de4088defeb976c1839221a5acc5a97aea9549bb.

39. Clinical response

Claim identity: SFT-MED-RESPONSE-001

Question and exact theorem. A response is a recorded change from baseline after an exposure or intervention, with controls required before the temporal relation becomes causal.

patient-baseline-and-follow-up-state__change-after-exposure-or-intervention__multidomain-course-with-controls__outcome-method-and-time__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001

- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-EXPOSURE-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	patient-baseline-and-follow-up-state	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	change-after-exposure-or-intervention	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	multidomain-course-with-controls	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
observation	observation-boundary-unrecorded	outcome-method-and-time	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: patient-baseline-and-follow-up-state__change-after-exposure-or-intervention__multidomain-course-with-controls__outcome-method-and-time__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces a response is a recorded change from baseline after an exposure or intervention, with controls required before the temporal relation becomes causal. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Dose, exposure, response, adverse event and toxicity are separately generated relations. Route, formulation, schedule, duration, organ or function, causality assessment and recovery remain held; one fitted curve cannot erase adverse or nonmonotone evidence.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:6de18fb7c0d16ee5789c6f7e495326a54c8adcb854d7d97f1420800624f5d15f.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:6d9e4323cdf7a37ffd3033c94aae85a4e80239d0eb19749197b48ee2965c2e34.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:5985192047b866790fe4077f1d992340a4637e01ab8f0e72c637e19a1f68a98e.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 8c2ee1c76696a8d2dd0e8d38fe40394ac9785acbed37bd8488aebb59edd32b4a.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256 : e40e0f00f30ed6de80d6a53cdfd185a78d223008f1141ff7aab7642259a03942; certificate
sha256 : 74059547f59949c3258c1dcf9faebf50c0cf751392608f8466eb2b1cdd64c56d; external-validation
sha256 : ad13c193b0f60ad0db2b77f4605044b4c5281f977b0f6c2be7f5b95c583a78f2.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 3 purpose-matched discriminators with ordered identity
sha256 : 38cf5e92df18017e07a01c865eeec419db25dd90dd56aaa6ce71b2d5130f17ac. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- MED-FDA-EXPOSURE-RESPONSE-001 - United States Food and Drug Administration; <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/exposure-response-relationships-study-design-data-analysis-and-regulatory-applications>; snapshot
experiments/external_sources/medicine/snapshots/MED-FDA-EXPOSURE-RESPONSE-001.html;
sha256 : 748ca63404e58ef7f56d321395e646aa918f1ba0e679dbab98e62be95bc40291; scope: dose, exposure, response, benefit and risk study boundary.
- MED-ICH-E6R3-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E6R3-001.pdf;
sha256 : e6ce19e36ce7d2e294f89ee89492b9e035178c3cca48984392bbd92eec9b002c; scope: final Good Clinical Practice: rights, safety, wellbeing, consent, protocol, allocation, blinding, adverse events, confidentiality and essential records.
- MED-FDA-BIOMARKER-001 - United States Food and Drug Administration;
<https://www.fda.gov/drugs/biomarker-qualification-program/about-biomarkers-and-qualification>; snapshot
experiments/external_sources/medicine/snapshots/MED-FDA-BIOMARKER-001.html;
sha256 : 93c5ee4b5e3480ed27aa805a7739f4acf7108701fc6f429ae5d4e5a1ad9d3d31; scope: normal and pathogenic processes; risk, diagnostic, monitoring, prognostic, response and safety biomarkers; method and context.

Comparison record:

- sft-med-response-001-authority-correspondence: predicted medicine:patient-baseline-and-follow-up-state__change-after-exposure-or-intervention__multidomain-course-with-controls__outcome-method-and-time; source-derived medicine:patient-baseline-and-follow-up-state__change-after-exposure-or-intervention__multidomain-course-with-controls__outcome-method-and-time; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage

- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:d214345dddb22982c2d270af4e73e8d354944224a033205adb3d24b6bcce3e11; derivation seal sha256:514c25fee7d315249a985faa06efc2153d83890829e08a3a8be95285c8326aab; engine receipt sha256:3a051896a773bd06407d83654f7e55a5f64aca695299eace57d1e33edef69367 at receipts/engine/model_admitted/SFT-MED-RESPONSE-001-3a051896a773bd06.json; empirical validation sha256:5946b5471c6bccac74ac04971fbd174887d99570c24308d3f09bc88f09090b0f; measurement receipt sha256:fd37d7cbe426db451060bf5fc76f963eb5a5a847ccb8a54f64c3c1629f47ced9; isolation sha256:724d1c4538a1122186aaf21a5153fbb617dfff4364935e3c5838cd877ff84eab; custody sha256:fb63fa99802a2f8beae1e9b3c91a5ff2ff63ed60f50643e24778f6ed4e70036e.

40. Dose-response relation

Claim identity: SFT-MED-DOSE-RESPONSE-001

Question and exact theorem. A dose-response relation retains every administered dose group, outcome, route, interval and nonmonotone or adverse pattern; no fitted curve is the law.

ordered-dose-groups-and-outcomes__outcome-incidence-by-dose__complete-dose-suppo
rt-with-adverse-shapes__population-route-and-time__complete-protocol-state-trans
ition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor
-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001

- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-RESPONSE-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	ordered-dose-groups-and-outcomes	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	outcome-incidence-by-dose	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	complete-dose-support-with-adverse-shapes	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	population-route-and-time	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: ordered-dose-groups-and-outcomes__outcome-incidence-by-dose__complete-dose-support-with-adverse-shapes__population-route-and-time__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed true.
- **relation:** relation and organization remain separately reconstructible; passed true.
- **boundary:** observation boundary cannot silently become the clinical law; passed true.

Scientific meaning. Every administered group and outcome remains in the exact support, including nonmonotone and adverse shapes. A fitted exposure-response curve may summarize declared data but cannot select or replace the Fold law.

Dose, exposure, response, adverse event and toxicity are separately generated relations. Route, formulation, schedule, duration, organ or function, causality assessment and recovery remain held; one fitted curve cannot erase adverse or nonmonotone evidence.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : e9e3f5abc5c25a6efe7e0532e255fe1284f40d21246c7c5d0a1deb3475e98752.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 7868c55a0ada5b81ff8194be25efdf3fc786edc54b3d28ef159ca86e189154b0.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 87973af1e5d598f7ea5b96d35791492b9c72f1e2b34ac6b207f36ad26c7b3362.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : aef2b6b22b6e56722987d3cc2ef0ea5228c406794cd4ce9469b6c1c9c13728dc.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256 : 48b999fdb1dd8a1ee2849a9941cc98493456fac24e93df59cb91dc81e0244d27; certificate
sha256 : 9556967c7057e3c7064e69b4fd0c47185682c4ab9cf17b8f64c260961aa0b6f0; external-validation
sha256 : 42e2e9d775e212439c7fc818a2dad495c799d85284bdfb8c48fd996c9adeadb9.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 3 purpose-matched discriminators with ordered identity

sha256 : 38cf5e92df18017e07a01c865eeec419db25dd90dd56aaa6ce71b2d5130f17ac. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- **MED-FDA-EXPOSURE-RESPONSE-001** - United States Food and Drug Administration; <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/exposure-response-relationships-study-design-data-analysis-and-regulatory-applications>; snapshot
experiments/external_sources/medicine/snapshots/MED-FDA-EXPOSURE-RESPONSE-001.html; sha256 : 748ca63404e58ef7f56d321395e646aa918f1ba0e679dbab98e62be95bc40291; scope: dose, exposure, response, benefit and risk study boundary.
- **MED-ICH-E6R3-001** - International Council for Harmonisation; https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf; snapshot

experiments/external_sources/medicine/snapshots/MED-ICH-E6R3-001.pdf;
 sha256:e6ce19e36ce7d2e294f89ee89492b9e035178c3cca48984392bbd92eec9b002c; scope: final Good Clinical Practice: rights, safety, wellbeing, consent, protocol, allocation, blinding, adverse events, confidentiality and essential records.

- MED-FDA-BIOMARKER-001 - United States Food and Drug Administration;
<https://www.fda.gov/drugs/biomarker-qualification-program/about-biomarkers-and-qualification>; snapshot experiments/external_sources/medicine/snapshots/MED-FDA-BIOMARKER-001.html;
 sha256:93c5ee4b5e3480ed27aa805a7739f4acf7108701fc6f429ae5d4e5a1ad9d3d31; scope: normal and pathogenic processes; risk, diagnostic, monitoring, prognostic, response and safety biomarkers; method and context.

Comparison record:

- sft-med-dose-response-001-authority-correspondence: predicted medicine:ordered-dose-groups-and-outcomes__outcome-incidence-by-dose__complete-dose-support-with-adverse-shapes__population-route-and-time; source-derived medicine:ordered-dose-groups-and-outcomes__outcome-incidence-by-dose__complete-dose-support-with-adverse-shapes__population-route-and-time; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:9ace8b7f287bbfe10c48eb9facf905792cae9ceb7a3a64f7f6b1bd75cbc152cb; derivation seal sha256:37ee5f29e32b0a4e8251ad5c05757f3b66e7bf80a6394997efda57fd6859112f; engine receipt sha256:e36d17521cbda0f61edc1297b12eabaa07ccc6c7bd50fd95ac57f9d5fd4020cd at receipts/engine/model_admitted/SFT-MED-DOSE-RESPONSE-001-e36d17521cbda0f6.json; empirical validation sha256:cf24900e72890c8929741a324ee27a2b593d7531345bef4d341cb167ee1db364; measurement receipt sha256:42cbc4fa8d872989f9b9da403888a759aa7f778cb773a548335e5bafb555fe91; isolation sha256:a05768b9b06c05da4223bf365849735ee0cd6472dae8a96241bf96290356450b; custody sha256:bc981ca9259740bfc826f08a2d0411e9b9f12141269d6ad7bb9b2fc86a4f737c.

41. Adverse event

Claim identity: SFT-MED-ADVERSE-EVENT-001

Question and exact theorem. Every unfavorable event after care remains recorded with seriousness, severity, timing, relatedness assessment and final outcome, whether or not causally attributed.

participant-unfavorable-event__event-after-exposure-or-care__seriousness-severity-relatedness-outcome__solicitation-time-and-follow-up__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-DOSE-RESPONSE-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	participant-unfavorable-event	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	event-after-exposure-or-care	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	seriousness-severity-relatedness-outcome	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	solicitation-time-and-follow-up	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: participant-unfavorable-event__event-after-exposure-or-care__seriousness-severity-relatedness-outcome__solicitation-time-and-follow-up__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the clinical law; passed **true**.

Scientific meaning. Every unfavorable event after care is preserved regardless of attribution. Timing, seriousness, severity, relatedness assessment and final outcome are separate labels, preventing causally inconvenient harm from disappearing.

Dose, exposure, response, adverse event and toxicity are separately generated relations. Route, formulation, schedule, duration, organ or function, causality assessment and recovery remain held; one fitted curve cannot erase adverse or nonmonotone evidence.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 89767ea513a595fa1c2ff18d56325b4b460420539190400bc007b88822095dc5.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 9cf1e5b80fde05432d4203e792d98d65892707dc4cc4d59084f436c66235e50a.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 4c8ce97fadf0e798a974dc11af8e3fc744f9028906304df92b08bea09542d63b.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: d4e0739ac5333f65d35df5eaa2f2dc61e24bc349f9e6bca471d68bbdcc2a1525.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256: 36c35b1f2704b1652d6a3745806d5c07767127f770388b550b6b3cc3ca4d3131; certificate
sha256: 10fc6c12286f03a034e9e0dc2c85add5e9ee3bfec21dc97b2752f29b1e95b3b4; external-validation
sha256: d251f7c6ec18fa557fe51a70bd9c34ca1752a04224cf638ad34c1a60bd3505cc.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 3 purpose-matched discriminators with ordered identity
sha256: 38cf5e92df18017e07a01c865eeec419db25dd90dd56aaa6ce71b2d5130f17ac. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- MED-FDA-EXPOSURE-RESPONSE-001 - United States Food and Drug Administration; <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/exposure-response-relationships-study-design-data-analysis-and-regulatory-applications>; snapshot
experiments/external_sources/medicine/snapshots/MED-FDA-EXPOSURE-RESPONSE-001.html;
sha256: 748ca63404e58ef7f56d321395e646aa918f1ba0e679dbab98e62be95bc40291; scope: dose, exposure, response, benefit and risk study boundary.
- MED-ICH-E6R3-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E6R3-001.pdf;
sha256: e6ce19e36ce7d2e294f89ee89492b9e035178c3cca48984392bbd92eec9b002c; scope: final Good Clinical Practice: rights, safety, wellbeing, consent, protocol, allocation, blinding, adverse events, confidentiality and essential records.
- MED-FDA-BIOMARKER-001 - United States Food and Drug Administration;
<https://www.fda.gov/drugs/biomarker-qualification-program/about-biomarkers-and-qualification>; snapshot
experiments/external_sources/medicine/snapshots/MED-FDA-BIOMARKER-001.html;
sha256: 93c5ee4b5e3480ed27aa805a7739f4acf7108701fc6f429ae5d4e5a1ad9d3d31; scope: normal and pathogenic processes; risk, diagnostic, monitoring, prognostic, response and safety biomarkers; method and context.

Comparison record:

- sft-med-adverse-event-001-authority-correspondence: predicted medicine:participant-unfavorable-event__event-after-exposure-or-care__seriousness-severity-relatedness-outcome__solicitation-time-and-follow-up; source-derived medicine:participant-unfavorable-event__event-after-exposure-or-care__seriousness-severity-relatedness-outcome__solicitation-time-and-follow-up; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:84093f38e3e398d7ad8dcb90cbb75b50a7fe43b2b474207d1f55bce36ea39bb7; derivation seal sha256:cbefac0eb22927142c96a354ee5153f7c4925d629261b1d773483b12a2baf9de; engine receipt sha256:78905c3c202917110d452a8e1f3e36297d06a9ca41227d0bf6daf96f71bc184a at receipts/engine/model_admitted/SFT-MED-ADVERSE-EVENT-001-78905c3c20291711.json; empirical validation sha256:8d6fa5684b78a47ea9a8a16e2191c9dc434d64e09450c90ea6b06012fa4bb986; measurement receipt sha256:f8d7e4ac001071ca75ee23a91bd467e120650202fddd24878e2ed89379c1b4a2; isolation sha256:72af5c9676d98e638fe289c7222aaab79bc4cc0f370ffe80a7fd7183c63d569f; custody sha256:8171916e882e34fbd62180cefde96b3fff99cad19b3f2e5b8bb82a9b6850f7fa.

42. Clinical toxicity

Claim identity: SFT-MED-TOXICITY-001

Question and exact theorem. Toxicity requires an exposure-linked harmful transition with dose, organ or function, timing, recovery and alternative causes retained.

exposure-and-harm-state__exposure-linked-dysfunction-or-injury__organ-course-dose-and-recovery__method-population-and-time__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001

- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-ADVERSE-EVENT-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	exposure-and-harm-state	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	exposure-linked-dysfunction-or-injury	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	organ-course-dose-and-recovery	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	method-population-and-time	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: exposure-and-harm-state__exposure-linked-dysfunction-or-injury__organ-course-dose-and-recovery__method-population-and-time__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces toxicity requires an exposure-linked harmful transition with dose, organ or function, timing, recovery and alternative causes retained. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Dose, exposure, response, adverse event and toxicity are separately generated relations. Route, formulation, schedule, duration, organ or function, causality assessment and recovery remain held; one fitted curve cannot erase adverse or nonmonotone evidence.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:1525cb8967bec58924b018dfd7ac911782b47ebfa3acbc235dd016b2cb72545.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:b4ed398810088de83f00d246480aae1894c0aa7fcda4de4a4017dcbe22b82503.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:78878fcb7fcee6817ce36508e87c3ecbbe512476cfa0c37ccc3177493af3b382.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:188f0a626dbd653f8183fdbbf6ad579b8339913fe8573a2773fecfba41238667.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation sha256: 84ac28d9444c4832a6edf915ec1f3eba90c05cacd9d967f70b456378a8447829; certificate sha256: 3ff2b2b12d3572c87ff422167383a301388063265b9558716e3c7bc0d13fa8f9; external-validation sha256: b00d21a6f874ee50e81af84bf59e0fd6d5bad681bb400d318ebe33443c66557d.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 3 purpose-matched discriminators with ordered identity sha256: 38cf5e92df18017e07a01c865eeec419db25dd90dd56aaa6ce71b2d5130f17ac. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- MED-FDA-EXPOSURE-RESPONSE-001 - United States Food and Drug Administration; <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/exposure-response-relationships-study-design-data-analysis-and-regulatory-applications>; snapshot experiments/external_sources/medicine/snapshots/MED-FDA-EXPOSURE-RESPONSE-001.html; sha256: 748ca63404e58ef7f56d321395e646aa918f1ba0e679dbab98e62be95bc40291; scope: dose, exposure, response, benefit and risk study boundary.
- MED-ICH-E6R3-001 - International Council for Harmonisation; https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf; snapshot experiments/external_sources/medicine/snapshots/MED-ICH-E6R3-001.pdf; sha256: e6ce19e36ce7d2e294f89ee89492b9e035178c3cca48984392bbd92eec9b002c; scope: final Good Clinical Practice: rights, safety, wellbeing, consent, protocol, allocation, blinding, adverse events, confidentiality and essential records.
- MED-FDA-BIOMARKER-001 - United States Food and Drug Administration; <https://www.fda.gov/drugs/biomarker-qualification-program/about-biomarkers-and-qualification>; snapshot experiments/external_sources/medicine/snapshots/MED-FDA-BIOMARKER-001.html; sha256: 93c5ee4b5e3480ed27aa805a7739f4acf7108701fc6f429ae5d4e5a1ad9d3d31; scope: normal and pathogenic processes; risk, diagnostic, monitoring, prognostic, response and safety biomarkers; method and context.

Comparison record:

- sft-med-toxicity-001-authority-correspondence: predicted medicine:exposure-and-harm-state__exposure-linked-dysfunction-or-injury__organ-course-dose-and-recovery__method-population-and-time; source-derived medicine:exposure-and-harm-state__exposure-linked-dysfunction-or-injury__organ-course-dose-and-recovery__method-population-and-time; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result

- prior answer as premise

Immutable evidence identities. Pre-source seal
experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest
sha256:b736f260a6b2b0709a85dc853ca64f1e60040dce9427aaecebd0466beb8ea783; derivation seal
sha256:1ae3be2f31029639b662a517005bf50c0bcf5e4225a856ffc34ca86622b4c0c6; engine receipt
sha256:829bb46bb023e38579070b20897606b964d882dbf05322a2c70cd09c983644df at
receipts/engine/model_admitted/SFT-MED-TOXICITY-001-829bb46bb023e385.json; empirical validation
sha256:4f3448096748d18a608fb520d100937b0daec10a83dee6d6a1e1aba633bcde94; measurement receipt
sha256:fd77dfb202c5d2dc02ed1ea52c8029e5a3269cbdf42e77bfaf3431e4b75fbbe0; isolation
sha256:7fcddb6bd03c8d53f82f2cbadd96f826dd5ee2e5201fe75e927ea03d77679426; custody
sha256:43854e03353485fed434dd6764741a1e1e09bec22c783428ad577406b51b3f3.

17. Efficacy Effectiveness Safety Benefit Harm

Efficacy under explanatory conditions and effectiveness in usual care are not synonyms. Safety is a complete adverse-outcome record, not proof of harmlessness, and benefit-harm composition cannot erase patient-valued components.

43. Efficacy

Claim identity: SFT-MED-EFFICACY-001

Question and exact theorem. Efficacy is a complete intervention-comparator outcome contrast under declared explanatory trial conditions.

controlled-intervention-comparison__outcome-contrast-under-explanatory-condition
s__assigned-arms-and-complete-outcomes__protocol-population-and-horizon__complet
e-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__pos
itive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001

- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-TOXICITY-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	controlled-intervention-comparison	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	outcome-contrast-under-explanatory-conditions	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	assigned-arms-and-complete-outcomes	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	protocol-population-and-horizon	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
extension	free-fit-exception -opaque-model-or-e xtra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: controlled-intervention-comparison__outcome-contrast-under-explanatory-conditions__assigned-arms-and-complete-outcomes__protocol-population-and-horizon__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.
- **complete-arm-outcome-table:** two arms crossed with two retained outcome labels generate four cells exactly once; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces efficacy is a complete intervention-comparator outcome contrast under declared explanatory trial conditions. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Efficacy under explanatory conditions and effectiveness in usual care are not synonyms. Safety is a complete adverse-outcome record, not proof of harmlessness, and benefit-harm composition cannot erase patient-valued components.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 2833574db310b6439bf3af74d07c08123e581da883c1b9897e031cbd434e328.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 8b2a8f778eb68fcdac9f8b36e75cf82649afea4771ccced7b89c5bb9b935fbc3.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 662d28a584272e966e9bb043be5592015592d47bfe6d5b0cba2c221b9b58eb2c.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: f80d2937f1a4f96ca1137822e128a353feea210467d503b41205af1141a19270.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256: f534cd37f43c983cb684df1dd338ebb602f5cc269d215ff898c98a40c48d1584; certificate
sha256: bdf723c6a3d04981a36377e443b0eba7fdcf4cb9cd6a6090d9ad704f2b44dca0; external-validation
sha256: 73f7a1a0472cfadd826bc82bf333d90cc06879ec71460951c65c6bd1b1353323.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 3 purpose-matched discriminators with ordered identity
sha256: 51958244b37fdb35376015b8a6f0f11eac756f37131352f5efbac249c2695729. Target opened after derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- MED-ICH-E6R3-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf; snapshot experiments/external_sources/medicine/snapshots/MED-ICH-E6R3-001.pdf; sha256:e6ce19e36ce7d2e294f89ee89492b9e035178c3cca48984392bbd92eec9b002c; scope: final Good Clinical Practice: rights, safety, wellbeing, consent, protocol, allocation, blinding, adverse events, confidentiality and essential records.
- MED-ICH-E9R1-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/E9-R1_Step4_Guideline_2019_1203.pdf; snapshot experiments/external_sources/medicine/snapshots/MED-ICH-E9R1-001.pdf; sha256:f7471f411f1c87ee76783d5b2b9faeeca31d01d530213f71b50d136b39e3b0d9; scope: final estimand, population, endpoint, intercurrent-event, missing-data and sensitivity-analysis distinctions.
- MED-COCHRANE-HANDBOOK-001 - Cochrane; <https://www.cochrane.org/authors/handbooks-and-manuals/handbook/current>; snapshot experiments/external_sources/medicine/snapshots/MED-COCHRANE-HANDBOOK-001.html; sha256:31f56118fc994e14f9703298b798983c2039c6328ee4c5f983f53bdef2bc0004; scope: current intervention-review scope, study selection, bias, adverse effects and interpretation.

Comparison record:

- sft-med-efficacy-001-authority-correspondence: predicted medicine:controlled-intervention-comparison__outcome-contrast-under-explanatory-conditions__assigned-arms-and-complete-outcomes__protocol-population-and-horizon; source-derived medicine: controlled-intervention-comparison__outcome-contrast-under-explanatory-conditions__assigned-arms-and-complete-outcomes__protocol-population-and-horizon; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:2172bd8cf97b1520e941e680e4e42e8dc6de22cafc0cfe7c9f1b02bcd926acc0; derivation seal sha256:56c5e7930429bc1d767cb90320a2787770bf1a226020078d257d1448af27817; engine receipt sha256:fad071eef2c92861ab7b6ef138adaa62012830e8c3708a35eb630d0946a4093c at receipts/engine/model_admitted/SFT-MED-EFFICACY-001-fad071eef2c92861.json; empirical validation sha256:2bfd18c30cd3e3c1670ec755fa4b87a0b8cbffcf284dc5e723f89519c81aeb25; measurement receipt sha256:3a39385da13fc54690da5109cc01054f8dd4bc8ce8475e1b1b21436d11477bfb0; isolation sha256:71cfaaa38776dcb1f7148198f6f2f518bba3c640b072d997d9d732570efe3f20; custody

sha256:323d6e9fd49e9a92e6a66c9ac87193c1a65b8e8aa0a1c5afc0a5cbaf922581f9.

44. Effectiveness

Claim identity: SFT-MED-EFFECTIVENESS-001

Question and exact theorem. Effectiveness is the corresponding contrast under retained real-care delivery, adherence, population and setting variation.

practice-intervention-comparison__outcome-contrast-under-usual-care__delivery-adherence-context-variation__practice-population-and-horizon__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001

- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-EFFICACY-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	practice-intervention-comparison	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	outcome-contrast-under-usual-care	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	delivery-adherence-context-variation	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	practice-population-and-horizon	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: practice-intervention-comparison__outcome-contrast-under-usual-care__delivery-adherence-context-variation__practice-population-and-horizon__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier: clinical carrier is explicit and nonempty; passed true.
- relation: relation and organization remain separately reconstructible; passed true.

- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces effectiveness is the corresponding contrast under retained real-care delivery, adherence, population and setting variation. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Efficacy under explanatory conditions and effectiveness in usual care are not synonyms. Safety is a complete adverse-outcome record, not proof of harmlessness, and benefit-harm composition cannot erase patient-valued components.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : d55310d24dceade98a62dc356a94be0979b4e7d273f40a879f1a5ee3b0679eda.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 93bb2a1a2cec9b5c18cc47fd6d975be2a03903741aacab8d554ec37f72fad038.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 852a9d09cd72766b09b59e725a05f7de33497e4a3e718e97151444cd8dfa414b.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : f237c080b5916a601e80687506307dedc2b321ae768f6c93a902794815ed8f0a.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256 : b8c8ec2ac0ea4df4f97387ced5433b1931e0b063143b11b40830898bf24c98ec; certificate
sha256 : 8647f6be9e63aec5aedcb91aa58898f44102ac7903d51a8054953718df39681b; external-validation
sha256 : f84f8ff7a5a4000ed3df3c523d11555d7fe5092f9eac5df6e8784939273cab48.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 3 purpose-matched discriminators with ordered identity
sha256 : 51958244b37fdb35376015b8a6f0f11eac756f37131352f5efbac249c2695729. Target opened after derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- MED-ICH-E6R3-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E6R3-001.pdf;
sha256 : e6ce19e36ce7d2e294f89ee89492b9e035178c3cca48984392bbd92eec9b002c; scope: final Good Clinical Practice: rights, safety, wellbeing, consent, protocol, allocation, blinding, adverse events, confidentiality and essential records.
- MED-ICH-E9R1-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/E9-R1_Step4_Guideline_2019_1203.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E9R1-001.pdf;
sha256 : f7471f411f1c87ee76783d5b2b9faeca31d01d530213f71b50d136b39e3b0d9; scope: final estimand, population, endpoint, intercurrent-event, missing-data and sensitivity-analysis distinctions.
- MED-COCHRANE-HANDBOOK-001 - Cochrane; <https://www.cochrane.org/authors/handbooks-and-manuals/handbook/current>;
snapshot experiments/external_sources/medicine/snapshots/MED-COCHRANE-HANDBOOK-001.html;
sha256 : 31f56118fc994e14f9703298b798983c2039c6328ee4c5f983f53bdef2bc0004; scope: current intervention-review scope, study selection, bias, adverse effects and interpretation.

Comparison record:

- sft-med-effectiveness-001-authority-correspondence: predicted medicine:practice-intervention-comparison__outcome-contrast-under-usual-care__delivery-adherence-context-variation__practice-population-and-horizon; source-derived medicine:practice-intervention-comparison__outcome-contrast-under-usual-care__delivery-adherence-context-variation__practice-population-and-horizon

zon; exact match True

- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:e9bf75469a37f07de9d94b62f33c22e1da1c25a48dcfa9cbaddc8a771292b584; derivation seal sha256:e5c018c4c069e46a2e269ff9e3c2ef1ab198c69f329ae58b1d42c9f64d19ec97; engine receipt sha256:ec467dd4d013aa9d8322d5c1f82886b7de39ac767f26d6201a8278129ef983ce at receipts/engine/model_admitted/SFT-MED-EFFECTIVENESS-001-ec467dd4d013aa9d.json; empirical validation sha256:305b9962673c0f78705444ee0da390e4eaa5eb723b144f5f5b16b83792d44b66; measurement receipt sha256:1ee82f78209739fe35e93ae8e36195188c6d1a20d4725716661fb613d6e1a871; isolation sha256:354da003b1a9213c5b4c9495a25e8b5905b3557947fa62cb449ed03bda4fb463; custody sha256:eea76092424e4ef2f22abde92a5f322b25d41c049266fbe4322e6741a51d6981.

45. Safety evidence

Claim identity: SFT-MED-SAFETY-001

Question and exact theorem. Safety is a complete comparative adverse-outcome record at a declared exposure and follow-up boundary, never proof that harm is absent.

all-exposed-and-comparator-participants__favorable-and-unfavorable-event-incidence__complete-event-severity-time-support__exposure-duration-and-follow-up__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001

- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-EFFECTIVENESS-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	all-exposed-and-comparator-participants	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	favorable-and-unfavorable-event-incidence	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	complete-event-severity-time-support	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
observation	observation-boundary-unrecorded	exposure-duration-and-follow-up	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: `all-exposed-and-comparator-participants__favorable-and-unfavorable-event-incidence__complete-event-severity-time-support__exposure-duration-and-follow-up__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule`.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. Safety means a complete comparative adverse-outcome record at a declared exposure and follow-up boundary. It never means an intervention is proven incapable of harm.

Efficacy under explanatory conditions and effectiveness in usual care are not synonyms. Safety is a complete adverse-outcome record, not proof of harmlessness, and benefit-harm composition cannot erase patient-valued components.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:b02c815ffd045530fcc613f102ac2d3b521a21863c4a115ca6f2d4470bcb0d6e.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:9398233b37a4f0d8696f78885577dbaf4aac4f7f094b1c61821f7e10cff820e0.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:4b27f1795cd8ad66bdd66edfcb92685e959686a4fc4876b5d63b80e337123e35.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:3b367aaf91e605e24fd21f291aee145163c33980599189b900d325829b1fdd92.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
 sha256:7bb1abf150b7d73cdc2b8d7a8d22f390cd96621fc25c2f190ba3a4b60bb91e41; certificate
 sha256:b84ac74ede02d89de3dee731364c1947ec36e06cf8929b9df59e7ec8975459ea; external-validation
 sha256:02e2ec0231435c6966feaef185de5d0ee4aa4fdd0178f6989990b69bb7ec6f6e.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 3 purpose-matched discriminators with ordered identity
 sha256:51958244b37fdb35376015b8a6f0f11eac756f37131352f5efbac249c2695729. Target opened after
 derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- MED-ICH-E6R3-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf; snapshot
 experiments/external_sources/medicine/snapshots/MED-ICH-E6R3-001.pdf;
 sha256:e6ce19e36ce7d2e294f89ee89492b9e035178c3cca48984392bbd92eec9b002c; scope: final Good
 Clinical Practice: rights, safety, wellbeing, consent, protocol, allocation, blinding, adverse events, confidentiality and essential
 records.
- MED-ICH-E9R1-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/E9-R1_Step4_Guideline_2019_1203.pdf; snapshot
 experiments/external_sources/medicine/snapshots/MED-ICH-E9R1-001.pdf;
 sha256:f7471f411f1c87ee76783d5b2b9faaeca31d01d530213f71b50d136b39e3b0d9; scope: final
 estimand, population, endpoint, intercurrent-event, missing-data and sensitivity-analysis distinctions.
- MED-COCHRANE-HANDBOOK-001 - Cochrane; <https://www.cochrane.org/authors/handbooks-and-manuals/handbook/current>;
 snapshot experiments/external_sources/medicine/snapshots/MED-COCHRANE-HANDBOOK-001.html;
 sha256:31f56118fc994e14f9703298b798983c2039c6328ee4c5f983f53bdef2bc0004; scope: current
 intervention-review scope, study selection, bias, adverse effects and interpretation.

Comparison record:

- sft-med-safety-001-authority-correspondence: predicted medicine:all-exposed-and-comparator-participants__favorable-and-unfa
 vorable-event-incidence__complete-event-severity-time-support__exposure-duration-and-follow-up; source-derived medicine:al
 l-exposed-and-comparator-participants__favorable-and-unfavorable-event-incidence__complete-event-severity-time-support__e
 xposure-duration-and-follow-up; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated
 form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is
 erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result

- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:c24f4f858bf14e7b6e07d3dcca232dab9dff34a79f6f4aa7a0a8f0f9d45661fc; derivation seal sha256:bdd302cb8ba9dbf2ca8f939bda27ce378125329087ff7ffcc508dd064b529be9; engine receipt sha256:d11984f1d2f2cf293239f79421b260618766fdee59141d743c3ba784a8b3611a at receipts/engine/model_admitted/SFT-MED-SAFETY-001-d11984f1d2f2cf29.json; empirical validation sha256:692beb947bb2721103ec1dded75774f7eebf14cdcd23f9af1d2b1fa46f78bc90; measurement receipt sha256:64d83c4b40a4c009c4322ae4b8497ff6169c424c1d0af4bce85f00e4903cc23e; isolation sha256:a1d73980f1278565ed5d58a86c399d359eb79966a30b65265d78c61f81b6463a; custody sha256:7e04ed3177f1f1aecf1b5b6e7b20b2ecfb2e03ad381b4e53593c018e219af397.

46. Benefit-harm composition

Claim identity: SFT-MED-BENEFIT-HARM-001

Question and exact theorem. Benefit and harm remain separate outcome components and may compose only under an explicit population, horizon and value boundary.

joint-benefit-and-harm-outcomes__noncollapsed-outcome-comparison__patient-valued-components-held__population-horizon-and-preference-boundary__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001

- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-SAFETY-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	joint-benefit-and-harm-outcomes	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	noncollapsed-outcome-comparison	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	patient-valued-components-held	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	population-horizon-and-preference-boundary	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: joint-benefit-and-harm-outcomes__noncollapsed-outcome-comparison__patient-valued-components-held__population-horizon-and-preference-boundary__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces benefit and harm remain separate outcome components and may compose only under an explicit population, horizon and value boundary. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Efficacy under explanatory conditions and effectiveness in usual care are not synonyms. Safety is a complete adverse-outcome record, not proof of harmlessness, and benefit-harm composition cannot erase patient-valued components.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 8780b3bfec6a77c7ab4f4a658c4055e1690efb43fc3d9b51d15456ef8d2fe6c1.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 1bf0c27050318c77291ac2d3a6fe520e01865c6c2ebcb5e2c08c9850c425a2e3.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: c389220a1facd25bee24e1c7913d5897ab8bb7c488d387134acfdafabf3e12e94.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 47846a5b899e796d7524d5a5f5051e0a21b95ba6b785eaac73f170ef989c85c1.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256: 16299df5a8975089a2f9e9e3c274c4f36581a98dd5f2771e9bc95e1b40b5b58e; certificate
sha256: 67099f97d444da03f035927d3210f57e813866f6895a782ed84e02bc8640b03c; external-validation
sha256: 3f20f40534362de794b28c91c7d04a57841885fb3908d15f583e1f76d97c0ffff.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 3 purpose-matched discriminators with ordered identity
sha256: 51958244b37fdb35376015b8a6f0f11eac756f37131352f5efbac249c2695729. Target opened after derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- MED-ICH-E6R3-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E6R3-001.pdf;
sha256: e6ce19e36ce7d2e294f89ee89492b9e035178c3cca48984392bbd92eec9b002c; scope: final Good Clinical Practice: rights, safety, wellbeing, consent, protocol, allocation, blinding, adverse events, confidentiality and essential records.
- MED-ICH-E9R1-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/E9-R1_Step4_Guideline_2019_1203.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E9R1-001.pdf;
sha256: f7471f411f1c87ee76783d5b2b9faeca31d01d530213f71b50d136b39e3b0d9; scope: final estimand, population, endpoint, intercurrent-event, missing-data and sensitivity-analysis distinctions.

- MED-COCHRANE-HANDBOOK-001 - Cochrane; https://www.cochrane.org/authors/handbooks-and-manuals/handbook/current;snapshot/experiments/external_sources/medicine/snapshots/MED-COCHRANE-HANDBOOK-001.html; sha256:31f56118fc994e14f9703298b798983c2039c6328ee4c5f983f53bdef2bc0004; scope: current intervention-review scope, study selection, bias, adverse effects and interpretation.

Comparison record:

- sft-med-benefit-harm-001-authority-correspondence: predicted medicine:joint-benefit-and-harm-outcomes__noncollapsed-outcome-comparison__patient-valued-components-held__population-horizon-and-preference-boundary; source-derived medicine:joint-benefit-and-harm-outcomes__noncollapsed-outcome-comparison__patient-valued-components-held__population-horizon-and-preference-boundary; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:22d6aa8e004126d878f2105ca327f78a4cef06177b38f4c58ac3c81a19a562f0; derivation seal sha256:3f0647cd585ce9db3069ed6084ef5af808a125cd62799450d7ae7180212c0fa0; engine receipt sha256:fdcc94f8f9d80f1cb1a1a62fdf23f9971938b470e0986f3c7c8d135223c65f1b at receipts/engine/model_admitted/SFT-MED-BENEFIT-HARM-001-fdcc94f8f9d80f1c.json; empirical validation sha256:56309477d30239f69c5613de3f892e7740a15d2fca2fbc31ba2b40304afe8200; measurement receipt sha256:5244c440861f8d12ea51f328f2578ff47bdbb9ec617b385070f211104c415d7c; isolation sha256:e6580e8e9486ef45a98a5f25b1a7fcf57a7ab0bc5e27a69cde112fd768e93ebf; custody sha256:9c622019b81af5f0beb22dcd0580a6cf252c04111b76bf08ae371a3015417729.

47. Absolute and relative effects

Claim identity: SFT-MED-ABSOLUTE-RELATIVE-EFFECT-001

Question and exact theorem. Absolute and relative effects are distinct exact readings of the same complete arm-by-outcome table; neither may replace the underlying counts.

arm-specific-outcome-counts__absolute-and-relative-comparison__complete-two-arm-event-table__population-and-horizon__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-BENEFIT-HARM-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	arm-specific-outcome-counts	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	absolute-and-relative-comparison	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	complete-two-arm-event-table	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	population-and-horizon	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: arm-specific-outcome-counts__absolute-and-relative-comparison__complete-two-arm-event-table__population-and-horizon__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the clinical law; passed **true**.
- **complete-arm-outcome-table:** two arms crossed with two retained outcome labels generate four cells exactly once; passed **true**.

Scientific meaning. The same complete four-cell arm-by-outcome table yields absolute and relative comparisons. Neither representation is privileged and neither may hide the arm counts, baseline incidence or horizon.

Efficacy under explanatory conditions and effectiveness in usual care are not synonyms. Safety is a complete adverse-outcome record, not proof of harmlessness, and benefit-harm composition cannot erase patient-valued components.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:1d68569633deada46ed1a5baed93a9e956f2ca11c94f17a7d0cfbbc19163d9db.

- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:0512b43027df568675287c01bffaa94535b3e6d9bfb43ade73d8fae42f2ebbf4.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:d6dccfffb743c65cbe0c9c99cb0f6d2686db86e1494d23b5ac0962729f24e0fe3.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:21148b35ed825d7b75daedb1da9f3462ac95b6a70e31328a5ce4f4c2506b4798.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:dd181805865ff70aad676fc3430f785b5d4db1056ee42af04570cdea392c9266; certificate

sha256:9c6dc389f416a80f0c48fa4bf98f865044ae52e23d78d033568f5c7b0d4dda62; external-validation

sha256:6ad992a3531445c89027def6d8f6310c2b275aeb70388a4987d303693dd75f11.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 3 purpose-matched discriminators with ordered identity

sha256:51958244b37fdb35376015b8a6f0f11eac756f37131352f5efbac249c2695729. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- MED-ICH-E6R3-001 - International Council for Harmonisation; https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf; snapshot experiments/external_sources/medicine/snapshots/MED-ICH-E6R3-001.pdf; sha256:e6ce19e36ce7d2e294f89ee89492b9e035178c3cca48984392bbd92eec9b002c; scope: final Good Clinical Practice: rights, safety, wellbeing, consent, protocol, allocation, blinding, adverse events, confidentiality and essential records.
- MED-ICH-E9R1-001 - International Council for Harmonisation; https://database.ich.org/sites/default/files/E9-R1_Step4_Guideline_2019_1203.pdf; snapshot experiments/external_sources/medicine/snapshots/MED-ICH-E9R1-001.pdf; sha256:f7471f411f1c87ee76783d5b2b9faeaca31d01d530213f71b50d136b39e3b0d9; scope: final estimand, population, endpoint, intercurrent-event, missing-data and sensitivity-analysis distinctions.
- MED-COCHRANE-HANDBOOK-001 - Cochrane; <https://www.cochrane.org/authors/handbooks-and-manuals/handbook/current>; snapshot experiments/external_sources/medicine/snapshots/MED-COCHRANE-HANDBOOK-001.html; sha256:31f56118fc994e14f9703298b798983c2039c6328ee4c5f983f53bdef2bc0004; scope: current intervention-review scope, study selection, bias, adverse effects and interpretation.

Comparison record:

- sft-med-absolute-relative-effect-001-authority-correspondence: predicted medicine:arm-specific-outcome-counts__absolute-and-relative-comparison__complete-two-arm-event-table__population-and-horizon; source-derived medicine:arm-specific-outcome-counts__absolute-and-relative-comparison__complete-two-arm-event-table__population-and-horizon; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity

- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:74551efdd7346884ee715417584dd034140bd308062b485e8b81a97f6299136a; derivation seal sha256:36755eaba7365fa63394287b761b5da82fb40623408e8519b5e09e67780f9419; engine receipt sha256:9935031e091e83e4d59f404b161de2f6e6f2eae61b99b5ef03bbe1b3f444151c at receipts/engine/model_admitted/SFT-MED-ABSOLUTE-RELATIVE-EFFECT-001-9935031e091e83e4.json; empirical validation sha256:42de9d3a64e660edc443ecdb952821bdec0a63d1b7891be546d21887d50c32cb; measurement receipt sha256:1d7feaa415980d2e5c741391492b133694d49a43a75e246277818fed41129ed2; isolation sha256:068748c4265770ae79637bb4d6a94b2232d8b0323b09605ef09fa9b423c432e7; custody sha256:3f7894056a7cc990f9e184e9fe9de98616b88b7d5c035ca8b6f3e378b2087908.

48. Treatment-effect heterogeneity

Claim identity: SFT-MED-TREATMENT-HETEROGENEITY-001

Question and exact theorem. Treatment heterogeneity requires the complete joint stratum, arm and outcome support with predeclared comparisons and multiplicity retained.

predeclared-patient-strata__arm-contrast-by-stratum__joint-stratum-arm-outcome__multiplicity-and-scale-boundary__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001

- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-ABSOLUTE-RELATIVE-EFFECT-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	predeclared-patient-strata	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	arm-contrast-by-stratum	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	joint-stratum-arm-outcome	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	multiplicity-and-scale-boundary	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
extension	free-fit-exception -opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: predeclared-patient-strata__arm-contrast-by-stratum__joint-stratum-arm-outcome__multiplicity-and-scale-boundary__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces treatment heterogeneity requires the complete joint stratum, arm and outcome support with predeclared comparisons and multiplicity retained. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Efficacy under explanatory conditions and effectiveness in usual care are not synonyms. Safety is a complete adverse-outcome record, not proof of harmlessness, and benefit-harm composition cannot erase patient-valued components.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:137ec7034fa3281f72ee7b9ac70e416230a83108744142bea5101e0b697f29c3.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:10fa2a80e44f9cae5e7bb4a60a73e75ed4baeba0367e438f2d9f92dc13cdf5a2.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:2e4751accd1e194b829c7f6d28f2dd4b11d4ca3b21e75e2202a345f2eb270fa6.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:09206a856c76c8261abdd55f7e5f2de49cc46abdcbb6153e77ddcf29e11aa355.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:a1b7506b60fe1ae2bba67a58c164b919547b5014e0ba9a0984970d11b9bf8bbe; certificate
sha256:674a6a4a42e9d4ba74d46bb4db1ea8e1537831de089eb9ee01ffcdce5b12dd75; external-validation
sha256:97a74d07bdbde208cf32de20269d4aefe149352ea89254afc3256add7308281f.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 3 purpose-matched discriminators with ordered identity
sha256:51958244b37fdb35376015b8a6f0f11eac756f37131352f5efbac249c2695729. Target opened after derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- MED-ICH-E6R3-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E6R3-001.pdf;
sha256:e6ce19e36ce7d2e294f89ee89492b9e035178c3cca48984392bbd92eec9b002c; scope: final Good
Clinical Practice: rights, safety, wellbeing, consent, protocol, allocation, blinding, adverse events, confidentiality and essential
records.
- MED-ICH-E9R1-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/E9-R1_Step4_Guideline_2019_1203.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E9R1-001.pdf;
sha256:f7471f411f1c87ee76783d5b2b9faeeca31d01d530213f71b50d136b39e3b0d9; scope: final
estimand, population, endpoint, intercurrent-event, missing-data and sensitivity-analysis distinctions.
- MED-COCHRANE-HANDBOOK-001 - Cochrane; <https://www.cochrane.org/authors/handbooks-and-manuals/handbook/current>;
snapshot experiments/external_sources/medicine/snapshots/MED-COCHRANE-HANDBOOK-001.html;
sha256:31f56118fc994e14f9703298b798983c2039c6328ee4c5f983f53bdef2bc0004; scope: current
intervention-review scope, study selection, bias, adverse effects and interpretation.

Comparison record:

- sft-med-treatment-heterogeneity-001-authority-correspondence: predicted
medicine:predeclared-patient-strata__arm-contrast-by-stratum__joint-stratum-arm-outcome__multiplicity-and-scale-boundary;
source-derived
medicine:predeclared-patient-strata__arm-contrast-by-stratum__joint-stratum-arm-outcome__multiplicity-and-scale-boundary;
exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest
sha256:9dce54fae27b85699f4d34678c38b1af0680b4e3e14ae163b40eaa9d5b901d82; derivation seal
sha256:3393cbfe554e02e8688ab8b8e1df04c2d93369f9e01535c095b907d9202ca10e; engine receipt
sha256:2c2ed8e3aa4b025c627d17df97ea84d966f9e0378b34cc2e2e8843769438a3eb at receipts/engine/model_admitted/SFT-MED-TREATMENT-HETEROGENEITY-001-2c2ed8e3aa4b025c.json; empirical
validation sha256:c81624ed6a25c2525baaab7010d2bee63cda6bd6f640abe030d6bd3767ea5a37;
measurement receipt sha256:ec523a35c18e2858afb6dc1b1a52963eb018b34d6e8b0700a6f4956f001c3fd3;
isolation sha256:8b91e09d4cc1985497be12a117b5850e12bae41ce6d16e6f88693ef8670ef917; custody

sha256:0085fe2a4585893eef95013126321c34d949fd887d03885cf16e5aae84b85b8c.

18. Trial Observational Synthesis

Trials, observational studies and evidence synthesis are distinct process grammars. Protocol, allocation, blinding, sampling, confounders, missingness, study dependence and selective reporting remain inspectable from registration to interpretation.

49. Clinical trial

Claim identity: SFT-MED-CLINICAL-TRIAL-001

Question and exact theorem. A trial is a prospective finite process retaining protocol, eligibility, allocation, arm flow, outcomes, deviations and all analyses.

eligible-participant-traces__prospective-allocation-and-outcome__protocol-arms-f
low-and-analysis__registered-setting-time-and-endpoints__complete-protocol-state
-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-suc
cessor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001

- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-TREATMENT-HETEROGENEITY-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	eligible-participant-traces	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	prospective-allocation-and-outcome	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	protocol-arms-flow-and-analysis	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	registered-setting-time-and-endpoints	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: eligible-participant-traces__prospective-allocation-and-outcome__protocol-arms-flow-and-analysis__registered-setting-time-and-endpoints__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces a trial is a prospective finite process retaining protocol, eligibility, allocation, arm flow, outcomes, deviations and all analyses. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Trials, observational studies and evidence synthesis are distinct process grammars. Protocol, allocation, blinding, sampling, confounders, missingness, study dependence and selective reporting remain inspectable from registration to interpretation.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: d66d66f062bbebf26c6833177fb32af48e2ae644fb89cb162147d3a1cf9048a1.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 8f735324fbc01f2b3ae9bcb9d510cb3cc9949cdd378047aaa4153584a03b2cd6.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 809b70d29074b5ee6192be70f4a4bca181751345a5e8158d90b7212d7157cb7e.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 9b70b880becbdf244bcc6893f86a068b0ca93e23418bedb74bc2fd14e0adaa8e.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256: ccfa72ce2a8e9854ce03bfca2117fca0220383807bc09737b08ca559aceef0b0; certificate
sha256: 8068d3abd474c0ef732bc0c0fa4980096282001460aa8d849310ca182fa32aa6; external-validation
sha256: cff1e05fb4539423391aab8fb858548b216b57e1f0fdffa8f283e934d31b9426.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 4 purpose-matched discriminators with ordered identity
sha256: 354b4ab87ed9bf2d98f9f83e30265bb2fbb11b91c4e22fa032e89ec0e0d8957e. Target opened after derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- MED-ICH-E6R3-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E6R3-001.pdf;
sha256: e6ce19e36ce7d2e294f89ee89492b9e035178c3cca48984392bbd92eec9b002c; scope: final Good Clinical Practice: rights, safety, wellbeing, consent, protocol, allocation, blinding, adverse events, confidentiality and essential records.
- MED-STROBE-001 - STROBE Initiative; <https://www.strobe-statement.org/>; snapshot
experiments/external_sources/medicine/snapshots/MED-STROBE-001.html;
sha256: 420432c148d01e7aaaab8f7bcb4bcb4412d028aa50cd7814fc1b058faf1cf7c; scope:
observational-study planning, conduct, findings and interpretation reporting.
- MED-COCHRANE-MISSING-EVIDENCE-001 - Cochrane;
<https://www.cochrane.org/authors/handbooks-and-manuals/handbook/current/chapter-13>; snapshot experiments/external_sources/medicine/snapshots/MED-COCHRANE-MISSING-EVIDENCE-001.html;
sha256: 9ebd35b49c689f1b12fa3919051ea9a5ed315b168d9eac7b1eb8860317d07bb; scope: selective dissemination, missing evidence, benefit overestimation and harm underestimation.

Comparison record:

- sft-med-clinical-trial-001-authority-correspondence: predicted medicine:eligible-participant-traces__prospective-allocation-and-outcome__protocol-arms-flow-and-analysis__registered-setting-time-and-endpoints; source-derived medicine:eligible-participant-traces__prospective-allocation-and-outcome__protocol-arms-flow-and-analysis__registered-setting-time-and-endpoints; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:c06020ac2e8ab8cf3674c54e939e5129dd49c0a4010274b5344001dee0847c5e; derivation seal sha256:9296ab5bbae54807f69573ad8d95754e37121122249aa64bb4b43070448231a7; engine receipt sha256:b2207bbc4ad20a8369f19ed3fb999460dc87fc7dea9c248774b3063f7cccc7fa at receipts/engine/model_admitted/SFT-MED-CLINICAL-TRIAL-001-b2207bbc4ad20a83.json; empirical validation sha256:1abde6376ace42236e138cb18a552e294f52d4c304e3992271a444f14024e04d; measurement receipt sha256:083ca18abc118ad6a916909119616a8b7b4059f172ac7fcecc489abcbaf28817; isolation sha256:f8c77b8d31a7f3bfbdef9cae96ed9f6e903ca2e906110f3175518c7471859bbe; custody sha256:c764c862e548591fb0f663f859e2b8f685be21fc919d4d35d677f6a9957329e9.

50. Randomized allocation in a deterministic model

Claim identity: SFT-MED-RANDOMIZATION-001

Question and exact theorem. Randomization is a reproducible held-label allocation process whose future labels are concealed from selectors; it imports no nondeterministic ontology.

eligible-participants-and-held-generator-state__concealed-unpredictable-assignment__reconstructible-sequence-and-custody__generation-concealment-release-boundary__complete-protocol-state-transition-adverse-null-record__root-bound-forward-folding__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001

- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-CLINICAL-TRIAL-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	eligible-participants-and-held-generator-state	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
relation	relation-imported-fitted-or-erased	concealed-unpredictable-assignment	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	reconstructible-sequence-and-custody	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	generation-concealment-release-boundary	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: `eligible-participants-and-held-generator-state__concealed-unpredictable-assignment__reconstructible-sequence-and-custody__generation-concealment-release-boundary__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule`.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier`: clinical carrier is explicit and nonempty; passed `true`.
- `relation`: relation and organization remain separately reconstructible; passed `true`.
- `boundary`: observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. Randomization is closed without ontological nondeterminism. A deterministic held-label generator produces a reproducible allocation sequence; concealment prevents selectors from observing future labels. The full generator, custody, assignment and release trace remains available for audit after the decision boundary.

Trials, observational studies and evidence synthesis are distinct process grammars. Protocol, allocation, blinding, sampling, confounders, missingness, study dependence and selective reporting remain inspectable from registration to interpretation.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:0364ac60a13654734feb44538dd6459304e7e73f20ba5c9c4ab7dabd0d7e8ce6`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:b1230e16059b1bcd9873008a430b332315b7a015144a67c205961f2d840fd41`.

- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:3cb264dae79c407942ec812bd32e28fdfeb1e9bf48f11a1deb0aa39eb5450f95.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:64d6d13431729f073e54df2f0bc0ca77daa05676efcd82796e101c321097ed7c.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:190ebd4b0c7e2d241353611f8f0a3c1ded686c34d2621dcd787031885fd0defa; certificate
sha256:86ca3d06e70125e19e9cb1c24fbb84886a0d9eb5abfe678165492a1f1839c6aa; external-validation
sha256:7346de0508c961ae8e2341171272300ff610942aaa8a59be6c1e7d0532441f27.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 4 purpose-matched discriminators with ordered identity
sha256:354b4ab87ed9bf2d98f9f83e30265bb2fbb11b91c4e22fa032e89ec0e0d8957e. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- MED-ICH-E6R3-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E6R3-001.pdf;
sha256:e6ce19e36ce7d2e294f89ee89492b9e035178c3cca48984392bbd92eec9b002c; scope: final Good Clinical Practice: rights, safety, wellbeing, consent, protocol, allocation, blinding, adverse events, confidentiality and essential records.
- MED-STROBE-001 - STROBE Initiative; <https://www.strobe-statement.org/>; snapshot
experiments/external_sources/medicine/snapshots/MED-STROBE-001.html;
sha256:420432c148d01e7aaaab8f7bcb4bcb4412d028aa50cd7814fc1b058faf1cf7c; scope:
observational-study planning, conduct, findings and interpretation reporting.
- MED-COCHRANE-MISSING-EVIDENCE-001 - Cochrane;
<https://www.cochrane.org/authors/handbooks-and-manuals/handbook/current/chapter-13>; snapshot experiments/external_sources/medicine/snapshots/MED-COCHRANE-MISSING-EVIDENCE-001.html;
sha256:9ebd35b49c689f1b12fa3919051ea9a5ed315b168d9eaec7b1eb8860317d07bb; scope: selective dissemination, missing evidence, benefit overestimation and harm underestimation.

Comparison record:

- sft-med-randomization-001-authority-correspondence: predicted medicine:eligible-participants-and-held-generator-state__concealed-unpredictable-assignment__reconstructible-sequence-and-custody__generation-concealment-release-boundary; source-derived medicine:eligible-participants-and-held-generator-state__concealed-unpredictable-assignment__reconstructible-sequence-and-custody__generation-concealment-release-boundary; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff

- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:b9a5665a2bd1a7dd668bc7ccc4db07758ea9627326368f4ec0c41add5077a22; derivation seal sha256:cbabe3d7e0e0e1080c46026f8e90f636a2c73961cb01e9397433021d1f704c5d; engine receipt sha256:3699c5d458f2c3427c92193951e86e2976685527a59c01128d8d9bb12d3ad1ae at receipts/engine/model_admitted/SFT-MED-RANDOMIZATION-001-3699c5d458f2c342.json; empirical validation sha256:c36a29bf34ada468979d1cafea3a8f827d7ebdb021fcdd0abd06d506e40709b5; measurement receipt sha256:474815f4c62201f2694fc6568a3aaf931670f194281ca9563b2e6c9159ab0582; isolation sha256:72173c40696c43539bd1c9b3bc8a5a1c72410ef4bac2858f9c8e616763f3fd76; custody sha256:f6e6d51892d418dcda138db370cca4910e253ea3c55a5518b1917f39dee66314.

51. Clinical blinding

Claim identity: SFT-MED-BLINDING-001

Question and exact theorem. Blinding is an exact information-access restriction by role and stage with every planned or accidental unblinding event retained.

participant-carer-assessor-role-word__role-to-allocation-access-restriction__access-and-unblinding-trace__stage-role-and-information-boundary__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001

- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-RANDOMIZATION-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	participant-carer-assessor-role-word	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	role-to-allocation-access-restriction	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	access-and-unblinding-trace	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	stage-role-and-information-boundary	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
extension	free-fit-exception -opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: participant-carer-assessor-role-word__role-to-a-lllocation-access-restriction__access-and-unblinding-trace__stage-role-and-information-boundary__complete-protocol-state-transition-adverse-null-record__root-bound-forward-for-cing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the clinical law; passed **true**.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces blinding is an exact information-access restriction by role and stage with every planned or accidental unblinding event retained. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Trials, observational studies and evidence synthesis are distinct process grammars. Protocol, allocation, blinding, sampling, confounders, missingness, study dependence and selective reporting remain inspectable from registration to interpretation.

Adverse controls.

- **false_premise**: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:4214f3bbf9014501112511d6299abae119284040c65c22f8b4790d30a2920617.
- **tampered_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:0a4f5eaff51d7ec6d75a5f9465e0e0caeae97e50eaa5dbeddbb3b33ef9f9bea72.
- **tampered_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:cbb1dfff3899ee720a063e37149c27ec01f418e8b2cfce62bd4de42dbbd26299.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:25eea8e60fc855c0a1d15ec0534dd587b8772f36408e9501b234d501e555e17b.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

```
sha256:9994b67f46d960da97430be0214b7182fe160c36b3453ae226d11e4b818e76d2; certificate
sha256:53c76c366e562301ab8701bccce0831afd1c9f3d7a7be3ae8638af8713ef818a5; external-validation
sha256:e297f35b9879c8c89d8057f56a4f1bbbbce2f56b0a56d7a696de6d38c8125bed.
```

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 4 purpose-matched discriminators with ordered identity

sha256: 354b4ab87ed9bf2d98f9f83e30265bb2fbb11b91c4e22fa032e89ec0e0d8957e. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- MED-ICH-E6R3-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf; snapshot experiments/external_sources/medicine/snapshots/MED-ICH-E6R3-001.pdf; sha256:e6ce19e36ce7d2e294f89ee89492b9e035178c3cca48984392bbd92eec9b002c; scope: final Good Clinical Practice: rights, safety, wellbeing, consent, protocol, allocation, blinding, adverse events, confidentiality and essential records.
- MED-STROBE-001 - STROBE Initiative; <https://www.strobe-statement.org/>; snapshot experiments/external_sources/medicine/snapshots/MED-STROBE-001.html; sha256:420432c148d01e7aaaab8f7bcb4bcb4412d028aa50cd7814fc1b058faf1cf7c; scope: observational-study planning, conduct, findings and interpretation reporting.
- MED-COCHRANE-MISSING-EVIDENCE-001 - Cochrane;
<https://www.cochrane.org/authors/handbooks-and-manuals/handbook/current/chapter-13>; snapshot experiments/external_sources/medicine/snapshots/MED-COCHRANE-MISSING-EVIDENCE-001.html; sha256:9ebd35b49c689f1b12fa3919051ea9a5ed315b168d9eaec7b1eb8860317d07bb; scope: selective dissemination, missing evidence, benefit overestimation and harm underestimation.

Comparison record:

- sft-med-blinding-001-authority-correspondence: predicted medicine:participant-carer-assessor-role-word__role-to-allocation-access-restriction__access-and-unblinding-trace__stage-role-and-information-boundary; source-derived medicine:participant-carer-assessor-role-word__role-to-allocation-access-restriction__access-and-unblinding-trace__stage-role-and-information-boundary; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:df854df8964fd38205c6ec7200846bc3787f1e62364fea919192d1ecb5992898; derivation seal sha256:336decf243f8cc1132a0345f75f36f11f9f135443183dc07b4e9c7983ed86805; engine receipt sha256:02ee97e420fd490447b38cfa12c0234827e469d9ed5280749776035075167585 at receipts/engine/model_admitted/SFT-MED-BLINDING-001-02ee97e420fd4904.json; empirical validation sha256:238858b7f18a18bed09b7108ef7956afd83e906dc8d4a65cf3c40d5cf628c152; measurement receipt sha256:bfc23b8e30b1bd8fc127713b6e5861e7946347c6a592e67a7ff3df6a62e5f207; isolation sha256:bfc23b8e30b1bd8fc127713b6e5861e7946347c6a592e67a7ff3df6a62e5f207; custody sha256:0705a3b4a58bbf3fbf63d8975cd988f023531ad70ffdb015d4a5b46c71eea940.

52. Observational clinical study

Claim identity: SFT-MED-OBSERVATIONAL-STUDY-001

Question and exact theorem. An observational study retains source population, sampling, exposure, outcomes, confounders, missingness and follow-up and does not silently become an intervention.

sampled-person-records__observed-exposure-outcome-relation__sampling-selection-and-confounders__source-population-time-and-follow-up__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001

- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-BLINDING-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	sampled-person-records	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	observed-exposure-outcome-relation	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	sampling-selection-and-confounders	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	source-population-time-and-follow-up	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: sampled-person-records__observed-exposure-outcome-relation__sampling-selection-and-confounders__source-population-time-and-follow-up__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier: clinical carrier is explicit and nonempty; passed true.
- relation: relation and organization remain separately reconstructible; passed true.
- boundary: observation boundary cannot silently become the clinical law; passed true.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces an observational study retains source population, sampling, exposure, outcomes, confounders, missingness and follow-up and does not silently become an intervention. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Trials, observational studies and evidence synthesis are distinct process grammars. Protocol, allocation, blinding, sampling, confounders, missingness, study dependence and selective reporting remain inspectable from registration to interpretation.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 05747abe3fe915e68faab73895cfca6324c094f7d099aaaf9f4bc3df153432bf.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: c71e594f3695e09fce66fca2fd52a053ddd42f55efc43ad512b50c30c1af96d9.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 203e994f4403b64c225489889064e5fd41d8e547ffe2863153e882884dfa49af.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: cc84761c203e47f59adef7bbbffaf86c45f4b9cef63fa0ebaa6b2fc53ac2dfe9.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256: fd10c0cc4cb16d16792d3bfb340aa707f9eb8f68125790d2c6867ef9d82d64d6; certificate
sha256: 4af8e03f9a6f912c07a3def2acd8a1d9799d69e4e0d4280f37cb1ce8189002c1; external-validation
sha256: 95b499afa491223b7c467ba03551e9f371911584734537a96ab3096e8b633afa.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 4 purpose-matched discriminators with ordered identity
sha256: 354b4ab87ed9bf2d98f9f83e30265bb2fbb11b91c4e22fa032e89ec0e0d8957e. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- MED-ICH-E6R3-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E6R3-001.pdf;
sha256: e6ce19e36ce7d2e294f89ee89492b9e035178c3cca48984392bbd92eec9b002c; scope: final Good Clinical Practice: rights, safety, wellbeing, consent, protocol, allocation, blinding, adverse events, confidentiality and essential records.
- MED-STROBE-001 - STROBE Initiative; <https://www.strobe-statement.org/>; snapshot
experiments/external_sources/medicine/snapshots/MED-STROBE-001.html;
sha256: 420432c148d01e7aaaab8f7bcb4bcb4412d028aa50cd7814fc1b058faf1cf7c; scope: observational-study planning, conduct, findings and interpretation reporting.
- MED-COCHRANE-MISSING-EVIDENCE-001 - Cochrane;
<https://www.cochrane.org/authors/handbooks-and-manuals/handbook/current/chapter-13>; snapshot experiments/external_sources/medicine/snapshots/MED-COCHRANE-MISSING-EVIDENCE-001.html;
sha256: 9ebd35b49c689f1b12fa3919051ea9a5ed315b168d9eaec7b1eb8860317d07bb; scope: selective dissemination, missing evidence, benefit overestimation and harm underestimation.

Comparison record:

- sft-med-observational-study-001-authority-correspondence: predicted medicine:sampled-person-records__observed-exposure-outcome-relation__sampling-selection-and-confounders__source-population-time-and-follow-up; source-derived medicine:sampled-person-records__observed-exposure-outcome-relation__sampling-selection-and-confounders__source-population-time-and-follow-up; exact match True

- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:c45983d1004cb8b5ca261547d858909f1d1ee78f47efd934d6387c148723e6bd; derivation seal sha256:7e85393f3c8cd5f10fbf6627cedf1c916819642ac34463bdb56c4e0b06a36e7b; engine receipt sha256:7a80fb51f8716b7b95fc3eabf2591ed77bb49c2b3ea96140162ceec838ba2015 at receipts/engine/model_admitted/SFT-MED-OBSERVATIONAL-STUDY-001-7a80fb51f8716b7b.json; empirical validation sha256:188f7baf95348f6acd587c3e01981b0dc86fcd71e4ba846c64acfc795641ed0c; measurement receipt sha256:bc52f422697615c0e4f10b5632177753e4397f42e48b2ae904434db8c558fd5d; isolation sha256:33713e4704827dc6c88f266e861ca82ae6730ca964cfafedc1c02a25d70c808f; custody sha256:b250bd48874600833458246a31006daa93a344d94c2b9a28277aad2ff95bdd8c.

53. Clinical evidence synthesis

Claim identity: SFT-MED-EVIDENCE-SYNTHESIS-001

Question and exact theorem. Evidence synthesis composes purpose-matched study carriers while retaining design differences, dependence, heterogeneity, bias, adverse and absent results.

study-level-evidence-carriers__question-matched-composition__heterogeneity-bias-and-dependence-held__protocol-search-selection-and-date__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-succesor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001

- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-OBSERVATIONAL-STUDY-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	study-level-evidence-carriers	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	question-matched-composition	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	heterogeneity-bias-and-dependence-held	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
observation	observation-boundary-unrecorded	protocol-search-selection-and-date	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: study-level-evidence-carriers__question-matched-composition__heterogeneity-bias-and-dependence-held__protocol-search-selection-and-date__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces evidence synthesis composes purpose-matched study carriers while retaining design differences, dependence, heterogeneity, bias, adverse and absent results. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Trials, observational studies and evidence synthesis are distinct process grammars. Protocol, allocation, blinding, sampling, confounders, missingness, study dependence and selective reporting remain inspectable from registration to interpretation.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:8ae5df82d62875e651ef2988bdaba1cf2670e3a4053d8617b46f87eefae379a6.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:b1095a6fa2200fe39802004a457aab1e150a671f777ba92ecc21ea0a19f4fd1b.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:f40ac99004d93bb120da8294ad6f15289a87d38d3c86f0d0046098e69ece27b2.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 219e8a66c9bf46600e55dc388d2358f9b4635ff71ee6e2c5ddcf046e3066177.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256 : 1244dc5b1f2575e833b1fc227b6eee1130d50d5b10e56c6caa172225c43c525a; certificate
sha256 : a81010dc6ab8bc4185d432b396e2195839a20325fe41274969df31fa21f61299; external-validation
sha256 : 8e1e27cf02e15f2e67e0cb592d299700129d93272e7d7a9555f02de5bea30921.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 4 purpose-matched discriminators with ordered identity
sha256 : 354b4ab87ed9bf2d98f9f83e30265bb2fbb11b91c4e22fa032e89ec0e0d8957e. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- MED-ICH-E6R3-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E6R3-001.pdf;
sha256 : e6ce19e36ce7d2e294f89ee89492b9e035178c3cca48984392bbd92eec9b002c; scope: final Good Clinical Practice: rights, safety, wellbeing, consent, protocol, allocation, blinding, adverse events, confidentiality and essential records.
- MED-STROBE-001 - STROBE Initiative; <https://www.strobe-statement.org/>; snapshot
experiments/external_sources/medicine/snapshots/MED-STROBE-001.html;
sha256 : 420432c148d01e7aaaab8f7bcb4bcb4412d028aa50cd7814fc1b058faf1cf7c; scope:
observational-study planning, conduct, findings and interpretation reporting.
- MED-COCHRANE-MISSING-EVIDENCE-001 - Cochrane;
<https://www.cochrane.org/authors/handbooks-and-manuals/handbook/current/chapter-13>; snapshot
experiments/external_sources/medicine/snapshots/MED-COCHRANE-MISSING-EVIDENCE-001.html;
sha256 : 9ebd35b49c689f1b12fa3919051ea9a5ed315b168d9eac7b1eb8860317d07bb; scope: selective dissemination, missing evidence, benefit overestimation and harm underestimation.

Comparison record:

- sft-med-evidence-synthesis-001-authority-correspondence: predicted medicine:study-level-evidence-carriers__question-matched-composition__heterogeneity-bias-and-dependence-held__protocol-search-selection-and-date; source-derived medicine:study-level-evidence-carriers__question-matched-composition__heterogeneity-bias-and-dependence-held__protocol-search-selection-and-date; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor

- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:69df84dbc0331b6d9c594866d5eae655c288cbd2d8813c5f68024a7829305643; derivation seal sha256:b8b6f3a039c12b1f36da6c22ca532cd5d7ee3cdedd87665938fe0d16a8234d0c; engine receipt sha256:91a1c5af78ab9f5415d436ec54c4fa6683fb9dbd364cbbb7d40d2d999ec01a8a at receipts/engine/model_admitted/SFT-MED-EVIDENCE-SYNTHESIS-001-91a1c5af78ab9f54.json; empirical validation sha256:f2274a1cc18eaec45aff0cf02b726f7a86b720ebbe0a2756fdbe0a2f0cad54ed; measurement receipt sha256:719533e7352b427635030bdfba44f65d7776f28617e9d18fc8191aec79b8691d; isolation sha256:0370d4a17993c20baf24d0c9138d4c45117b76f792a7948cf32120a95bedbd76; custody sha256:e8dbeea5058750c08f00f60beeeb5fb5aab0d6b4a66001f6a7252b5980ba6a44.

54. Missing and selective reporting

Claim identity: SFT-MED-MISSING-REPORTING-001

Question and exact theorem. A registered outcome or study absent from the visible report remains an explicit missing distinction and cannot be interpreted as a favorable absence.

registered-and-observed-study-results__registration-publication-correspondence__reported-unreported-support__registry-search-and-cutoff-date__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001

- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-EVIDENCE-SYNTHESIS-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	registered-and-observed-study-results	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	registration-publication-correspondence	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	reported-unreported-support	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	registry-search-and-cutoff-date	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: registered-and-observed-study-results__registration-publication-correspondence__reported-unreported-support__registry-search-and-cutoff-date__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. Registered outcomes and studies remain present even when results are unavailable. Cochrane's external evidence records how selective dissemination can overstate benefit and understate harm; the SFT law prevents the missing row from being rewarded as a favorable result.

Trials, observational studies and evidence synthesis are distinct process grammars. Protocol, allocation, blinding, sampling, confounders, missingness, study dependence and selective reporting remain inspectable from registration to interpretation.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 66b8d8b93ea4db788b999176bcbfa8e111d89aa6d3c5de66e9a15c885a93e5a1.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 1f0e54b37a7e30f709936c96343f1c9bf62be5f0fd97d99aaea3f53aed2a4c95.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: aef59a7035e0321830b6eff5d54f22c45e1a657a75a6539ca4f3ffd48c9c0f56.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: bfb0039149c94dab687d0579d9cd9c9286b0cbf023197b83e84638c3ff17cd52.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256: d0bada684032eebf5c4938abcc3bfff2b68d09575c5bc27bc190b93d60257849; certificate
sha256: 7c9909eebfdb3e859b2de5248ef1003c9764cfbae752572a21773638b938795e; external-validation
sha256: 3a2709e87f38e32020a5d96b592a6727a7f359931cf33165081e3e1d0661dae4.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 4 purpose-matched discriminators with ordered identity
sha256: 354b4ab87ed9bf2d98f9f83e30265bb2fbb11b91c4e22fa032e89ec0e0d8957e. Target opened after derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- MED-ICH-E6R3-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E6R3-001.pdf;
sha256: e6ce19e36ce7d2e294f89ee89492b9e035178c3cca48984392bbd92eec9b002c; scope: final Good Clinical Practice: rights, safety, wellbeing, consent, protocol, allocation, blinding, adverse events, confidentiality and essential records.

- MED-STROBE-001 - STROBE Initiative; <https://www.strobe-statement.org/>; snapshot experiments/external_sources/medicine/snapshots/MED-STROBE-001.html; sha256:420432c148d01e7aaaab8f7bcb4bcbcb4412d028aa50cd7814fc1b058faf1cf7c; scope: observational-study planning, conduct, findings and interpretation reporting.
- MED-COCHRANE-MISSING-EVIDENCE-001 - Cochrane; <https://www.cochrane.org/authors/handbooks-and-manuals/handbook/current/chapter-13>; snapshot experiments/external_sources/medicine/snapshots/MED-COCHRANE-MISSING-EVIDENCE-001.html; sha256:9ebd35b49c689f1b12fa3919051ea9a5ed315b168d9eaec7b1eb8860317d07bb; scope: selective dissemination, missing evidence, benefit overestimation and harm underestimation.

Comparison record:

- sft-med-missing-reporting-001-authority-correspondence: predicted medicine:registered-and-observed-study-results__registration-publication-correspondence__reported-unreported-support__registry-search-and-cutoff-date; source-derived medicine:registered-and-observed-study-results__registration-publication-correspondence__reported-unreported-support__registry-search-and-cutoff-date; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:068db3d0c259de641a433bb8eb40f2d6ffacfb11f9a1a279e979da6a4e45f0c2e; derivation seal sha256:bf5d20e14cbd6ba90776a4e3cdf382114ce17eaaa02eaafe9eb129caca3d7aef; engine receipt sha256:cdaf826443e597536993c81365b2978a96a2e4a68ca1d8cbd496694b47a8d484 at receipts/engine/model_admitted/SFT-MED-MISSING-REPORTING-001-cdaf826443e59753.json; empirical validation sha256:cffaa8d0297e7cda61ee69270e47e2ee642cc7d54ce58b7d715a6a97eb44f032; measurement receipt sha256:fe6b03de2f9d68aa1d51a015b4788885dfde3a607ef78d47c180df23781577d4; isolation sha256:7a18df5c66e591fdf7477ccbddcf212e189f6600ec897c060fd93bd684385e68; custody sha256:ac177a34c0576a82ccc1829fe20fbd6c1a672d8312124aca0bc6b897fbd79dd4.

19. Screening Prevention Treatment Rehabilitation

Screening is the complete pathway from an apparently healthy population through confirmation and care, including false results and overdiagnosis. Prevention, treatment, monitoring, stopping and rehabilitation retain their different timing, purpose and outcome structures.

55. Screening

Claim identity: SFT-MED-SCREENING-001

Question and exact theorem. Screening is the whole test-to-confirmation-to-care pathway in a declared asymptomatic population, including false results, overdiagnosis and downstream harms.

asymptomatic-eligible-population__test-to-follow-up-care-path__benefit-harm-and-overdiagnosis-chain__population-interval-and-program__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001

- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-MISSING-REPORTING-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	asymptomatic-eligible-population	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	test-to-follow-up-care-path	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	benefit-harm-and-overdiagnosis-chain	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	population-interval-and-program	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: asymptomatic-eligible-population__test-to-follow-up-care-path__benefit-harm-and-overdiagnosis-chain__population-interval-and-program__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier: clinical carrier is explicit and nonempty; passed true.
- relation: relation and organization remain separately reconstructible; passed true.
- boundary: observation boundary cannot silently become the clinical law; passed true.

Scientific meaning. The theorem covers the entire screening chain, not test accuracy alone: population, test, confirmation, intervention, outcomes, false results, overdiagnosis, burden and harm. WHO correspondence is opened only after this chain is sealed.

Screening is the complete pathway from an apparently healthy population through confirmation and care, including false results and overdiagnosis. Prevention, treatment, monitoring, stopping and rehabilitation retain their different timing, purpose and outcome structures.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : f9a7366c40bf135618039266dd98b8965c65e60811f33a18fa394fe799c9aa80.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 96df2a4985ebaa39b2eba5863b582d918c2b1b8b6dfd636f9565dfc5606f733d.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 087283a05b6bfda98a224d6362cb9aa2b3509cab716ce0a8518e4667ac4995aa.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 880fb8c534584496bfad38173bebc8e34e8f33c8edb91640bf5838a0d023fe0d.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256 : 478293d4889cc68c475ffba5b800cddebc388e8804a86efc62b61d798f3b2bac; certificate
sha256 : 4added181425fd2bbcb62c955b9a3619a599d1eb3ea618cae5df40c3ea2ff1807; external-validation
sha256 : e03e507b70ac1c2e2c2b8adeb3415df52d4dda433671416d8f24ba7c868ec9dd.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 3 purpose-matched discriminators with ordered identity
sha256 : e7e1ef013da606dc7319d8154cfadd9356428e8b339c4b5740a1bba855cffb4f. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- **MED-WHO-SCREENING-001** - World Health Organization Regional Office for Europe;
<https://www.who.int/publications/i/item/9789289054782>; snapshot
experiments/external_sources/medicine/snapshots/MED-WHO-SCREENING-001.html;
sha256 : b56f491a97d2efbe4ff4d59e304649dd1fb1d551a2ca6b88a9bbcf43d385b5f7; scope: screening population, risk, early intervention, benefits, harms and quality assurance.
- **MED-NICHD-REHABILITATION-001** - Eunice Kennedy Shriver National Institute of Child Health and Human Development;
<https://www.nichd.nih.gov/health/topics/rehabilitation-medicine>; snapshot
experiments/external_sources/medicine/snapshots/MED-NICHD-REHABILITATION-001.html;
sha256 : c1b54134e6cbcfb9542a7c012236395eb609a67a789818f201c5b42202dc8cae; scope: rehabilitation function, participation, independence, quality of life and outcomes.

Comparison record:

- **sft-med-screening-001-authority-correspondence:** predicted medicine:asymptomatic-eligible-population__test-to-follow-up-care-path__benefit-harm-and-overdiagnosis-chain__population-interval-and-program; source-derived medicine:asymptomatic-eligible-population__test-to-follow-up-care-path__benefit-harm-and-overdiagnosis-chain__population-interval-and-program; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:485ab3d052d72a9dd5891e77d56693ece485963f624287d9de2f76cd0a574919; derivation seal sha256:5173a0bddd590eb819f02ce919709c2af0b52141d81e38110b1d42a56d78d107; engine receipt sha256:770e80dec3643f760475ced8d8d7b7ebb4b08f18f2fce76579dcf6bebd004168 at receipts/engine/model_admitted/SFT-MED-SCREENING-001-770e80dec3643f76.json; empirical validation sha256:740c7ddf653ed94f84aed277c36f261da2a629b7b8c1bc2e94871491e83204ca; measurement receipt sha256:277a53b5f069e75b573a1c3def2940fe74932cbb80dc43e5e43c080e2f887a03; isolation sha256:eb830c69842913a8b84661c8826b2702931ce80f932ed86163ef9be8081c10c2; custody sha256:dd8ea243859fb8c1c6745417ced28414be00ad513f9a9bb792a315e4c489b982.

56. Prevention

Claim identity: SFT-MED-PREVENTION-001

Question and exact theorem. Prevention is an action preceding a declared outcome in an at-risk carrier, with prevention level, comparator and all later outcomes retained.

at-risk-person-or-population__action-before-target-outcome__primary-secondary-tertiary-boundary__risk-frame-time-and-outcomes__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001

- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-SCREENING-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	at-risk-person-or-population	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	action-before-target-outcome	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	primary-secondary-tertiary-boundary	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	risk-frame-time-and-outcomes	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: `at-risk-person-or-population__action-before-target-outcome__primary-secondary-tertiary-boundary__risk-frame-time-and-outcomes__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule`.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces prevention is an action preceding a declared outcome in an at-risk carrier, with prevention level, comparator and all later outcomes retained. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Screening is the complete pathway from an apparently healthy population through confirmation and care, including false results and overdiagnosis. Prevention, treatment, monitoring, stopping and rehabilitation retain their different timing, purpose and outcome structures.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 5ff6ff2678f67ec9cb0e4d9cf1a252f825ac86ca9fd9b296b6f75d739d78d3d8.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 878b011e7debc7617e211b9db2c8df9cc6b65fd721020b09eb6f9deec87746d2.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 4ff97b18fd3e8186f5cd96e092ac0b140ddcb34f3e111fe903924d9626d772f6.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 24947de1cdad390d74633843a5674b025e8d8eed9ac877907dd3d4dc2340a0b8.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation sha256:b932c5b6710dcb4c9f91ba18ed1a1c8c6cd9c3b4caba1e7f06671051789e9ae4; certificate sha256:3ced2d7c7a5303571c3ed8f29dd0076a1e0e37e9d5e9bea5e5989d774ed1d818; external-validation sha256:15440972004fad52f39a45a814f21d5a17ba713dc2963ed5f11c40d3149bd10d.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 3 purpose-matched discriminators with ordered identity sha256:e7e1ef013da606dc7319d8154cfadd9356428e8b339c4b5740a1bba855cffb4f. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- MED-WHO-SCREENING-001 - World Health Organization Regional Office for Europe; <https://www.who.int/publications/i/item/9789289054782>; snapshot experiments/external_sources/medicine/snapshots/MED-WHO-SCREENING-001.html; sha256:b56f491a97d2efbe4ff4d59e304649dd1fb1d551a2ca6b88a9bbcf43d385b5f7; scope: screening population, risk, early intervention, benefits, harms and quality assurance.
- MED-NICHD-REHABILITATION-001 - Eunice Kennedy Shriver National Institute of Child Health and Human Development; <https://www.nichd.nih.gov/health/topics/rehabilitation-medicine>; snapshot experiments/external_sources/medicine/snapshots/MED-NICHD-REHABILITATION-001.html; sha256:c1b54134e6cbcfb9542a7c012236395eb609a67a789818f201c5b42202dc8cae; scope: rehabilitation function, participation, independence, quality of life and outcomes.

Comparison record:

- sft-med-prevention-001-authority-correspondence: predicted medicine:at-risk-person-or-population__action-before-target-outcome__primary-secondary-tertiary-boundary__risk-frame-time-and-outcomes; source-derived medicine:at-risk-person-or-population__action-before-target-outcome__primary-secondary-tertiary-boundary__risk-frame-time-and-outcomes; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:d4e6d6ebc5d213f40f2655568cf2d7fdb94d5020f61642b71caad2b8e9bb24e; derivation seal sha256:33753c864f77dcc67fe11ae752b0f3757cd0caa3822af809e84ceef97f9520ba; engine receipt sha256:b5f5c829fd5d0973df088228ef932b37b6d5b8b52e8f75976985d4075882e759 at

receipts/engine/model_admitted/SFT-MED-PREVENTION-001-b5f5c829fd5d0973.json; empirical validation sha256:c2e77b86e664390c8cc35c1a255d3e58de9312573b58d7a3cb65ad6550c7cb03; measurement receipt sha256:451d108380d499c552fb45654612657654c0c5cd65892564ca915e810c3c9fe0; isolation sha256:f388d5157acbec2eb432f98cfcf97646b9512244fdc8b28acfd1b7d0988ef8b1; custody sha256:6c47d560bb38eca49d2002ca3cdce1462d2db291830a17cfa081ee71a4d47ace.

57. Treatment

Claim identity: SFT-MED-TREATMENT-001

Question and exact theorem. Treatment binds a declared indication to delivered therapeutic action, comparator, response, harm and follow-up rather than to intent alone.

patient-with-declared-condition__therapeutic-action-to-course__content-delivery-response-and-harm__indication-setting-time-and-follow-up__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001

- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-PREVENTION-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	patient-with-declared-condition	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	therapeutic-action-to-course	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	content-delivery-response-and-harm	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	indication-setting-time-and-follow-up	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: patient-with-declared-condition__therapeutic-action-to-course__content-delivery-response-and-harm__indication-setting-time-and-follow-up__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces treatment binds a declared indication to delivered therapeutic action, comparator, response, harm and follow-up rather than to intent alone. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Screening is the complete pathway from an apparently healthy population through confirmation and care, including false results and overdiagnosis. Prevention, treatment, monitoring, stopping and rehabilitation retain their different timing, purpose and outcome structures.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:14a15e66b069c83b554df070f982a033f2c717d2deabaa58994bf7c8572717e3.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:19bbe8527397bca33145567001053191bf38e783f9d5b38c6c807dab72d2cfab.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:0fbeb7e786153fa7e78fe1e21b78fe1243a3bb686cf5184aa1535cf97848e231.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:31fb1fcd70db370fd1410dfce1e04cdea2d64a34207cbca07b2d25df821a01f6.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:e40a4cbf6ba8fae05a9496c52ee77524ec592dea870f982ae392a2d2b2051583; certificate
sha256:fbe3717726724308880ca5d8b6b4f2ae5c23b89bb80604dfcfa21ac471291663; external-validation
sha256:3b8f988a8e94d91e3378927c412192fc2fde2f477fff7fe5836a19b9007095ad.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 3 purpose-matched discriminators with ordered identity
sha256:e7e1ef013da606dc7319d8154cfadd9356428e8b339c4b5740a1bba855cffb4f. Target opened after derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- **MED-WHO-SCREENING-001** - World Health Organization Regional Office for Europe;
<https://www.who.int/publications/i/item/9789289054782>; snapshot
[experiments/external_sources/medicine/snapshots/MED-WHO-SCREENING-001.html](https://www.who.int/publications/i/item/9789289054782);
sha256:b56f491a97d2efbe4ff4d59e304649dd1fb1d551a2ca6b88a9bbcf43d385b5f7; scope: screening population, risk, early intervention, benefits, harms and quality assurance.
- **MED-NICHD-REHABILITATION-001** - Eunice Kennedy Shriver National Institute of Child Health and Human Development;
<https://www.nichd.nih.gov/health/topics/rehabilitation-medicine>; snapshot
[experiments/external_sources/medicine/snapshots/MED-NICHD-REHABILITATION-001.html](https://www.nichd.nih.gov/health/topics/rehabilitation-medicine);
sha256:c1b54134e6cbcfb9542a7c012236395eb609a67a789818f201c5b42202dc8cae; scope: rehabilitation function, participation, independence, quality of life and outcomes.

Comparison record:

- **sft-med-treatment-001-authority-correspondence:** predicted medicine:patient-with-declared-condition__therapeutic-action-to-course__content-delivery-response-and-harm__indication-setting-time-and-follow-up; source-derived medicine:patient-with-declared-condition__therapeutic-action-to-course__content-delivery-response-and-harm__indication-setting-time-and-follow-up; exact

match True

- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:2b4cc2bd2b1d4cd5788cd2e4bb0fe681cf81f148f60ded4559100d7fa0bf5a89; derivation seal sha256:91081c264d6f2101ed37b92bb5d2fad28de702070d86d23a10319555d820ee7f; engine receipt sha256:400c1338756c2f4c002d95b68607cb71f6e97a573e3a553d9106fde8e21c4c68 at receipts/engine/model_admitted/SFT-MED-TREATMENT-001-400c1338756c2f4c.json; empirical validation sha256:0518eab891b4eca3a62163358584d8454d07421ca825059bf4daad157d60a29c; measurement receipt sha256:cbad984179c9907b6b3d98ab60512fd78290587e9932036fdd9743c01ba5b17a; isolation sha256:6a8ad708c38e3cf5953c8c78b2b78b6c78d8de80ee520ab94b016d84a6844f38; custody sha256:e29a76161e8103d643f0544a83bde17bd8d7970c4680c5f51f0278832f8b62b3.

58. Rehabilitation

Claim identity: SFT-MED-REHABILITATION-001

Question and exact theorem. Rehabilitation is an iterated, goal-bound relation among intervention, function, participation and lived context across retained timepoints.

person-function-and-participation-state__iterated-action-to-capacity-change__goals-interventions-context-and-course__baseline-follow-up-and-lived-setting__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001

- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-TREATMENT-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	person-function-and-participation-state	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	iterated-action-to-capacity-change	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	goals-interventions-context-and-course	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
observation	observation-boundary-unrecorded	baseline-follow-up-and-lived-setting	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: `person-function-and-participation-state__iterated-action-to-capacity-change__goals-interventions-context-and-course__baseline-follow-up-and-lived-setting__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule`.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. Rehabilitation is an iterated person-centred relation among goals, interventions, function, participation and lived context. It is not reducible to one impairment score or one encounter.

Screening is the complete pathway from an apparently healthy population through confirmation and care, including false results and overdiagnosis. Prevention, treatment, monitoring, stopping and rehabilitation retain their different timing, purpose and outcome structures.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : a4713ec5943449b25d12827d020a7e17260808bc17cd6a671abf9a9358f18b37.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 211ee05ca0faa27e7be6765b3fd2b441cb37fee5fcc8f8acf96868886b2b79bf.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 0e706cc6429172dd35f65a53399feee636b3f51bd668488f95f33a733e2406a6.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt

sha256:38b42ea42d4f92615463d6808b2fc806da8568e285bf41071f4f5f6cf6c9ea12.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation sha256:75e9db15b0ca24ea8796eab4d10aaea376c7b7081b90841dd5a2f18f131d8c1e; certificate sha256:796b07d3dbfeccdd93f7c4d40df2db8fe4ce5e15a77dd3f02979a6d04205c437; external-validation sha256:08fa5ecb93446e9f8bb8b8a46eb44eb5c58a0f2ea2550a63714a9da2b185367b.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 3 purpose-matched discriminators with ordered identity sha256:e7e1ef013da606dc7319d8154cfadd9356428e8b339c4b5740a1bba855cffb4f. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- MED-WHO-SCREENING-001 - World Health Organization Regional Office for Europe; <https://www.who.int/publications/i/item/9789289054782>; snapshot experiments/external_sources/medicine/snapshots/MED-WHO-SCREENING-001.html; sha256:b56f491a97d2efbe4ff4d59e304649dd1fb1d551a2ca6b88a9bbcf43d385b5f7; scope: screening population, risk, early intervention, benefits, harms and quality assurance.
- MED-NICHD-REHABILITATION-001 - Eunice Kennedy Shriver National Institute of Child Health and Human Development; <https://www.nichd.nih.gov/health/topics/rehabilitation-medicine>; snapshot experiments/external_sources/medicine/snapshots/MED-NICHD-REHABILITATION-001.html; sha256:c1b54134e6cbcfb9542a7c012236395eb609a67a789818f201c5b42202dc8cae; scope: rehabilitation function, participation, independence, quality of life and outcomes.

Comparison record:

- sft-med-rehabilitation-001-authority-correspondence: predicted medicine:person-function-and-participation-state__iterated-action-to-capacity-change__goals-interventions-context-and-course__baseline-follow-up-and-lived-setting; source-derived medicine: person-function-and-participation-state__iterated-action-to-capacity-change__goals-interventions-context-and-course__baseline-follow-up-and-lived-setting; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:1de03205fe13fc72cc23bcb3973cf89e6e7a31c6a393f34961c82408afe75078; derivation seal

sha256:9324a4926640a39b10067b6b29023c2f9542b2e55eedf3caa0d6b91763038f93; engine receipt sha256:248785ef1ba749a5a71984ec0bcdfed368de479cee3f665aff962ee786cd87eb at receipts/engine/model_admitted/SFT-MED-REHABILITATION-001-248785ef1ba749a5.json; empirical validation sha256:a542004e97d2e01b8ec0e622a3a1336058fba3926c9057b918eb41b16576b963; measurement receipt sha256:f9dcf954ff7ac65c75ecfe40d986524e5b1a2585e79b0af1e8c23fda29c1cd93; isolation sha256:5abdf27521250ce8f6c350b621762cc3be0ef0df427999207655ff0aa27e87fc; custody sha256:5e7d82b292b195a7f26892bb9eff4be49f091b7d63f407fc763ff9e8c97c03cf.

59. Clinical monitoring

Claim identity: SFT-MED-MONITORING-001

Question and exact theorem. Monitoring is a predeclared sequence of measurements and action rules; the measurement history may not be replaced by the latest value.

ordered-patient-measurements__change-to-decision-relation__schedule-threshold-action-trace__method-interval-and-care-context__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001

- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-REHABILITATION-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	ordered-patient-measurements	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	change-to-decision-relation	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	schedule-threshold-action-trace	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	method-interval-and-care-context	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: ordered-patient-measurements__change-to-decision-relation__schedule-threshold-action-trace__method-interval-and-care-context__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier`: clinical carrier is explicit and nonempty; passed `true`.
- `relation`: relation and organization remain separately reconstructible; passed `true`.
- `boundary`: observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces monitoring is a predeclared sequence of measurements and action rules; the measurement history may not be replaced by the latest value. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Screening is the complete pathway from an apparently healthy population through confirmation and care, including false results and overdiagnosis. Prevention, treatment, monitoring, stopping and rehabilitation retain their different timing, purpose and outcome structures.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: b82a61078436c0f8f312bf2c2b78e0f40ecb4929f718e1c4eb8c232c8e54cd36.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: c738716f2c0c9ad56e508e4c4aea4c4051dbe8b82490521f8510247b9888bbfe.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 2b05368fdcbafc16c62ee361ce13efd215f60e67c8b0313ce9d0b2bf9cabf402.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: f49ce0d83ddf38a85f44ced4dd6fa42c81fc8fe0b2dd3ddd7f2ab565ec4aeffe.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256: ee877406ba620a7dc5e767f6b9b6d254ebd242673abd8934d36e6a41cfdcf89b; certificate
sha256: 41d4b90fec9dc196167077ed1e0394d37b3ffae02a91bcbcd6536186daddca15; external-validation
sha256: de8b3edef056ee51be605823c4bd0e7275e5e0982d8d2f80c73870b67bad9b1b.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 3 purpose-matched discriminators with ordered identity

sha256: e7e1ef013da606dc7319d8154cfadd9356428e8b339c4b5740a1bba855cffb4f. Target opened after derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- MED-WHO-SCREENING-001 - World Health Organization Regional Office for Europe;
<https://www.who.int/publications/i/item/9789289054782>; snapshot
experiments/external_sources/medicine/snapshots/MED-WHO-SCREENING-001.html;
sha256: b56f491a97d2efbe4ff4d59e304649dd1fb1d551a2ca6b88a9bbcf43d385b5f7; scope: screening population, risk, early intervention, benefits, harms and quality assurance.
- MED-NICHD-REHABILITATION-001 - Eunice Kennedy Shriver National Institute of Child Health and Human Development;
<https://www.nichd.nih.gov/health/topics/rehabilitation-medicine>; snapshot
experiments/external_sources/medicine/snapshots/MED-NICHD-REHABILITATION-001.html;
sha256: c1b54134e6cbcfb9542a7c012236395eb609a67a789818f201c5b42202dc8cae; scope: rehabilitation function, participation, independence, quality of life and outcomes.

Comparison record:

- sft-med-monitoring-001-authority-correspondence: predicted medicine:ordered-patient-measurements__change-to-decision-relation__schedule-threshold-action-trace__method-interval-and-care-context; source-derived medicine:ordered-patient-measurement

s__change-to-decision-relation__schedule-threshold-action-trace__method-interval-and-care-context; exact match True

- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:6befa89dcd0804aa2b1e83767dd77aa5a389da939e0e393b45df3506514f05e2; derivation seal sha256:56d64264f087400d1997d327312d3b5fa64c49deed4f16d4d4419674af1be0f3; engine receipt sha256:2940988359a18d6bcd422e2a4008e6b504f8d13208542f7b8c4701191f4cd163 at receipts/engine/model_admitted/SFT-MED-MONITORING-001-2940988359a18d6b.json; empirical validation sha256:a9607dd1f6fc912109ec3bb9666947af9e298a69e37dd4712aff9d825888c601; measurement receipt sha256:05b5404b3e083bcaa5f6f88b9458e43dec6947867e6845e8c3644498623817f0; isolation sha256:df1dc960e98dacf07df5f19d0c1d85383dd84ee6f617f75620c19c9d417665f6; custody sha256:ff41f850e235fe18ccc49963c80945a586853a8236a6692b966f68f3516ebaf4.

60. Stopping and withdrawal

Claim identity: SFT-MED-STOPPING-WITHDRAWAL-001

Question and exact theorem. Stopping or withdrawal is a recorded care transition with reason, alternative, taper or aftercare and post-transition outcomes held.

ongoing-care-and-patient-state__stop-continue-change-decision__reason-transition-and-aftercare__decision-time-and-follow-up__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001

- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-MONITORING-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	ongoing-care-and-patient-state	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	stop-continue-change-decision	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	reason-transition-and-aftercare	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
observation	observation-boundary-unrecorded	decision-time-and-follow-up	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: ongoing-care-and-patient-state__stop-continue-change-decision__reason-transition-and-aftercare__decision-time-and-follow-up__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces stopping or withdrawal is a recorded care transition with reason, alternative, taper or aftercare and post-transition outcomes held. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Screening is the complete pathway from an apparently healthy population through confirmation and care, including false results and overdiagnosis. Prevention, treatment, monitoring, stopping and rehabilitation retain their different timing, purpose and outcome structures.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:97a06bd91181413db5b493cf142d7e4cba057d7e9f7a566d71c6b455013381b3.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:5bdd5d9a149259e38b831dad94932811cc05c84a3e13922932b24f987448e780.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:949c9566686e19d50194e686b4f59c3a21466069585e3635613f9b1df9fbc531.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 8c344d3cbb52f1b850fa262d856ea430d1f9cde2bde3aad1a8d43bfff451d78bc.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256 : 69460571f626dacd7e10dafb0dc711b88e1617cb7ea1891c52c3eb3eafd1efa3; certificate
sha256 : 3013a8bec0652a9ee8dd4c411b22404acc60a4a930a0185536f8cda6abe8438a; external-validation
sha256 : 3a89a83c21779601aa75b3b36664f53905268bf66c6b9297a7249adbdb954734.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 3 purpose-matched discriminators with ordered identity
sha256 : e7e1ef013da606dc7319d8154cfadd9356428e8b339c4b5740a1bba855cffb4f. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- MED-WHO-SCREENING-001 - World Health Organization Regional Office for Europe;
<https://www.who.int/publications/i/item/9789289054782>; snapshot
experiments/external_sources/medicine/snapshots/MED-WHO-SCREENING-001.html;
sha256 : b56f491a97d2efbe4ff4d59e304649dd1fb1d551a2ca6b88a9bbcf43d385b5f7; scope: screening population, risk, early intervention, benefits, harms and quality assurance.
- MED-NICHD-REHABILITATION-001 - Eunice Kennedy Shriver National Institute of Child Health and Human Development;
<https://www.nichd.nih.gov/health/topics/rehabilitation-medicine>; snapshot
experiments/external_sources/medicine/snapshots/MED-NICHD-REHABILITATION-001.html;
sha256 : c1b54134e6cbcfb9542a7c012236395eb609a67a789818f201c5b42202dc8cae; scope: rehabilitation function, participation, independence, quality of life and outcomes.

Comparison record:

- sft-med-stopping-withdrawal-001-authority-correspondence: predicted medicine:ongoing-care-and-patient-state__stop-continue-change-decision__reason-transition-and-aftercare__decision-time-and-follow-up; source-derived medicine:ongoing-care-and-patient-state__stop-continue-change-decision__reason-transition-and-aftercare__decision-time-and-follow-up; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal
experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest

sha256:3bbc218bd7fe5c9dbc7d9c085daea38883afd3d400feb33279f423556d288acd; derivation seal sha256:07df3be49815637eb302a02919d1884448793fecdd21412f254545e3d09adee5; engine receipt sha256:0f8125348a367df9c09b5efc52277455b1e2c4a560cb78784ef598f16444f157 at receipts/engine/model_admitted/SFT-MED-STOPPING-WITHDRAWAL-001-0f8125348a367df9.json; empirical validation sha256:1cf1026147757e0bc8eefbfff8900fa5bcc719ecfee86e46827bf570c1b3cfd62; measurement receipt sha256:4871f3c09ede616558ff7af14f64d8241347a05db26574769a0382639f85b837; isolation sha256:1d20d3550f74684c44ee72d52b963ee77c194a090b0f1b44a02e5d0303cbe9a0; custody sha256:e45e1b9d8eaadf5c2468bb768ff66d13e0fb73a723f81cf7f22c93cf736b0353.

20. Individual Population Inference

Population evidence does not uniquely determine an individual decision. Transport, subgroup multiplicity, patient context, consequence, preference and unresolved alternatives are retained at every handoff.

61. Individual clinical inference

Claim identity: SFT-MED-INDIVIDUAL-INFERENCE-001

Question and exact theorem. Individual inference conditions only on the available patient evidence and preserves unsupported alternatives, values, context and uncertainty.

one-patient-evidence-word__evidence-to-patient-support__alternatives-and-uncertainty-held__patient-values-context-and-time__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001

- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-STOPPING-WITHDRAWAL-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	one-patient-evidence-word	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	evidence-to-patient-support	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	alternatives-and-uncertainty-held	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	patient-values-context-and-time	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: one-patient-evidence-word__evidence-to-patient-support__alternatives-and-uncertainty-held__patient-values-context-and-time__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite

te-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the clinical law; passed **true**.

Scientific meaning. A population result is evidence available to an individual inference, not the individual conclusion. Patient evidence, preferences, context, alternatives and uncertainty remain held.

Population evidence does not uniquely determine an individual decision. Transport, subgroup multiplicity, patient context, consequence, preference and unresolved alternatives are retained at every handoff.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : a9465d8323b2f64836af1dd4b79cf1e2a77fbb7f5bb3efc953563d1257d1ea89.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : af2ce43a2838cacfc0df914fa0c1111aed1c3a2cbc0d12bd84cd3ed58a8612d0.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : c3cbcfcdc4d69fb481fa6dd120179bae9dcd2c6637baaa09eedd84eb35392c17f.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : a8b8f359490e6002e6c3049457c8da7d52e2e6b8b7682560d26da74701cd3b97.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256 : 8ab3b64934245b9d7efd44f5a2368e5fccd4acf346dedc4c48af56bb7da9fe94; certificate
sha256 : b4a069fc4bb660eac3c0cc7f26acd4daf7be9b5faab55b135b56959e4b93fce9; external-validation
sha256 : a3a598a3b4d64d05919cdf4c565dda7100b009d50c747deb082434b8dc3e3fda.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 4 purpose-matched discriminators with ordered identity
sha256 : 1e9252391026d19c11cdc794c6f083d609f9f03b5fd96876d30bc5ebe8a8b9d3. Target opened after derivation seal: **true**. All rows preserved: **true**. Exact comparison: **true**. A deliberately changed observation was rejected.

Sources:

- MED-ICH-E9R1-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/E9-R1_Step4_Guideline_2019_1203.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E9R1-001.pdf;
sha256 : f7471f411f1c87ee76783d5b2b9faeca31d01d530213f71b50d136b39e3b0d9; scope: final
estimand, population, endpoint, intercurrent-event, missing-data and sensitivity-analysis distinctions.
- MED-CDC-EPIDEMIOLOGY-001 - United States Centers for Disease Control and Prevention;
https://archive.cdc.gov/www_cdc_gov/csels/dsepd/ss1978/lesson1/summary.html; snapshot
experiments/external_sources/medicine/snapshots/MED-CDC-EPIDEMIOLOGY-001.html;
sha256 : 430c2d885670c185491ab9b84d1c703eeaff0eece3ddb69fc855c001fb93478b; scope: population,
comparison, frequency, pattern, cause, disease and injury epidemiology.

Comparison record:

- sft-med-individual-inference-001-authority-correspondence: predicted medicine:one-patient-evidence-word__evidence-to-patient-support__alternatives-and-uncertainty-held__patient-values-context-and-time; source-derived medicine:one-patient-evidence-word__evidence-to-patient-support__alternatives-and-uncertainty-held__patient-values-context-and-time; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256: 4617e3c8d651fb542c03e68ef797375dffafbc3e7dd691b7b73c53ab2e7af37c; derivation seal sha256: 5bcc6db1c700a89c2abb7a44bb27796e6d7428b8bb7eb888b25f1c409537a819; engine receipt sha256: 4ff1ffffee484941acd7ec9e3fb6e4392c7d42b1a8e5b236a003e0584d6817dd3 at receipts/engine/model_admitted/SFT-MED-INDIVIDUAL-INFERENC-001-4ff1ffffee484941a.json; empirical validation sha256: 1ed54979afee4d65a940f1515cac94ba8de500329868e839a7554a287d9f58f1; measurement receipt sha256: b51859c88588b1edc091f6e02b39db70bbf303aa71dc522e1377d4d6c69444de; isolation sha256: 46fb688ca07b14ed6316297915a6c0b9d8d361b1d5b42ede8e058602497b5a05; custody sha256: 53bc630a446c196410b2354d93c99890569d56297a97f34957be95f1acadd64d.

62. Population clinical inference

Claim identity: SFT-MED-POPULATION-INFERENC-001

Question and exact theorem. Population inference requires an explicit sample-to-source relation with selection, missingness, eligibility, place and time retained.

sample-and-source-population__sample-to-population-relation__selection-weights-and-missingness__eligibility-place-time-and-method__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001

- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-INDIVIDUAL-INFERENCE-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	sample-and-source-population	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	sample-to-population-relation	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
organization	organization-collapsed	selection-weights-and-missingness	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	eligibility-place-time-and-method	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: `sample-and-source-population__sample-to-population-relation__selection-weights-and-missingness__eligibility-place-time-and-method__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule`.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces population inference requires an explicit sample-to-source relation with selection, missingness, eligibility, place and time retained. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Population evidence does not uniquely determine an individual decision. Transport, subgroup multiplicity, patient context, consequence, preference and unresolved alternatives are retained at every handoff.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:05ded0761680cbfd2c96c9f2192f0188f0098ce07de704c1f72d9c4e57187690`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:af89e01eac9975ff1e76ce67d8ea341282b9e3f2807b00e35d29d087f39d1e1b`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:b93cd716d57af21c2549c727a129f4cd7b0d48b40aee0698cee8253866e24f5a.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:9bcb4df02787b1063dbf8d7af2ded1dbeb2e657df0085730eef5741528af67a.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:84d7658468bce213c4d36724425e6fce2c409cb5bcfdca03dee7549977044226; certificate
sha256:07e23cc5e842d4074f6f785dcfba13aee881edfeb1fba441cdfbe5b2b4818f0d; external-validation
sha256:3c2cef4debcf0973e081bb88be716c5c12678036a331903d5f08bec9d6681605.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 4 purpose-matched discriminators with ordered identity

sha256:1e9252391026d19c11cdc794c6f083d609f9f03b5fd96876d30bc5ebe8a8b9d3. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- MED-ICH-E9R1-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/E9-R1_Step4_Guideline_2019_1203.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E9R1-001.pdf;
sha256:f7471f411f1c87ee76783d5b2b9faeeca31d01d530213f71b50d136b39e3b0d9; scope: final
estimand, population, endpoint, intercurrent-event, missing-data and sensitivity-analysis distinctions.
- MED-CDC-EPIDEMIOLOGY-001 - United States Centers for Disease Control and Prevention;
https://archive.cdc.gov/www_cdcgov/csels/dsepd/ss1978/lesson1/summary.html; snapshot
experiments/external_sources/medicine/snapshots/MED-CDC-EPIDEMIOLOGY-001.html;
sha256:430c2d885670c185491ab9b84d1c703eeaff0eece3ddb69fc855c001fb93478b; scope: population,
comparison, frequency, pattern, cause, disease and injury epidemiology.

Comparison record:

- sft-med-population-inference-001-authority-correspondence: predicted medicine:sample-and-source-population__sample-to-population-relation__selection-weights-and-missingness__eligibility-place-time-and-method; source-derived medicine:sample-and-source-population__sample-to-population-relation__selection-weights-and-missingness__eligibility-place-time-and-method;
exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:ab0b8c547935d39447daf9125101c0fdd1ff5a2a52e5666261db562bd3f8e7d2; derivation seal sha256:75c8aff1dc6c3de5249feb1ee88a9b1da75b48ca3bbfea215b8d597f44fa8024; engine receipt sha256:b289ddf62ddadfe5bafb3b725fd180d9c22a97175463d0d68d03973382c747da at receipts/engine/model_admitted/SFT-MED-POPULATION-INFERENCE-001-b289ddf62ddadfe5.json; empirical validation sha256:e99a86333a5791ba741d541ef7688fcad8f40a5049a5000eacb9aaf4f5b654bb; measurement receipt sha256:1fa63ec8edea91034eae1d549eb58b32eb1c7ae603358282c6d35e8d5d44fb36; isolation sha256:3d9200d5d85edac1b14f8ae5aca515ddb24885c75031430a2ffb92ca3361897a; custody sha256:3d6c7c4342ea41054cad623e9480b578e478f0ef8acffc01fca49df8d38da91c.

63. Clinical transportability

Claim identity: SFT-MED-TRANSPORTABILITY-001

Question and exact theorem. Transportability requires explicit source-target comparison and may carry only relations invariant under every retained population and setting difference.

source-and-target-populations__invariant-and-different-features__joint-population-treatment-outcome__setting-time-and-care-system__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001

- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-POPULATION-INFERENCE-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	source-and-target-populations	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	invariant-and-different-features	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	joint-population-treatment-outcome	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	setting-time-and-care-system	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: source-and-target-populations__invariant-and-different-features__joint-population-treatment-outcome__setting-time-and-care-system__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier`: clinical carrier is explicit and nonempty; passed `true`.
- `relation`: relation and organization remain separately reconstructible; passed `true`.
- `boundary`: observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces transportability requires explicit source-target comparison and may carry only relations invariant under every retained population and setting difference. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Population evidence does not uniquely determine an individual decision. Transport, subgroup multiplicity, patient context, consequence, preference and unresolved alternatives are retained at every handoff.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 55aab970a59fc72d506b60435f05b37ab5583285f8231c0fb8683ee76377fecb.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: f7aeed2dcc8b99096c01f22fa2ef59ff652fbc6ba0949aa845d364f9b2106cef.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: ebc8eacf03fcacf011c7075820da7a3a160318e8765d0e42e0f4d5f092c21cb3b.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 12ae75a16f3365d6d2648add4065ecb8fa43bfde0bc60cb1421dda2e5b570fca.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256: e14defcf1f3a6b1589e10cf5f714ba412189f430c6495d5696c3dc34a5513ddb; certificate
sha256: c2961f441a30b497ce714d4ab148cc4bf3ca6c82964ca30686cda59710550bce; external-validation
sha256: 230b420b94a6fb320f608053a263261000e94193db92ea9faaf0ed3e3ca8e4e6.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 4 purpose-matched discriminators with ordered identity

sha256: 1e9252391026d19c11cdc794c6f083d609f9f03b5fd96876d30bc5ebe8a8b9d3. Target opened after derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- MED-ICH-E9R1-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/E9-R1_Step4_Guideline_2019_1203.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E9R1-001.pdf;
sha256: f7471f411f1c87ee76783d5b2b9faeca31d01d530213f71b50d136b39e3b0d9; scope: final
estimand, population, endpoint, intercurrent-event, missing-data and sensitivity-analysis distinctions.
- MED-CDC-EPIDEMIOLOGY-001 - United States Centers for Disease Control and Prevention;
https://archive.cdc.gov/www_cdc_gov/csels/dsepd/ss1978/lesson1/summary.html; snapshot
experiments/external_sources/medicine/snapshots/MED-CDC-EPIDEMIOLOGY-001.html;
sha256: 430c2d885670c185491ab9b84d1c703eeaff0eece3ddb69fc855c001fb93478b; scope: population,
comparison, frequency, pattern, cause, disease and injury epidemiology.

Comparison record:

- sft-med-transportability-001-authority-correspondence: predicted medicine:source-and-target-populations__invariant-and-different-features__joint-population-treatment-outcome__setting-time-and-care-system; source-derived medicine:source-and-target-populations__invariant-and-different-features__joint-population-treatment-outcome__setting-time-and-care-system; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:18d06120b280e33a354edba87b0c753b6bbbc508122c07346161325f20e1e435; derivation seal sha256:6dfd0e5aac439c85a045c6378b8f88497387c28c4e6e14a36016e75522477e1e; engine receipt sha256:99f4642cbb83fb23bf102d691842d4df867e8df7b68a394f166fa040ebaf44f8 at receipts/engine/model_admitted/SFT-MED-TRANSPORTABILITY-001-99f4642cbb83fb23.json; empirical validation sha256:4da65d574bc816fa6296c2ec533b35f0fca9cb64271fd3b72fec3c304cf59a0f; measurement receipt sha256:318f7f2d436717c7ad2ed84228850cd31ecf48603d03935a249aa7cc7007c082; isolation sha256:3ca97ccb74f5943e328eb0e615f7bc35ff53b215a6102054c7cbca0bdf02ffba; custody sha256:11c9d9ec955acb00207f90f46d6a4c3dd501ad8dac053797b658ce71a3c8d8be.

64. Subgroups and multiplicity

Claim identity: SFT-MED-SUBGROUP-MULTIPLICITY-001

Question and exact theorem. Every generated subgroup and outcome comparison remains in one complete family so selection after outcomes cannot masquerade as prior inference.

generated-subgroup-comparisons__comparison-count-and-result-incidence__complete-tested-family__predeclaration-and-selection-boundary__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001

- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-TRANSPORTABILITY-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	generated-subgroup-comparisons	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	comparison-count-and-result-incidence	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
organization	organization-collapsed	complete-tested-family	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	predeclaration-and-selection-boundary	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: generated-subgroup-comparisons__comparison-count-and-result-incidence__complete-tested-family__predeclaration-and-selection-boundary__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces every generated subgroup and outcome comparison remains in one complete family so selection after outcomes cannot masquerade as prior inference. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Population evidence does not uniquely determine an individual decision. Transport, subgroup multiplicity, patient context, consequence, preference and unresolved alternatives are retained at every handoff.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:e6cba47a7e4882512f3915c5e8e5c9cfdd0d4f8e9884da8de01dadf0fc0483.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:ecee1e5cdd7fee0d0d6371a4e95fe69e510d8018a1b6a2be4e08be399cfb3bed.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:c7698b16c3de2a3e976657361eed0ec13d037a6f7ded5d58fb0cbba10578a500.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:52e592f869f531d75362ea3e1546783467403dff85b78db81c18632fc8fd31e9.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:fa26090c806799002469bee2df197e18bc9215f796f577c756eda8f3ffc4c104; certificate
sha256:cfbcb264024a5b88f879bbf12a73fd93c223960cc5181d777fb1d47aa90695295; external-validation
sha256:c9ca2be6f046c81932c28f9a527347d97774f9218609d3aeca5739808daa6013.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 4 purpose-matched discriminators with ordered identity

sha256:1e9252391026d19c11cdc794c6f083d609f9f03b5fd96876d30bc5ebe8a8b9d3. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- MED-ICH-E9R1-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/E9-R1_Step4_Guideline_2019_1203.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E9R1-001.pdf;
sha256:f7471f411f1c87ee76783d5b2b9faeca31d01d530213f71b50d136b39e3b0d9; scope: final
estimand, population, endpoint, intercurrent-event, missing-data and sensitivity-analysis distinctions.
- MED-CDC-EPIDEMIOLOGY-001 - United States Centers for Disease Control and Prevention;
https://archive.cdc.gov/www_cdc_gov/csels/dsepd/ss1978/lesson1/summary.html; snapshot
experiments/external_sources/medicine/snapshots/MED-CDC-EPIDEMIOLOGY-001.html;
sha256:430c2d885670c185491ab9b84d1c703eeaff0eece3ddb69fc855c001fb93478b; scope: population,
comparison, frequency, pattern, cause, disease and injury epidemiology.

Comparison record:

- sft-med-subgroup-multiplicity-001-authority-correspondence: predicted medicine:generated-subgroup-comparisons__comparison-count-and-result-incidence__complete-tested-family__predeclaration-and-selection-boundary; source-derived medicine:generated-subgroup-comparisons__comparison-count-and-result-incidence__complete-tested-family__predeclaration-and-selection-boundary; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:3764491ca11ca1b6688eed4cca037d40da22a8ccd2938a2a0d62a5b485b56c98; derivation seal sha256:abb4b9f332940d526e8533a3377d53af2d265ebcdd8718b5a65d3d2247be68fb; engine receipt sha256:eb9b3ff70f470dbdcf485e20f6c0924063e0af4f2ec14cc315d7caf442303679 at receipts/engine/model_admitted/SFT-MED-SUBGROUP-MULTIPLICITY-001-eb9b3ff70f470dbd.json; empirical validation sha256:8cc4266d35525e4bea40f70f683d7200f0bcaa06fe253a8c40c25eebe514d00f; measurement receipt sha256:b03d6e780a369c186687bb58fcd2d2141986885aefca7267d92c2a73b9fa5c5; isolation sha256:3b0f46403ea70e19a6c437857238f8d9c4c1d35f87ea864693d2ded9632befc4; custody sha256:fceb2898055c409b1dd88eb9aa8bbe83b58d2644bc2736948bc217e08c1caaf1.

65. Clinical decision boundary

Claim identity: SFT-MED-CLINICAL-DECISION-001

Question and exact theorem. A clinical decision composes evidence with explicit actions, consequences and patient values; a population average cannot uniquely select an individual action.

patient-action-alternatives__evidence-consequence-preference-relation__all-actions-outcomes-and-values__decision-maker-time-and-resource-boundary__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001

- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-SUBGROUP-MULTIPLICITY-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	patient-action-alternatives	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	evidence-consequence-preference-relation	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	all-actions-outcomes-and-values	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	decision-maker-time-and-resource-boundary	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: patient-action-alternatives__evidence-consequence-preference-relation__all-actions-outcomes-and-values__decision-maker-time-and-resource-boundary__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier`: clinical carrier is explicit and nonempty; passed `true`.
- `relation`: relation and organization remain separately reconstructible; passed `true`.
- `boundary`: observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces a clinical decision composes evidence with explicit actions, consequences and patient values; a population average cannot uniquely select an individual action. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Population evidence does not uniquely determine an individual decision. Transport, subgroup multiplicity, patient context, consequence, preference and unresolved alternatives are retained at every handoff.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 4949b683f9d2b39453a59df517d27530f042f7e66de2fef5c9ae0ca8affa2f70.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: c36b4ab322f70ecc2ab58543bd76f16b8b90b8d846a15f33583e661810458519.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 80b56a982bd0e8f6a12370ce30ff0f3816dafa6446aa7014c31b51565ba6dbd2.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 5d954137e770bb83133ac5f99c2c0d53666d16743b619da3d7e81eaf3d83008b.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256: ab6232be206adbdfc3661c5323198e9aa209ad9bdd36cd8f3843b847a0ef6261; certificate
sha256: 59c9ff0e4168de11c122c0982ce85202e9508d1a6e7baa881e18e33d0b5bd544; external-validation
sha256: d5135b571c602ada0db8258d4bbd2197046f6edfaa53e5cf1c0c45847e3a0c1c.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 4 purpose-matched discriminators with ordered identity

sha256: 1e9252391026d19c11cdc794c6f083d609f9f03b5fd96876d30bc5ebe8a8b9d3. Target opened after derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- MED-ICH-E9R1-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/E9-R1_Step4_Guideline_2019_1203.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E9R1-001.pdf;
sha256: f7471f411f1c87ee76783d5b2b9faeca31d01d530213f71b50d136b39e3b0d9; scope: final
estimand, population, endpoint, intercurrent-event, missing-data and sensitivity-analysis distinctions.
- MED-CDC-EPIDEMIOLOGY-001 - United States Centers for Disease Control and Prevention;
https://archive.cdc.gov/www_cdc_gov/csels/dsepd/ss1978/lesson1/summary.html; snapshot
experiments/external_sources/medicine/snapshots/MED-CDC-EPIDEMIOLOGY-001.html;
sha256: 430c2d885670c185491ab9b84d1c703eeaff0eece3ddb69fc855c001fb93478b; scope: population,
comparison, frequency, pattern, cause, disease and injury epidemiology.

Comparison record:

- sft-med-clinical-decision-001-authority-correspondence: predicted medicine:patient-action-alternatives__evidence-consequence-preference-relation__all-actions-outcomes-and-values__decision-maker-time-and-resource-boundary; source-derived medicine:patient-action-alternatives__evidence-consequence-preference-relation__all-actions-outcomes-and-values__decision-maker-time-and-resource-boundary; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:72ac8f79eab561599b21ddbb69f10a0e39f83ed8d08cc64378424a3e2e153c9d; derivation seal sha256:445767e369e3bd36bd14ef0ada73e4d99f443f54d09fde40e22bcc2a056fd52a; engine receipt sha256:10acbaa77b0fff08e501c884b6b91ac42f3a7883868308ffa2fa7c62979411d4 at receipts/engine/model_admitted/SFT-MED-CLINICAL-DECISION-001-10acbaa77b0fff08.json; empirical validation sha256:2217cef8bdf9a2afbc2f9ee159ac66e0b906c4f3a2e6afb0bc494959309fd2d4; measurement receipt sha256:3a3d90c5eb70709de3dfd212b026a8ffac52d1120aa507cad2c08b2512c21e68; isolation sha256:9b85699bc094c6094addc69257445675525cd440d98d6faf087833f97472a669; custody sha256:6a8f796c93b73db205472e2b35a7951f6ba722b62401b43f9bb3a0a8b984b0e9.

66. Clinical uncertainty

Claim identity: SFT-MED-CLINICAL-UNCERTAINTY-001

Question and exact theorem. Clinical uncertainty is the exact retained support after available evidence, including known missingness and unresolved alternatives, not an ontological nondeterminism claim.

retained-admissible-clinical-states__evidence-conditioned-support__known-missing-and-unknown-record__decision-and-observation-boundary__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001

- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-CLINICAL-DECISION-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	retained-admissible-clinical-states	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	evidence-conditioned-support	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
organization	organization-collapsed	known-missing-and-unknown-record	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	decision-and-observation-boundary	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: `retained-admissible-clinical-states__evidence-conditioned-support__known-missing-and-unknown-record__decision-and-observation-boundary__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule`.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces clinical uncertainty is the exact retained support after available evidence, including known missingness and unresolved alternatives, not an ontological nondeterminism claim. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Population evidence does not uniquely determine an individual decision. Transport, subgroup multiplicity, patient context, consequence, preference and unresolved alternatives are retained at every handoff.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:5ed333777ca19a10d1c403ecef849d211e57ee1e66b920cded26a3e1b1265c`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:461791ff003f5d0c6353a7ad41ce9b85e03304699a82a6b49e978dedb08bcd1f`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:05deabb051984a9654b173791569be94d13270dee198a18fca1220e818b275c9.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:f1a9bbce75e7caf9018eda8c07b1e12113767d0524e64bc8efe0cf1fe690db30.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:4b77f67c268c19502cca4aabb2c96fc2e8f1c061d2dbd6e445aaef679471828f; certificate
sha256:63af165b5fcefb583e58ed756aafb6eadbfb9e7d2764f0910c4831071c5a39c; external-validation
sha256:a7138b0f5c435b545aab083a6d89eb250aa311955bec4a6351a0111ed84c76cd.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 4 purpose-matched discriminators with ordered identity

sha256:1e9252391026d19c11cdc794c6f083d609f9f03b5fd96876d30bc5ebe8a8b9d3. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- MED-ICH-E9R1-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/E9-R1_Step4_Guideline_2019_1203.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E9R1-001.pdf;
sha256:f7471f411f1c87ee76783d5b2b9faeeca31d01d530213f71b50d136b39e3b0d9; scope: final
estimand, population, endpoint, intercurrent-event, missing-data and sensitivity-analysis distinctions.
- MED-CDC-EPIDEMIOLGY-001 - United States Centers for Disease Control and Prevention;
https://archive.cdc.gov/www_cdcgov/csels/dsepd/ss1978/lesson1/summary.html; snapshot
experiments/external_sources/medicine/snapshots/MED-CDC-EPIDEMIOLGY-001.html;
sha256:430c2d885670c185491ab9b84d1c703eeaff0eece3ddb69fc855c001fb93478b; scope: population,
comparison, frequency, pattern, cause, disease and injury epidemiology.

Comparison record:

- sft-med-clinical-uncertainty-001-authority-correspondence: predicted medicine:retained-admissible-clinical-states__evidence-conditioned-support__known-missing-and-unknown-record__decision-and-observation-boundary; source-derived medicine:retained-admissible-clinical-states__evidence-conditioned-support__known-missing-and-unknown-record__decision-and-observation-boundary; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal
experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:ad5cc5c446583ea8df9664a0f840b09a86459f77bb56cfa804e0c170eb2b28e5; derivation seal sha256:ad8d3ce70d45fef7d7c00253b480bdd6623e21b1e4ef74666e8ee7a7a365104f; engine receipt sha256:c86b7c45809b6536ad297ecd36cd5ea0f33cbdf639da79841e6694e96c71eb66 at receipts/engine/model_admitted/SFT-MED-CLINICAL-UNCERTAINTY-001-c86b7c45809b6536.json; empirical validation sha256:d1600cc6d4dac5ff1f3e901fa2231ff3b8fb8c983914479ac89b39500497b1e6; measurement receipt sha256:084295e1a86198c1a4ef06bba4b957db4eac1e44ad241cf67e43fbfadc1f7c00; isolation sha256:d1d7f1a78ba1ceb8ee0ef4f58b0acf34834325c1fab4a5cfe7381d272d46e4f0; custody sha256:fc34bd6b4de77aefc8bccd0f3929d58a077fef527c1286fc47f266ad0d4df645.

21. Consent Ethics Privacy Uncertainty

The clinical evidence system cannot erase the person who bears its risk. Consent, capacity, privacy, minimization, equipoise, stopping and reviewer custody are operational admission boundaries with complete records, not decorative ethics prose.

67. Informed consent boundary

Claim identity: SFT-MED-INFORMED-CONSENT-001

Question and exact theorem. A consent record is valid only for its declared person, information, understanding, voluntariness, purpose, scope and time, with refusal and withdrawal retained.

person-information-choice-record__understanding-voluntariness-authorization__versioned-scope-and-withdrawal__purpose-time-capacity-and-recorder__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001

- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-CLINICAL-UNCERTAINTY-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	person-information-choice-record	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	understanding-voluntariness-authorization	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	versioned-scope-and-withdrawal	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	purpose-time-capacity-and-recorder	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: person-information-choice-record__understanding-voluntariness-authorization__versioned-scope-and-withdrawal__purpose-time-capacity-and-recorder__complete-protocol-state-transition-adverse-null-record__root-bound-forward-folding__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier: clinical carrier is explicit and nonempty; passed true.
- relation: relation and organization remain separately reconstructible; passed true.
- boundary: observation boundary cannot silently become the clinical law; passed true.

Scientific meaning. Consent requires information, understanding, voluntariness, authorization, purpose, scope and time. Refusal and withdrawal are first-class outcomes; a signed field alone cannot substitute for the process.

The clinical evidence system cannot erase the person who bears its risk. Consent, capacity, privacy, minimization, equipoise, stopping and reviewer custody are operational admission boundaries with complete records, not decorative ethics prose.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:eca857ac7ecc1751e8beaae2e71d8e8fe8b3ee6eb66985744ab2b27a6a9baf59.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:e24fc0cb7da7168d23334a909e64cebe74d998c8b2d1bde947898d18dffcefacc.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:487bb2baf924a9ca0f63fb5aa0f12747b98802556ef16f010ade3bd3dddc47d6.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:34c73be10c0f5e1515a2d5d3ddb20c77084c714a6ec4cb093549476c11b529ee.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:10d8f6642412d26f8e07f12c016523244976559801953d3207e94e7d96e3fb40; certificate sha256:dd328342b9ebd038d6eae21dea45c17a108807d502cc34c26b55bdc8a078210f; external-validation sha256:d43329d93c78fe17430ec4f3da7426299ca5779d08db95ba86679ae4e51b83d5.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 4 purpose-matched discriminators with ordered identity

sha256:df62e80765b3e9eb1f3e41ecdbe47e1203d4585e7c5ed7dfe549fcde25b96ff9. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- MED-ICH-E6R3-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf; snapshot experiments/external_sources/medicine/snapshots/MED-ICH-E6R3-001.pdf;
sha256:e6ce19e36ce7d2e294f89ee89492b9e035178c3cca48984392bbd92eec9b002c; scope: final Good Clinical Practice: rights, safety, wellbeing, consent, protocol, allocation, blinding, adverse events, confidentiality and essential records.
- MED-COCHRANE-MISSING-EVIDENCE-001 - Cochrane;
<https://www.cochrane.org/authors/handbooks-and-manuals/handbook/current/chapter-13>; snapshot experiments/external

`l_sources/medicine/snapshots/MED-COCHRANE-MISSING-EVIDENCE-001.html;`
`sha256:9ebd35b49c689f1b12fa3919051ea9a5ed315b168d9eac7b1eb8860317d07bb;` scope: selective dissemination, missing evidence, benefit overestimation and harm underestimation.

Comparison record:

- `sft-med-informed-consent-001-authority-correspondence: predicted medicine:person-information-choice-record__understanding-voluntariness-authorization__versioned-scope-and-withdrawal__purpose-time-capacity-and-recorder;` source-derived medicine:person-information-choice-record__understanding-voluntariness-authorization__versioned-scope-and-withdrawal__purpose-time-capacity-and-recorder; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

`experiments/sealed_predictions/medicine_foundation_complete_pre_source.json;` source manifest `sha256:07ca9c27f036c14b3dbbfff90337f1a0609cc31d54de3ae923cdba226794c3092;` derivation seal `sha256:4ab8b1fa8d05b22469c14285699d4be3712870a33c81abc675130b679c915174;` engine receipt `sha256:39a8cd25cdabbff7d3f96ed129ad05b4f2ecd91b89ad4bd6a8c0bc2352b7d407` at `receipts/engine/model_admitted/SFT-MED-INFORMED-CONSENT-001-39a8cd25cdabbff7.json;` empirical validation `sha256:fc4121ed8ffe52d52626935fc2526e669e5d2aa6b017ec4daf308a9bfd54e734;` measurement receipt `sha256:8a6214011187f5d30258416c366243947859e2e060b1b96418643472ef72c80c;` isolation `sha256:a662082336e462d84a34e79c8b82d9dd878e71e587e02e95661518dc2aba49e9;` custody `sha256:b034b58bba367abf433542fe1e7afb20a8f4eeca9e275a94b49b2ab86ceb4f0.`

68. Decision capacity boundary

Claim identity: SFT-MED-DECISION-CAPACITY-001

Question and exact theorem. Capacity is decision- and time-specific evidence over retained abilities and assistance; diagnosis or status alone cannot erase the person's choice.

`person-decision-context__understand-retain-weigh-communicate__decision-specific-support-and-assistance__time-decision-and-assessment-method__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule`

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-INFORMED-CONSENT-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	person-decision-context	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	understand-retain-weigh-communicate	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	decision-specific-support-and-assistance	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	time-decision-and-assessment-method	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: `person-decision-context__understand-retain-weigh-communicate__decision-specific-support-and-assistance__time-decision-and-assessment-method__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule`.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces capacity is decision- and time-specific evidence over retained abilities and assistance; diagnosis or status alone cannot erase the person's choice. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

The clinical evidence system cannot erase the person who bears its risk. Consent, capacity, privacy, minimization, equipoise, stopping and reviewer custody are operational admission boundaries with complete records, not decorative ethics prose.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:1fae46f14e48d37d68a0fcceab1950a3f3e37d1993dcad34903fab93c0181239.

- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:884cf3a0c33dc51820a3934d95bd0b2cc8618eb4ed9551f7c3e61e479dfacd73.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:9e213c0db858f0bcf8e2f4aed8844ecbe60df83b731c69437a933e89d1236082.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:b1127db9e5dcc79a2e8a2fcc0c997fbafa6fbd5213253c52ad7fd55405030b62.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:6df0352da58ed26e22f966f594f0c5836e6b957c049ce8b22cc4d67b8ca954d6; certificate

sha256:f1e2d260159fc2928b5fb6ce59c07d4b9ee802e5ac4ba71c475bda9ad40d8915; external-validation

sha256:60eb2cb43784258500ddc701d1d42a106a8ef2afe759c043684ab5fcb19544c1.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 4 purpose-matched discriminators with ordered identity

sha256:df62e80765b3e9eb1f3e41ecdbe47e1203d4585e7c5ed7dfe549fcde25b96ff9. Target opened after

derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- MED-ICH-E6R3-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf; snapshot experiments/external_sources/medicine/snapshots/MED-ICH-E6R3-001.pdf;
sha256:e6ce19e36ce7d2e294f89ee89492b9e035178c3cca48984392bbd92eec9b002c; scope: final Good Clinical Practice: rights, safety, wellbeing, consent, protocol, allocation, blinding, adverse events, confidentiality and essential records.
- MED-COCHRANE-MISSING-EVIDENCE-001 - Cochrane;
<https://www.cochrane.org/authors/handbooks-and-manuals/handbook/current/chapter-13>; snapshot experiments/external_sources/medicine/snapshots/MED-COCHRANE-MISSING-EVIDENCE-001.html;
sha256:9ebd35b49c689f1b12fa3919051ea9a5ed315b168d9eaec7b1eb8860317d07bb; scope: selective dissemination, missing evidence, benefit overestimation and harm underestimation.

Comparison record:

- sft-med-decision-capacity-001-authority-correspondence: predicted medicine:person-decision-context__understand-retain-weigh-communicate__decision-specific-support-and-assistance__time-decision-and-assessment-method; source-derived medicine:person-decision-context__understand-retain-weigh-communicate__decision-specific-support-and-assistance__time-decision-and-assessment-method; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff

- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:79610d025ce895605e22833e8fbec1ee29fa71ecaeb6964c5d98377126f4bf45; derivation seal sha256:9926b0e581f7fc8dcab25f9f8df7dc22d2c2c56c282899e7bfeb53a6f5b857b5; engine receipt sha256:9de84a984e7b1bf1105c2ce95f7b363d79b21901a177fb9605e1c1a71004320f at receipts/engine/model_admitted/SFT-MED-DECISION-CAPACITY-001-9de84a984e7b1bf1.json; empirical validation sha256:ece407f31b829e0c9439a127db9006ebdd76410d74511e94c39379cf51073ab7; measurement receipt sha256:81669e08d1fdc7df9789bda8f0c36f2adcac52afa4a0e0b3eddcdf745c9191d; isolation sha256:ec20d9d082a06156213d9802f605c2872cf3f4e1360a9a51a756c9e453350167; custody sha256:d4795ba08460b1fe4e1b25eb4cd9b83b2352f11e8fa912b2819768b40e655abb.

69. Clinical privacy

Claim identity: SFT-MED-CLINICAL-PRIVACY-001

Question and exact theorem. Clinical privacy requires purpose-bound access, minimized disclosure and a complete identity-content-recipient-time trace; public evidence may not expose an unnecessary person link.

person-linked-health-record__purpose-bound-access-and-disclosure__identity-content-recipient-trace__consent-law-purpose-and-time__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001

- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-DECISION-CAPACITY-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	person-linked-health-record	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	purpose-bound-access-and-disclosure	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	identity-content-recipient-trace	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	consent-law-purpose-and-time	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
extension	free-fit-exception -opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: person-linked-health-record__purpose-bound-access-and-disclosure__identity-content-recipient-trace__consent-law-purpose-and-time__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. Privacy is purpose-bound access and disclosure with complete identity, content, recipient and time custody. The public proof can be maximally transparent without exposing an unnecessary patient link.

The clinical evidence system cannot erase the person who bears its risk. Consent, capacity, privacy, minimization, equipoise, stopping and reviewer custody are operational admission boundaries with complete records, not decorative ethics prose.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:5957518d7fe377a73a5783ef1757610a2c0e3d9bd614604591d44f5341ff7429.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:6df4a94b47278efcaf9e217db3b4cd4d4f9a90f06f5fde8ed6d46249c19b52c0.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:a6f54c3e1915d1dd932c7c98f7c643dae2658aa830be421b7124b52ea09b45b4.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:2e2f6b958db7278bc3cdc732f17ebe7c13e4b61ebdc5640e20f1ddac43a2a5ca.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:110ebe9fde8c8db1ed56dfb9716640ccb3e03a6d17aaf18445fea5c32c3cd7b3; certificate
sha256:8e2efe89bc27c468ce6790979b8be99d9290d8577aeba5367038c783fbf3a747; external-validation
sha256:840ed21e821cf11093c44b1fccde019665ef893971136e4b2960f047128ce674.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 4 purpose-matched discriminators with ordered identity

sha256:df62e80765b3e9eb1f3e41ecdbe47e1203d4585e7c5ed7dfe549fcde25b96ff9. Target opened after derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- MED-ICH-E6R3-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E6R3-001.pdf;

sha256:e6ce19e36ce7d2e294f89ee89492b9e035178c3cca48984392bbd92eec9b002c; scope: final Good Clinical Practice: rights, safety, wellbeing, consent, protocol, allocation, blinding, adverse events, confidentiality and essential records.

- MED-COCHRANE-MISSING-EVIDENCE-001 - Cochrane;
<https://www.cochrane.org/authors/handbooks-and-manuals/handbook/current/chapter-13>; snapshot experiments/external_sources/medicine/snapshots/MED-COCHRANE-MISSING-EVIDENCE-001.html;
 sha256:9ebd35b49c689f1b12fa3919051ea9a5ed315b168d9eaec7b1eb8860317d07bb; scope: selective dissemination, missing evidence, benefit overestimation and harm underestimation.

Comparison record:

- sft-med-clinical-privacy-001-authority-correspondence: predicted medicine:person-linked-health-record__purpose-bound-access-and-disclosure__identity-content-recipient-trace__consent-law-purpose-and-time; source-derived medicine:person-linked-health-record__purpose-bound-access-and-disclosure__identity-content-recipient-trace__consent-law-purpose-and-time; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:e0cc0fe7e1144c20a8f0648112863b8d5125a116db10cfad007dc94faba579f2; derivation seal sha256:d6b593eca9a08c3ebd3ad8536d95568891a600dc9c01a1f71a8b208c5c2a177b; engine receipt sha256:6780db2d12f9c570c9ee4cf2494ed173ba656f9731020f992d4971e517dcfb53 at receipts/engine/model_admitted/SFT-MED-CLINICAL-PRIVACY-001-6780db2d12f9c570.json; empirical validation sha256:0e8a7d5e55db9b59a00af22cc0019822d5af4c85d6cd52142ac75f776d8cf1be; measurement receipt sha256:fc832541b3d9a4e331b47acf3d0d010fadfa586430c3f118c14c059e84feefeb; isolation sha256:c6b47da1b78f08c8925f9c63a879acdca935f59c0db2475f1734c50d85e37b40; custody sha256:930c85658dde40a2ae796d44a9fd024334e3dd4f27d7b76783de757402558d8c.

70. Clinical data minimization

Claim identity: SFT-MED-DATA-MINIMIZATION-001

Question and exact theorem. A clinical dataset may retain only fields forced by its declared purpose while preserving an auditable record of exclusion, derivation, access and retention.

purpose-declared-data-fields__field-necessity-to-purpose__included-excluded-and-derived-fields__protocol-access-and-retention-period__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-CLINICAL-PRIVACY-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	purpose-declared-data-fields	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	field-necessity-to-purpose	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	included-excluded-and-derived-fields	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	protocol-access-and-retention-period	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: purpose-declared-data-fields__field-necessity-to-purpose__included-excluded-and-derived-fields__protocol-access-and-retention-period__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the clinical law; passed **true**.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces a clinical dataset may retain only fields forced by its declared purpose while preserving an auditable record of exclusion, derivation, access and retention. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

The clinical evidence system cannot erase the person who bears its risk. Consent, capacity, privacy, minimization, equipoise, stopping and reviewer custody are operational admission boundaries with complete records, not decorative ethics prose.

Adverse controls.

- **false_premise**: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:d9bb8bf1fcd08164be1e6302a85ff7850f451d05ee95e8cc38dd607888ac1b97.
- **tampered_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:c0d848c9cb3ac2ebfa5bfacbd3e8874006d2f1137153d32b0661acdc6df22ea3.
- **tampered_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:0a3dd92a432ee66847c6571ccdea442df367946dbe0594f163d318347b7fe98d.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:ccdaef61373aef30ffe74885da3c7a0033efe033fd94e06a7b7d33f73b40635a.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:3f777f0dbf5ceaf9e2fc5976c62d4114502aafeaa5ae51225b2493aebce6b411; certificate
sha256:bb1c305d3efcf93f7d79662a3a471667a4aaf31fecb7cb25c4b3be82be52781c; external-validation
sha256:1c58b58a66dd97c420abb2b04c69949d7a15e20ee53e49f5066c35d95574bcfb.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 4 purpose-matched discriminators with ordered identity
sha256:df62e80765b3e9eb1f3e41ecdbe47e1203d4585e7c5ed7dfe549fcde25b96ff9. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- MED-ICH-E6R3-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E6R3-001.pdf;
sha256:e6ce19e36ce7d2e294f89ee89492b9e035178c3cca48984392bbd92eec9b002c; scope: final Good Clinical Practice: rights, safety, wellbeing, consent, protocol, allocation, blinding, adverse events, confidentiality and essential records.
- MED-COCHRANE-MISSING-EVIDENCE-001 - Cochrane;
<https://www.cochrane.org/authors/handbooks-and-manuals/handbook/current/chapter-13>; snapshot experiments/external_sources/medicine/snapshots/MED-COCHRANE-MISSING-EVIDENCE-001.html;
sha256:9ebd35b49c689f1b12fa3919051ea9a5ed315b168d9eaec7b1eb8860317d07bb; scope: selective dissemination, missing evidence, benefit overestimation and harm underestimation.

Comparison record:

- sft-med-data-minimization-001-authority-correspondence: predicted medicine:purpose-declared-data-fields__field-necessity-to-purpose__included-excluded-and-derived-fields__protocol-access-and-retention-period; source-derived medicine:purpose-declared-data-fields__field-necessity-to-purpose__included-excluded-and-derived-fields__protocol-access-and-retention-period; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal
experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest
sha256:c867d30061f449ec83dc9be79df91b0fca5e8ebb433f46a33a7d8726967f61bd; derivation seal
sha256:fcf4eb64c9a4843b676f2cf070aa5dc8c8b8be1b76c30712776f145ced6054c2; engine receipt
sha256:dbcd403cde9af573b23fca628854e07ffa7483be141ed4274e9ac4ac281b86d8 at
receipts/engine/model_admitted/SFT-MED-DATA-MINIMIZATION-001-dbcd403cde9af573.json;
empirical validation sha256:e373850017d2b1e3d03f86027224af92173e664eca5ae31428927286865fed49;
measurement receipt sha256:e5abae35b9fd8a4223753d6de770adb8d3dc1459433f4545b0c3ecaca7b9be16;
isolation sha256:be97a0721cc5cecdc09f889ee64189a5cfa763ce6ff34251779b8fd59e5cd7e6; custody
sha256:d378efb24ed353190c6021f7fa45f2a1a1ff5ccb9d713f491906b8430851d94c.

71. Clinical equipoise and stopping boundary

Claim identity: SFT-MED-CLINICAL-EQUIPOISE-001

Question and exact theorem. A comparative study remains admissible only while its declared evidence support leaves the care alternatives unresolved and its harm-stopping rules remain active.

competing-care-support__unresolved-comparative-benefit-harm__monitoring-and-stop
ping-rules__evidence-date-population-and-oversight__complete-protocol-state-tran
sition-adverse-null-record__root-bound-forward-forcing__positive-finite-successo
r-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001

- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001
- SFT-MED-DATA-MINIMIZATION-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	competing-care-support	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	unresolved-comparative-benefit-harm	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	monitoring-and-stopping-rules	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	evidence-date-population-and-oversight	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
generality	single-patient-fav orable-study	positive-finite-successor -and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception -opaque-model-or-e xtra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: competing-care-support__unresolved-comparative-benefit-harm__monitoring-and-stopping-rules__evidence-date-population-and-oversight__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The result is an exact relational classification at the declared clinical boundary. It forces a comparative study remains admissible only while its declared evidence support leaves the care alternatives unresolved and its harm-stopping rules remain active. Patient-, population-, method-, setting- and time-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

The clinical evidence system cannot erase the person who bears its risk. Consent, capacity, privacy, minimization, equipoise, stopping and reviewer custody are operational admission boundaries with complete records, not decorative ethics prose.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:5c29fca29b447880eb5591e1f8bd7d12305c53e713be45c5d966b1bc872d741c.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:e3e4e3b228b980578e7e04f3499e47b924b809816cfce51aab126b0f4eea3bf7.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:d223a21baf273af64b7749363503716f06a738ec2a76d6510215f4580c395cfe.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:30ce5cfafae1e2ac4a0ca7dff02fd0ba5110e99d6f875846e0ab7b82e0b8503.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:12ff743f3e998f349ac1413d444d9eef2ca6a30fa821414565b845d289b1d27c; certificate
sha256:8a1eccd0dba196f09cf79e5808f100cd32e46f4b8055040ce24b5240ec0b5338; external-validation
sha256:539f7e19b75ae54fdd9b1c9e9be845d00dc3f3f000017f52c33a6d430329be13.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 4 purpose-matched discriminators with ordered identity
sha256:df62e80765b3e9eb1f3e41ecdbe47e1203d4585e7c5ed7dfe549fcde25b96ff9. Target opened after

derivation seal: **true**. All rows preserved: **true**. Exact comparison: **true**. A deliberately changed observation was rejected.

Sources:

- MED-ICH-E6R3-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf; snapshot experiments/external_sources/medicine/snapshots/MED-ICH-E6R3-001.pdf;
 sha256:e6ce19e36ce7d2e294f89ee89492b9e035178c3cca48984392bbd92eec9b002c; scope: final Good Clinical Practice: rights, safety, wellbeing, consent, protocol, allocation, blinding, adverse events, confidentiality and essential records.
- MED-COCHRANE-MISSING-EVIDENCE-001 - Cochrane;
<https://www.cochrane.org/authors/handbooks-and-manuals/handbook/current/chapter-13>; snapshot experiments/external_sources/medicine/snapshots/MED-COCHRANE-MISSING-EVIDENCE-001.html;
 sha256:9ebd35b49c689f1b12fa3919051ea9a5ed315b168d9eaec7b1eb8860317d07bb; scope: selective dissemination, missing evidence, benefit overestimation and harm underestimation.

Comparison record:

- sft-med-clinical-equipoise-001-authority-correspondence: predicted medicine:competing-care-support__unresolved-comparative-benefit-harm__monitoring-and-stopping-rules__evidence-date-population-and-oversight; source-derived medicine:competing-care-support__unresolved-comparative-benefit-harm__monitoring-and-stopping-rules__evidence-date-population-and-oversight; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:776fa27f7e68fc1733f1d92520c3630ffcce8a0a8baa5d5bdb0e154a7283d600; derivation seal sha256:052b42f807a8135c49858e2dc9d4a0cf21a21e81baf4288cfcea11396c0b4d03; engine receipt sha256:32be873f515a8ec20f5df36bc4cfa3d393c6defb5ace2fdeaf07b16d089dfdd at receipts/engine/model_admitted/SFT-MED-CLINICAL-EQUIPOISE-001-32be873f515a8ec2.json; empirical validation sha256:4a6ad44f929b7e31420b544860f95122615aa2422be04d543a005433c047584f; measurement receipt sha256:ad1ba2f0ba1d11f898986893a9431646045c2520ab99d4520cd69a6440984af4; isolation sha256:a72d0376de3ec982e281340cc2f325930dbbd2b2fc32df4e1c04cdbac19788c3; custody sha256:88a16171e438f02c5dc7fd811f1ff38a23672ec54c5161f2572c6ebd29559173.

72. Clinical evidence handoff

Claim identity: SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001

Question and exact theorem. A medical claim may cross to a reviewer only with reconstructible derivation, data provenance, code, adverse/null/unresolved rows, privacy boundary and custody record.

consented-deidentified-evidence-record__claim-to-reviewer-reconstruction__derivation-data-code-adverse-results__purpose-access-version-and-custody__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-SEM-SPECIFICATION-001
- SFT-COMP-SEM-VERIFICATION-001
- SFT-COMP-DIST-CAUSALITY-001
- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-MAT-MEAS-TRACEABILITY-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001
- SFT-BIO-PHYSIOLOGY-001
- SFT-BIO-HOMEOSTASIS-001
- SFT-BIO-POPULATION-001
- SFT-BIO-BIO-CONDITION-001
- SFT-BIO-BIO-ASSAY-001
- SFT-BIO-BIO-UNCERTAINTY-001
- SFT-BIO-BIO-CAUSALITY-001

• SFT-MED-CLINICAL-EQUIPOISE-001

The dependency graph independently reaches the premise-free root theorem; a branch name never substitutes for a receipt.

Generated grammar. Generate the Cartesian product of all registered binary preservation dimensions before opening any clinical source or outcome.

Boundary: Exactly eight binary dimensions and therefore 256 forms; closure is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-aggregate-or-answer-only	consented-deidentified-evidence-record	Erasing the patient, population, specimen or evidence carrier prevents clinical reconstruction. The complete generated clinical carrier is retained.
relation	relation-imported-fitted-or-erased	claim-to-reviewer-reconstruction	An imported, fitted or absent relation can manufacture a familiar medical answer. Only the generated transition or comparison relation is admitted.
organization	organization-collapsed	derivation-data-code-adverse-results	Collapsing person, arm, outcome, time or evidence organization merges clinically distinct states. Every required clinical organization distinction remains held.
observation	observation-boundary-unrecorded	purpose-access-version-and-custody	An unrecorded population, method, setting, interval or assessor cannot identify the observation class. The complete declared observation boundary is retained and cannot select the law.
record	favorable-result-without-complete-record	complete-protocol-state-transition-adverse-null-record	A favorable label without protocol, transitions, missingness, adverse outcomes and controls is not auditable. Initial state, transitions, protocol deviations, adverse, null, unresolved and missing rows are recorded.
provenance	authority-consensus-prior-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, prior SFT work or target outcomes may test but cannot select the Fold law. Every decision traces through admitted dependencies to There Is No Nothing.
generality	single-patient-favorable-study	positive-finite-successor-and-unfavorable-closure	One favorable patient, trial, population or analysis cannot close the generated clinical class. The base carrier, every supplied positive finite successor and all registered unfavorable controls preserve the relation at its stated boundary.
extension	free-fit-exception-opaque-model-or-extra-rule	no-extra-rule	A free coefficient, fitted cutoff, hidden exclusion, opaque model or exception can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.

Unique survivor and depth independence. Sole survivor: `consented-deidentified-evidence-record__claim-to-reviewer-reconstruction__derivation-data-code-adverse-results__purpose-access-version-and-custody__complete-protocol-state-transition-adverse-null-record__root-bound-forward-forcing__positive-finite-successor-and-unfavorable-closure__no-extra-rule`.

Base: The least positive finite clinical carrier retains every required identity, relation, boundary, outcome and proof record.

Successor: Adding one lawful finite successor preserves all earlier persons, arms, outcomes, times, adverse and missing distinctions and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- **carrier:** clinical carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the clinical law; passed `true`.

Scientific meaning. The final foundation law makes reproducibility and patient protection compositional: derivation, source custody, code, adverse/null/unresolved evidence and privacy boundaries must all survive transfer to an independent reviewer.

The clinical evidence system cannot erase the person who bears its risk. Consent, capacity, privacy, minimization, equipoise, stopping and reviewer custody are operational admission boundaries with complete records, not decorative ethics prose.

Adverse controls.

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : b942ca613dfbcfa988063023179f9042839514cd1368ee83855109aabc53bd31.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : a796b4665a69b4d224f76090439b6e918eba68c30e0f782e75bb6f5c72b0825f.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 331f6fc3a1fd58f3f9fd580ea2acd4e52a75d8cb2e75277b43eefb72e79b155b.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : c72a578d117a7c260869609d885b03110408675bbdb9296e2693f58c23d27b22.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256 : ca553717c6d3dff1e6e066b412e5d9c999645c82ea609ecf95de681f8bbaf7a0; certificate
sha256 : b1bf8159fa3d92755b14a78a3e6caa47ba47462c8371fdc1dd1ce09ce3e55cee; external-validation
sha256 : 79ff8df86d464f632b1cc20ec1b6cde5ab273e3d5ec4ae5cbf24d4048041aea3.

Post-seal external comparison. The entire 72-law prediction family was sealed before any source identity was chosen. This claim requires 4 purpose-matched discriminators with ordered identity

sha256 : df62e80765b3e9eb1f3e41ecdbe47e1203d4585e7c5ed7dfe549fcde25b96ff9. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- MED-ICH-E6R3-001 - International Council for Harmonisation;
https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf; snapshot
experiments/external_sources/medicine/snapshots/MED-ICH-E6R3-001.pdf;
sha256 : e6ce19e36ce7d2e294f89ee89492b9e035178c3cca48984392bbd92eec9b002c; scope: final Good Clinical Practice: rights, safety, wellbeing, consent, protocol, allocation, blinding, adverse events, confidentiality and essential records.
- MED-COCHRANE-MISSING-EVIDENCE-001 - Cochrane;
<https://www.cochrane.org/authors/handbooks-and-manuals/handbook/current/chapter-13>; snapshot experiments/external_sources/medicine/snapshots/MED-COCHRANE-MISSING-EVIDENCE-001.html;
sha256 : 9ebd35b49c689f1b12fa3919051ea9a5ed315b168d9eac7b1eb8860317d07bb; scope: selective dissemination, missing evidence, benefit overestimation and harm underestimation.

Comparison record:

- sft-med-clinical-evidence-handoff-001-authority-correspondence: predicted medicine:consented-deidentified-evidence-record__claim-to-reviewer-reconstruction__derivation-data-code-adverse-results__purpose-access-version-and-custody; source-derived medicine:consented-deidentified-evidence-record__claim-to-reviewer-reconstruction__derivation-data-code-adverse-results__purpose-access-version-and-custody; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative medical source contradicts a required relation, if another generated form preserves every dimension, or if any patient, arm, outcome, adverse, null, missing, consent or provenance distinction is

erased.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or cutoff
- external target leakage
- opaque predictor
- unrecorded patient or population
- erased adverse, null, absent or unresolved result
- prior answer as premise

Immutable evidence identities. Pre-source seal

experiments/sealed_predictions/medicine_foundation_complete_pre_source.json; source manifest sha256:f1b8aeac1e5a2a2de2396adf8bd367a6919f0c376a187d94af7afb9dc51256e6; derivation seal sha256:d43567a0f6e6d9b0b04d713e34ff708963b14d266fbf1d35b77e6d32f208100d; engine receipt sha256:55e545140e2d2716d836eb426154f3ac606229eff3cd7fd063b00bddd1847b01 at receipts/engine/model_admitted/SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001-55e545140e2d2716.json; empirical validation sha256:62179eec2dfe6e517a6805da24268a64b1850287198cb8cff8d435e5b525ec5f; measurement receipt sha256:fe57afceca941b797d5433bcd55d2af7b417559536d98b856e391c77236cf0a; isolation sha256:981d5cc390673a2129d184fa9f40b02c00e184857f4365c316c6509e0f1d9301; custody sha256:3f745be29a41a4f038addc8f9de7a099fb348214341744f1aa1ec6934a2a1eb6.

22. Integrated audit result

- Prior entries reviewed: 763.
- Medicine-relevant prior rows: 4.
- Medicine-owned atoms: 5.
- Same-strength V3-closed atoms: 5.
- Open atoms: 0.
- Corrected prior atoms: 5.
- Audit identity: sha256:0bbfbbb18f3c348bc2ed62273e68f745b87e3e0ba8892269ca6e6c302f371f75.

23. External-source ledger

- MED-WHO-HEALTH-001 — <https://www.who.int/about/frequently-asked-questions>; captured sha256:274c2ad78787f127a9f1cf8a479c691e1f89b263d0e21dd49b1753f7805d64f4; text sha256:4a895afb8bd430d6a619d8f07c5e45c199298dd67a16fd05fa254e47042a345d; health state and disease-absence distinction.
- MED-FDA-BIOMARKER-001 — <https://www.fda.gov/drugs/biomarker-qualification-program/about-biomarkers-and-qualification>; captured sha256:93c5ee4b5e3480ed27aa805a7739f4acf7108701fc6f429ae5d4e5a1ad9d3d31; text sha256:f82a712f63b6c8e0de9e2807038051f549a09deccaea480240f9f44e637023fa; normal and pathogenic processes; risk, diagnostic, monitoring, prognostic, response and safety biomarkers; method and context.
- MED-CDC-EPIDEMIOLOGY-001 — https://archive.cdc.gov/www_cdc_gov/csels/dsepd/ss1978/lesson1/summary.html; captured sha256:430c2d885670c185491ab9b84d1c703eeaff0eece3ddb69fc855c001fb93478b; text sha256:8a6116f8803b6a7f649053b0295feb8faabc17c3ad4513430f2fdda39fe9b7d3; population,

comparison, frequency, pattern, cause, disease and injury epidemiology.

- MED-ICH-E6R3-001 — https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf; captured sha256 : e6ce19e36ce7d2e294f89ee89492b9e035178c3cca48984392bbd92eec9b002c; text sha256 : 66bf30ef9101596bd9e70ae82e7d55bab519a554c50c77b99670f2294c949592; final Good Clinical Practice: rights, safety, wellbeing, consent, protocol, allocation, blinding, adverse events, confidentiality and essential records.
- MED-ICH-E9R1-001 — https://database.ich.org/sites/default/files/E9-R1_Step4_Guideline_2019_1203.pdf; captured sha256 : f7471f411f1c87ee76783d5b2b9faeaca31d01d530213f71b50d136b39e3b0d9; text sha256 : 95bcb99e56eb725947eb2acb0b9f0bfe4c6efedb9ffaba18b8232c700e7ff0f5; final estimand, population, endpoint, intercurrent-event, missing-data and sensitivity-analysis distinctions.
- MED-FDA-EXPOSURE-RESPONSE-001 — <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/exposure-response-relationships-study-design-data-analysis-and-regulatory-applications>; captured sha256 : 748ca63404e58ef7f56d321395e646aa918f1ba0e679dbab98e62be95bc40291; text sha256 : 38b68aea1735c0e5cee35f0383ea9c2a4959431cf3907c078d425583e87c4338; dose, exposure, response, benefit and risk study boundary.
- MED-COCHRANE-HANDBOOK-001 — <https://www.cochrane.org/authors/handbooks-and-manuals/handbook/current>; captured sha256 : 31f56118fc994e14f9703298b798983c2039c6328ee4c5f983f53bdef2bc0004; text sha256 : 784ffb274b550dcc55be18189542515ab5d2d8e73dd9d86e815ee0b00e91bf57; current intervention-review scope, study selection, bias, adverse effects and interpretation.
- MED-COCHRANE-MISSING-EVIDENCE-001 — <https://www.cochrane.org/authors/handbooks-and-manuals/handbook/current/chapter-13>; captured sha256 : 9ebd35b49c689f1b12fa3919051ea9a5ed315b168d9eaec7b1eb8860317d07bb; text sha256 : a0b274f7613fdc00437f68193541343ad00b26a145dcba905617a9e708f4c4e9; selective dissemination, missing evidence, benefit overestimation and harm underestimation.
- MED-WHO-SCREENING-001 — <https://www.who.int/publications/i/item/9789289054782>; captured sha256 : b56f491a97d2efbe4ff4d59e304649dd1fb1d551a2ca6b88a9bbcf43d385b5f7; text sha256 : 8c39fbef90e52cc44b728718e3b25de977c522f7be189d5d72e7415fcbab6708e; screening population, risk, early intervention, benefits, harms and quality assurance.
- MED-HHS-CONSENT-001 — <https://www.hhs.gov/ohrp/regulations-and-policy/guidance/faq/informed-consent/index.html>; **transport failed and preserved** sha256 : f2dda5323f2dfa71d16fad0fb1aa65c5d3ecf0c567603dba0d2076ca0517f291; prospective informed-consent identity; transport failure preserved and not used for a passing comparison.
- MED-HHS-PRIVACY-001 — <https://www.hhs.gov/hipaa/for-professionals/privacy/guidance/minimum-necessary-requirement/index.html>; **transport failed and preserved** sha256 : f2dda5323f2dfa71d16fad0fb1aa65c5d3ecf0c567603dba0d2076ca0517f291; minimum-necessary privacy identity; transport failure preserved and not used for a passing comparison.
- MED-NICHD-REHABILITATION-001 — <https://www.nichd.nih.gov/health/topics/rehabilitation-medicine>; captured sha256 : c1b54134e6cbcfb9542a7c012236395eb609a67a789818f201c5b42202dc8cae; text sha256 : cfecdd3cd39e0a9363ef0d039a740808cb545f8386f29d1f2b29760e394bd808; rehabilitation function, participation, independence, quality of life and outcomes.
- MED-STROBE-001 — <https://www.strobe-statement.org/>; captured sha256 : 420432c148d01e7aaaab8f7bcb4bcb4412d028aa50cd7814fc1b058faf1cf7c; text sha256 : 846530d8dfd2b38331defa6388d14ff7d9eeb4ddeef248f0435428549c18c996; observational-study planning, conduct, findings and interpretation reporting.

The failed HHS transports are evidence about transport, not evidence for the clinical laws. They remain visible, while ICH supplies independently captured purpose-matched content. No unavailable source is converted into a successful row.

24. Reproducibility and falsification

The cross-platform Medicine suite checks inventory/spec alignment, all 18,432 candidates, 72 unique survivors, dependency order, source and failure hashes, every purpose-matched fragment, the two exact four-cell certificates and prohibited blueprint fields. The immutable engine and verification-authority seals run before admission and again at completion. The publication gate was frozen before any Medicine receipt existed.

The branch is falsified at its declared boundary if any source hash changes without a new version, any required fragment is absent, any target becomes accessible before sealing, any adverse/null/missing/unresolved or failed row is omitted, any tampered target is accepted, any dependency loses its root path, any candidate census is incomplete, any law has other than one survivor, any independent validator fails, or any patient privacy boundary is violated.

25. Full-field Medicine and Health Sciences roadmap

This paper completes Layer One. Later versions extend in dependency order through anatomy, physiology and pathology at the medical boundary; epidemiology and causal inference; diagnostics, imaging, laboratory medicine and biomarkers; pharmacology, pharmacokinetics, toxicology and therapeutics; infectious and immune disease; genetic, metabolic and developmental disorders; cardiovascular, respiratory, renal, gastrointestinal and endocrine medicine; neurology, psychiatry and behavioural health with a strict Consciousness handoff; oncology and haematology; dermatology, musculoskeletal and autoimmune medicine; reproductive, maternal, neonatal, paediatric and ageing medicine; surgery, anaesthesia, critical care, trauma and rehabilitation; dentistry, vision and hearing; primary, emergency and integrated care; nursing and allied-health evidence; public, occupational, environmental and global health; nutrition and prevention; precision medicine; health-services and implementation evidence; clinical informatics and verified medical AI; devices and biomaterials; replication, adverse outcomes and limits of generalization.

The paper will be versioned as those families are independently completed. Applications, institutions, treatment preferences and regulatory outcomes may test the laws but may never select them.

26. Limitations

- The 72-law foundation is current-evidence closed inside its frozen grammar and remains extension-open; it is not the complete specialty-by-specialty reconstruction.
- Most results are exact relational and evidence-constitution laws, not universal patient magnitudes.
- Authority correspondence is a post-seal content test, not a premise and not a claim that every authority statement is correct.
- Two registered HHS endpoints were unavailable to automated capture and remain unfavorable transport rows.
- No clinical decision, diagnosis or treatment recommendation for a real person follows from this foundational paper.
- Biology retains living mechanism, Chemistry molecular identity, Materials devices and interfaces, and Consciousness subjective-experience laws; Medicine cites but does not reown them.
- Ethical and privacy claims here are operational evidence-admission boundaries. They do not claim to exhaust moral philosophy or every jurisdiction's law.

27. Conclusion

Foundational Medicine is current-evidence closed and extension-open: 72 required laws, 18,432 exact candidates and decisions, 72 sole survivors, 288 passing adverse controls, 72 independent reconstructions, 72 post-seal comparisons, 72 receipt-backed root traces, and five of five legacy Medicine atoms reconciled at stronger clinical boundaries.

The result is not a glossary and not a conventional clinical model decorated with Fold language. It is a public constitution that refuses to erase the patient, comparator, alternative diagnosis, competing event, adverse outcome, missing result, consent scope or privacy boundary. It derives auditable randomisation in a superdeterministic ontology, separates association from cause, separates safety evidence from harmlessness, and makes individual inference explicitly different from population inference.

That is the empirical gravity of this branch: a medical statement is admitted only when its exact carrier, alternatives, controls, evidence custody and root trace survive independent reconstruction. Open criticism remains unrestricted. Scientific admission remains earned.

28. Repository and publication status

- Canonical repository: <https://github.com/MettaMazza/ernos-labs-sft-platform>
- Zenodo DOI: not created; separate authorization required
- Author: Maria Smith, Ernos Labs
- Contact: Maria.Smith.Sftoe@gmail.com
- Submissions: <https://discord.gg/ucwGryVxGr>
- Current state: local prepublication; no remote action performed

29. References

- World Health Organization. Constitution and definition of health.
- United States Food and Drug Administration and National Institutes of Health. BEST biomarker terminology and context of use.
- Centers for Disease Control and Prevention. *Principles of Epidemiology in Public Health Practice*.
- International Council for Harmonisation. *E6(R3) Guideline for Good Clinical Practice*. Final version, 2025.
- International Council for Harmonisation. *E9(R1) Addendum on Estimands and Sensitivity Analysis in Clinical Trials*. Final version, 2019.
- United States Food and Drug Administration. *Exposure-Response Relationships: Study Design, Data Analysis, and Regulatory Applications*.
- Cochrane. *Handbook for Systematic Reviews of Interventions*, version 6.5.
- World Health Organization Regional Office for Europe. *Screening Programmes: A Short Guide*.
- Eunice Kennedy Shriver National Institute of Child Health and Human Development. Rehabilitation Medicine.
- STROBE Initiative. *Strengthening the Reporting of Observational Studies in Epidemiology*.
- Smith, Maria. *From Nothing to Fold*. doi:10.5281/zenodo.21515629.
- Smith, Maria. *From Fold to Mathematics*. doi:10.5281/zenodo.21516146.
- Smith, Maria. *From Distinction to Information*. doi:10.5281/zenodo.21516916.
- Smith, Maria. *From Fold to Physics*. doi:10.5281/zenodo.21520881.
- Smith, Maria. *From Fold to Chemistry*. doi:10.5281/zenodo.21531455.
- Smith, Maria. *From Fold to Life*. doi:10.5281/zenodo.21630203.

Open-science evidence supporting the institutional argument:

- Lundh A et al. *Conflicts of interest at the European Medicines Agency: a policy analysis*. 2018. <https://doi.org/10.1007/s00134-018-5293-7>.
- Fabbri A et al. *The influence of industry sponsorship on the research agenda*. 2018. <https://pubmed.ncbi.nlm.nih.gov/30252531/>.
- Gallo SA et al. *Reliability and fairness in peer review of research funding*. 2023. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10553257/>.
- Demicheli V and Di Pietrantonj C. *Peer review for improving the quality of grant applications*. Cochrane review and archive. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8973940/>.
- UNESCO. *Recommendation on Open Science*. 2021. <https://www.unesco.org/en/legal-affairs/recommendation-open-science>.
- NIST. *Artificial Intelligence Risk Management Framework 1.0*. <https://airc.nist.gov/airmf-resources/airmf/3-sec-characteristics/>.